Pierce County
Public Works and Utilities
Surface Water Management

Water Year 2011
Annual Stormwater Monitoring Report

NPDES Municipal Phase 1 Permit (Effective February 2007)

Section S8.D - Stormwater Characterization Monitoring

March 2012

Pierce County Public Works and Utilities
Surface Water Management Division
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1 Introduction

The NPDES Phase I Municipal Stormwater Permit (permit) regulates stormwater discharges from large and medium municipal separate storm sewer systems (MS4s), as established in the federal Clean Water Act (Title 40 Code of Federal Regulations, part 122.26). The current permit was issued by the Washington Department of Ecology (Ecology) with an effective date of February 16, 2007, and was modified on June 17, 2009 and September 1, 2010. Pierce County is one of the municipalities with a population exceeding 100,000 that is regulated under the permit.

Special Condition 8 (S8) of the permit requires municipalities to conduct four types of stormwater-related monitoring: S8.D Stormwater Characterization Monitoring; S8.E Targeted Stormwater Management Program Effectiveness Monitoring; S8.F Stormwater Treatment and Hydrologic Management Best Management Practice (BMP) Evaluation Monitoring, and S8.F Flow Reduction Monitoring.

Annual reporting is required under Section 8.H - Monitoring Program Report Requirements. Report contents are based on the *Stormwater Monitoring Report Guidance – Phase 1 Municipal Stormwater Permit Reporting Requirements for Special Condition 8* (Ecology Publication No. 10-10-028).

This report presents results of S8.D stormwater characterization monitoring for Water Year 2011 (October 1, 2010 – September 30, 2011). The goal of this monitoring effort is to characterize stormwater runoff as it relates to different land use types. It also allows analysis of pollutant loadings and changes in water quality conditions over time. Pierce County may use data from our study to gauge whether the County's Stormwater Management Program (SWMP) is making progress toward pollution reduction, and to help define any adaptive strategies for improving the SWMP.

A Quality Assurance Project Plan (QAPP) for the S8.D monitoring program was prepared for Pierce County by Herrera Environmental Consultants dated November 5, 2009: *Quality Assurance Project Plan. Pierce County Phase 1 Municipal Stormwater NPDES Permit Section S8.D -Stormwater Characterization* (Herrera, 2009). The QAPP was conditionally approved by Ecology on September 8, 2009, and monitoring commenced in spring, 2010. The monitoring work is being implemented by Pierce County Public Works and Utilities - Surface Water Management Division (SWM) staff or contractors under the direction of SWM.

The S8.D data for qualifying events collected in Water Year 2011 is not required to be submitted to Ecology's Environmental Information Management system but is included in appendices to this report.

An adequate number of qualifying samples have not been collected at any of the sites for annual loading calculations. This report presents the data collected thus far. It should be considered a status report. A final report including loading calculations will be provided next water year, when an adequate data set has been collected, or monitoring requirements have been terminated.

2 Monitoring Locations

The permit requires each Permittee to monitor stormwater runoff and sediment from three different land use types: low density residential, high density residential and commercial. The locations, associated catchment areas, and revisions to these areas are discussed below.

2.1 Low Density Residential Site (LORES)

Low density residential land use is being monitored near Sylvia Lake, in unincorporated Pierce County, west of Gig Harbor. The specific outfall monitoring station is located near the corner of 81st Ave. NW and 68th Street NW. The site captures runoff from residential subdivisions, roads, farms, and forested areas northeast of the monitoring site. Monitoring is occurring from a type 2 catch basin with a 36-inch inlet pipe and a 36-inch outlet pipe.

A Sigma A/V sensor and a Sigma Model 900 MAX refrigerated sampler are being used to monitor level, velocity and flow and to collect water quality samples. The Sigma A/V sensor is installed above the inlet pipe to the catch basin which monitors water levels and velocities in the pipe every 15 minutes. The sampler is protected in metal housing bolted to a concrete pad. A Sigma rain gauge is installed above the housing and is integrated with the data logger on the sampler. Precipitation is being monitored at 15-minute intervals.

Figure 1 shows the boundary, land uses, and surface drainages for the catchment area associated with the LORES site as shown in the QAPP. As a quality assurance check the rainfall depth was compared to the runoff depth throughout the monitoring year. For forested basins in the Pacific Northwest, the percent reduction (via evapotranspiration) between rainfall depth and runoff depths can be as high as 55 percent (Post and Jones 2001). For highly impervious basins, the percent reduction (via evapotranspiration) between rainfall depth and runoff depth should be approximately 5 percent (Herrera 2008). Percent reduction for low density residential land use was anticipated to be between these values, and closer to values for forested areas due to low residential density and high vegetation cover. However, the calculated values exceeded the expected range and were as high as 99% for some storm events.

In response, a Pierce County GIS specialist redelineated the catchment area for the LORES site. Analysis resulted in the catchment size for the LORES site being downsized from approximately 218.4 acres as depicted in the QAPP, to 107 acres. Also, the drainage is configured slightly differently. The revised catchment area for the LORES site is shown in Figure 2. The error was attributed to human error and/or the unavailability of more accurate LIDAR maps at the time the QAPP was written. The residential density within the revised drainage area is approximately one dwelling unit per acre, thus the existing monitoring site continues to represent low-density residential land use. The revised catchment area will be used for future annual loading calculations.

2.2 High Density Residential Site (HIRES)

High density residential land use is being monitored near the City of Puyallup, in the Clover Creek drainage system. The site is referred to as the HIRES (high density residential) site. The outfall monitoring station is located on a drainageway near the intersection of 128th Street East and 82nd Avenue Court East, at the end of the cul-de-sac on 82nd Avenue Court East. Figure 3 shows the catchment area associated with this site as shown in the QAPP; the QAPP estimated the catchment area to be 125 acres.

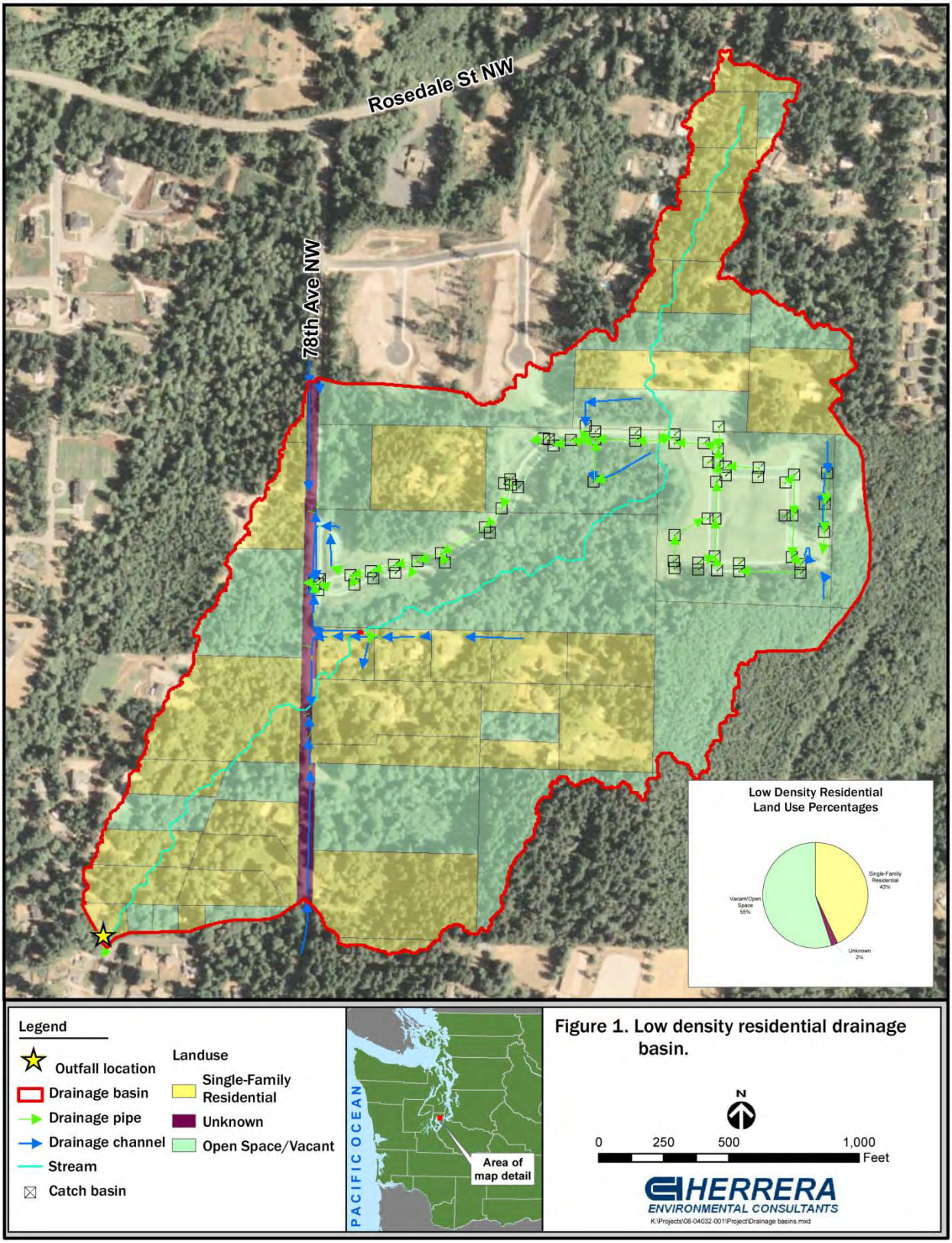

Figure 1 - Catchment Area for LORES Site Shown in QAPP

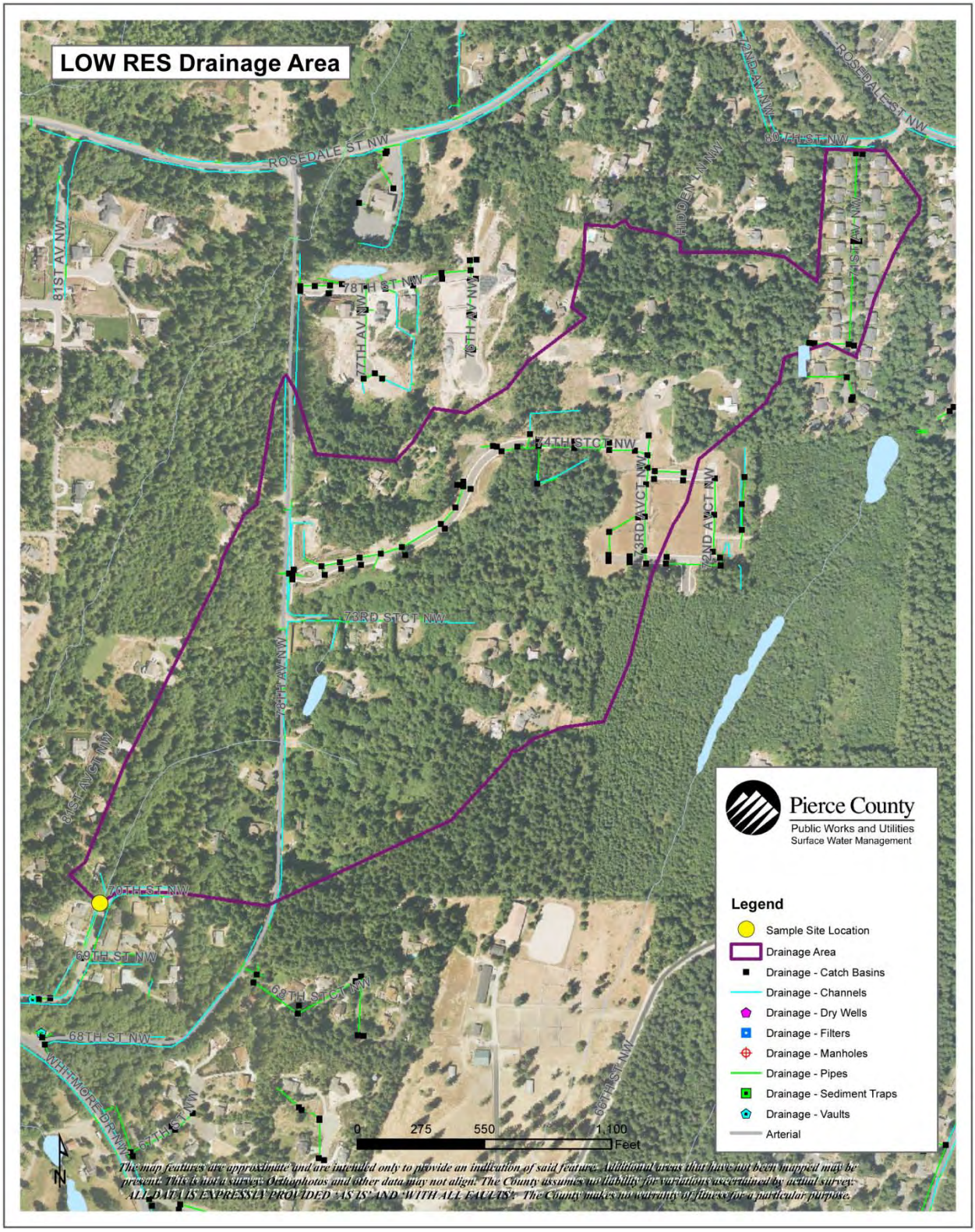

Figure 2 - Revised Catchment Area for LORES Site

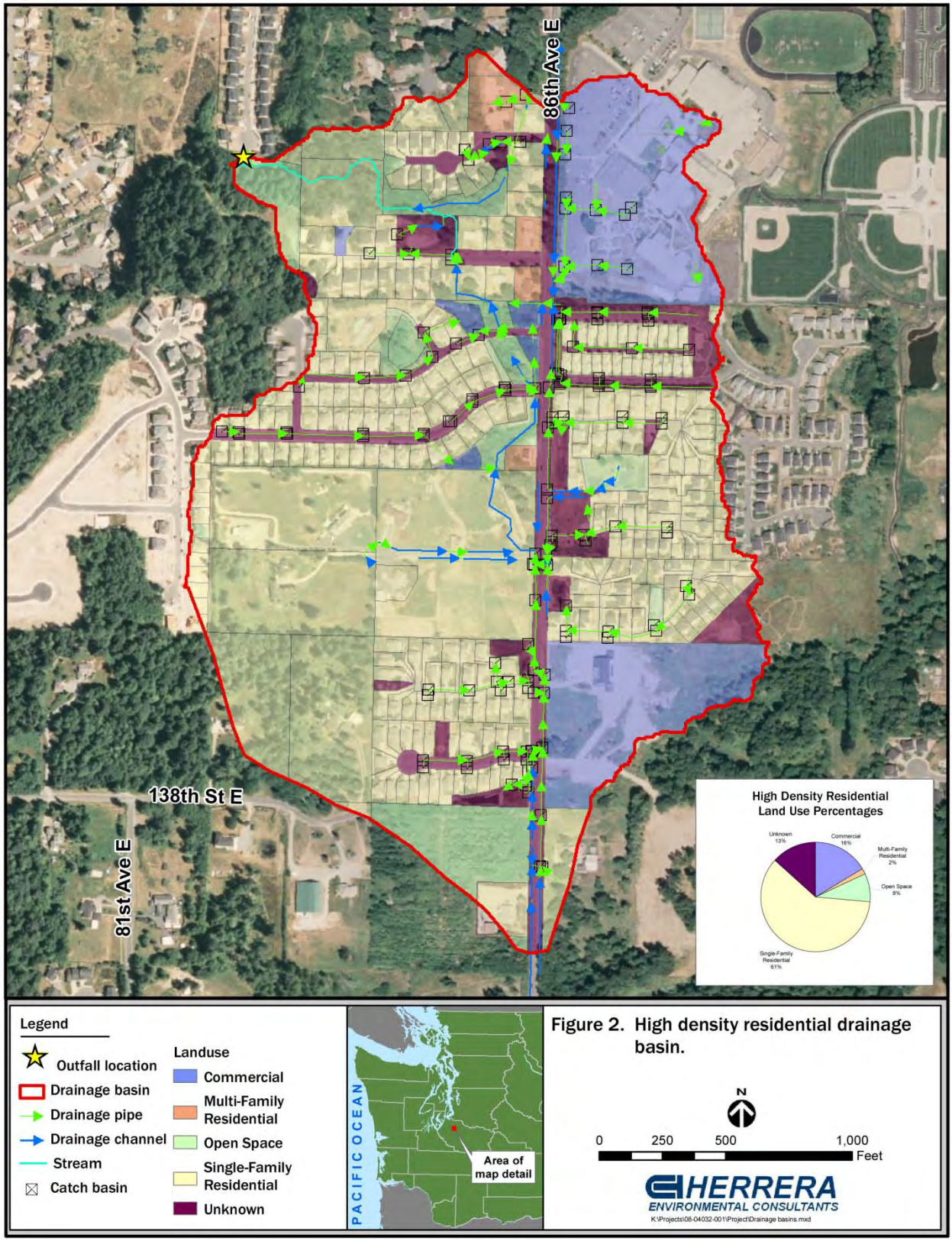

Figure 3 - Catchment Area for HIRES Site Shown in QAPP

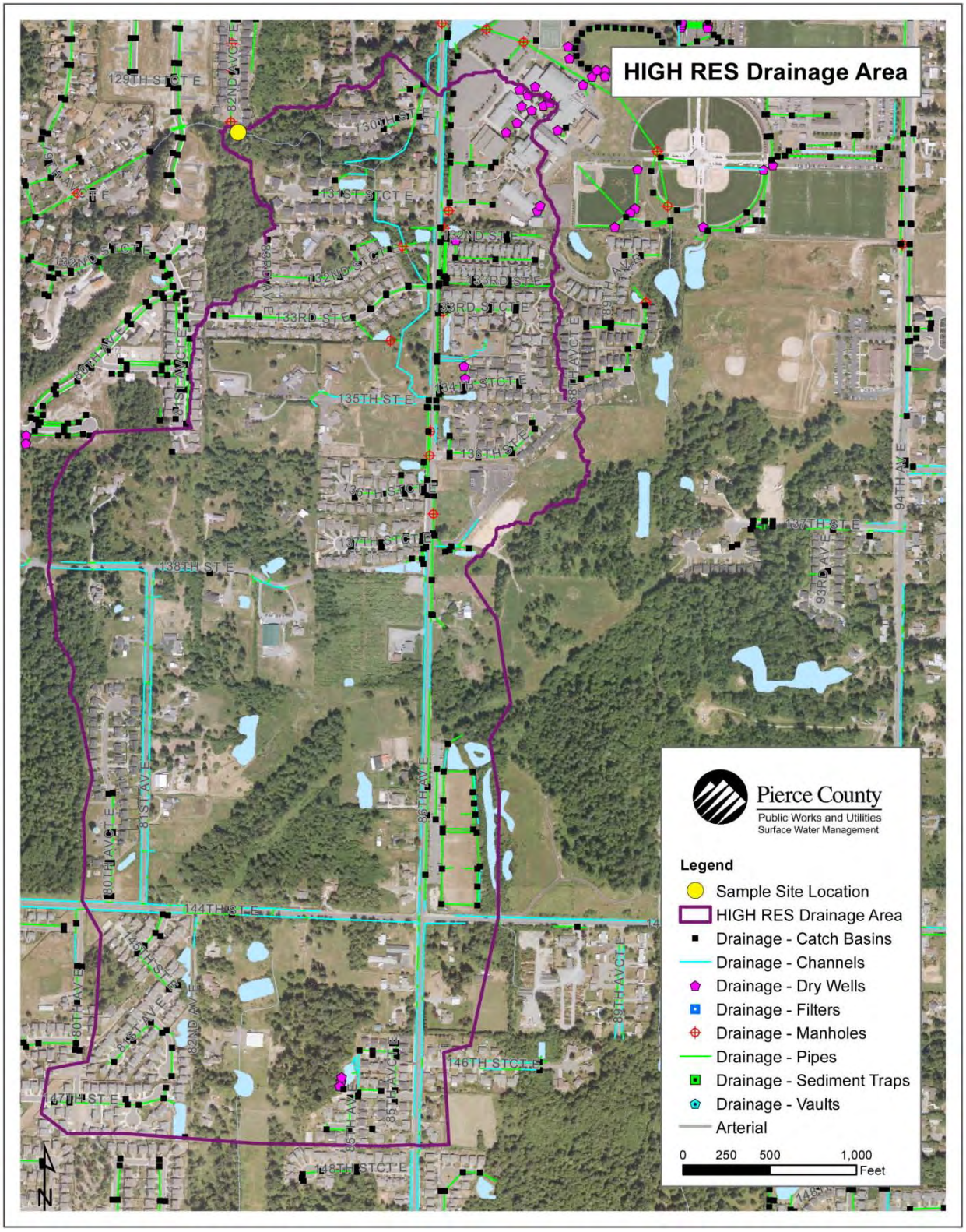

Figure 4 - Revised Catchment Area for HIRES Site

A Sigma A/V sensor and a Sigma Model 900 MAX refrigerated sampler are being used to monitor level, velocity and flow and to collect water quality samples. In this water year, flow was directed with sand wing walls through a 24-inch ADS pipe to improve monitoring. The Sigma A/V sensor is installed (slightly offset) within the ADS pipe, and monitors water levels and velocities in the pipe every 15 minutes. The sampling equipment is protected in metal housing bolted to a concrete pad. A Sigma rain gauge is installed above the housing and is integrated with the data logger on the sampler. Precipitation depth is being monitored at 15-minute intervals.

In response to the error found at the LORES site, a Pierce County GIS specialist redelineated the catchment area for the HIRES site based on LIDAR imagery and more recent GIS data on the county's drainage sytem. The catchment area was redelineated as shown in Figure 4. The revised are is approximately 313.5 acres in size. The resulting residential density is approximately 1.5 dwelling units per acre. The revised catchment area will be used for annual loading calculations.

2.3 Commercial Site (COMM)

The site selected and monitored as the commercial site is located near the intersection of 112th Street East and Canyon Road East. The area includes a bank, hardware store, and other retail businesses. The area is predominantly impervious, and consists of portions of 112th Street East and Canyon Road East, rooftops, and parking lots. Stormwater from the catchment area collects through a series of stormwater conveyance pipes, then flows into an 18-inch pipe that discharges to a type 2 catch basin with multiple orifices. This catch basin acts as a flow control structure for a downstream Contech StormFilter stormwater treatment system. The outfall monitoring station is located within the 18-inch pipe, to facilitate stormwater sampling prior to any treatment in the StormFilter system. The effectiveness of the Contech StormFilter system is being monitored under Section 8.F of the permit.

A Sigma A/V sensor and a Sigma Model 900 MAX refrigerated sampler are being used to monitor level, velocity and flow and to collect water quality samples. The Sigma A/V sensor is installed above the inlet pipe to the catch basin which monitors water levels and velocities in the pipe every 15 minutes. All equipment is housed within a fenced area protecting a Pierce County Public Works and Utilities –Roads StormFilter system. A Sigma rain gauge is installed above the housing and is integrated with the data logger on the sampler. Precipitation depth is being monitored at 15-minute intervals.

The QAPP estimated the catchment area for the COMM site to be 11.2 acres. In response to the error found at the LORES site, a Pierce County GIS specialist redelineated the catchment area for the COMM site based on LIDAR imagery and more recent GIS data on the county's drainage system. The catchment area was expanded to include additional portions of 112th Street East, Canyon Road East, and the rooftop and parking area of the Pierce County Library – Summit Branch. Figures 5 and 6 show the catchment area delineated in the QAPP and the revised catchment area, respectively. The acreage has been revised to 21.2 acres. The revised catchment area will be used for all loading calculations.

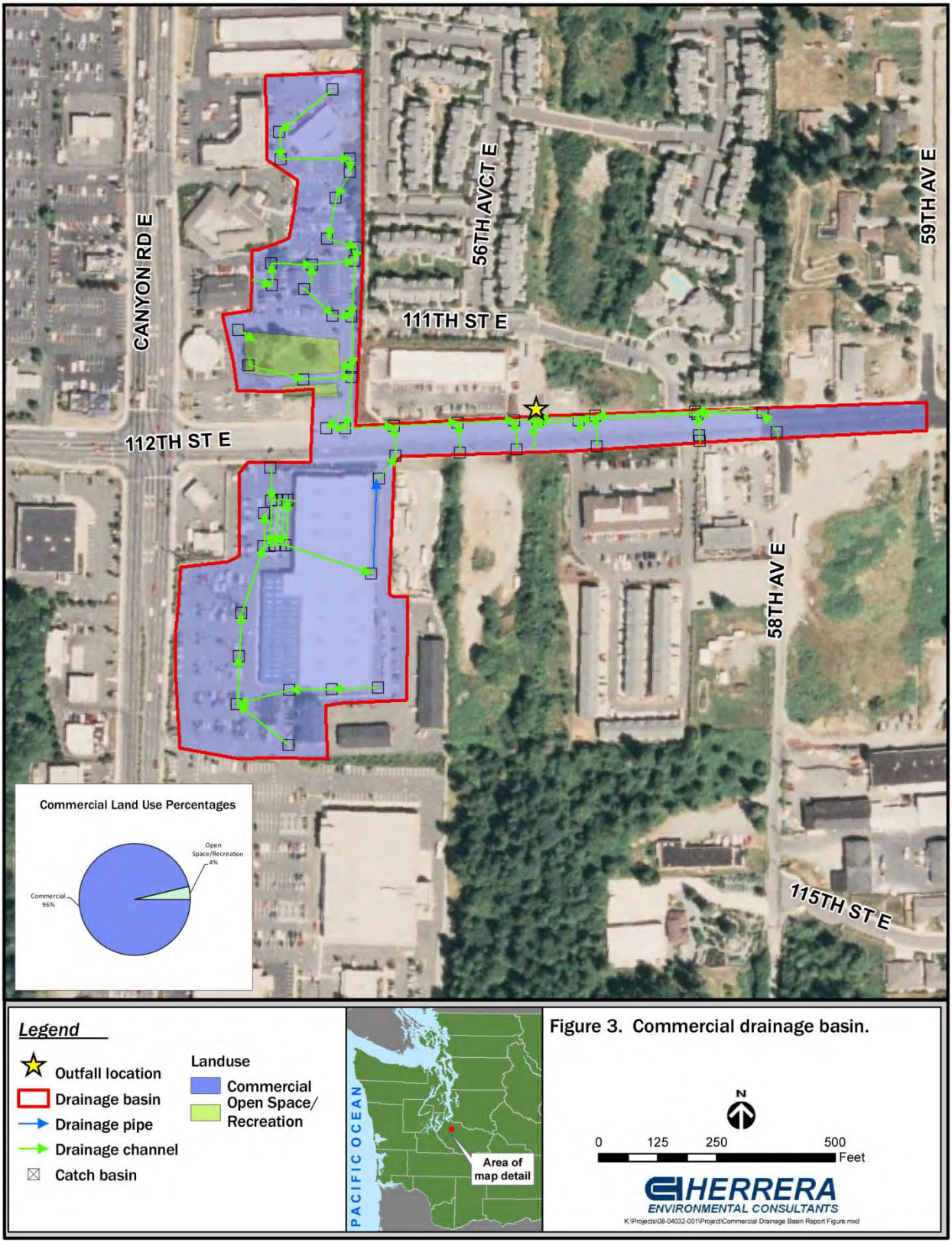

Figure 5 - Catchment Area for COMM Site Shown in QAPP

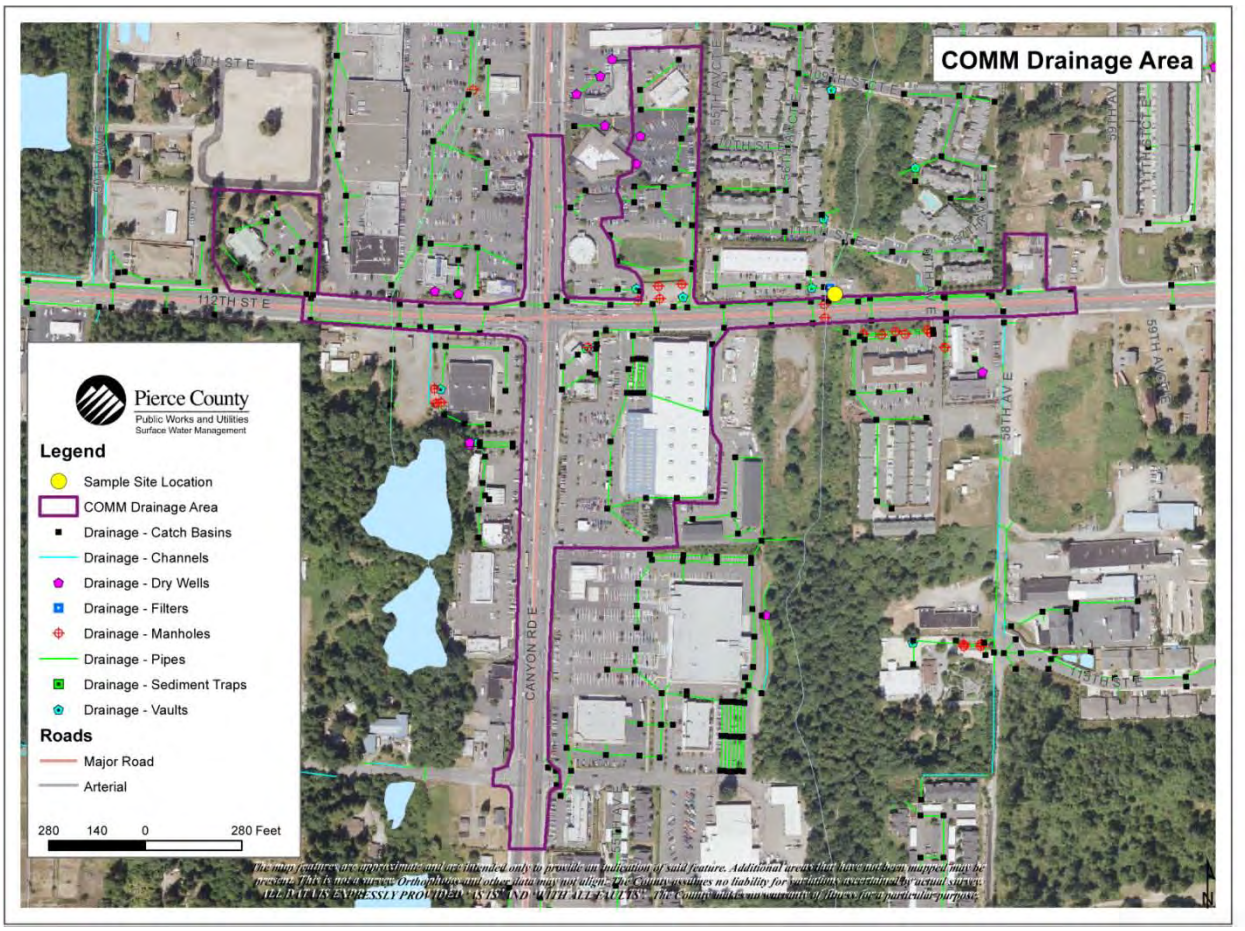

Figure 6 - Revised Catchment Area for COMM Site

Revised catchment area sizes for the three outfall sites are shown in Table 1 – Revised Catchment Sizes for S8.D Monitoring Sites

Table 1 – Revised Catchment Sizes for S8.D Monitoring Sites

Monitoring Site	Catchment Size per QAPP (acres)	Revised Catchment Size (acres)
LORES	218.5	107
HIRES	125	313.5
COMM	11.2	21.2

3 Data Collection

3.1 Storm Targeting

Storms were targeted for sampling from October 1, 2010 until September 30, 2011. The criteria defined in the QAPP for sampling and storm representativeness are presented in Table 2 - QAPP Criteria for Storm Representativeness

Weather was monitored each morning on the National Weather Service website and MxVision Weather Sentry Online. Per the QAPP, sampling was to be attempted if the storm had a 75% or greater probability of occurring, was likely to occur within the next 24 hours, would be at least 0.2 inches, met the antecedent dry period; and did not occur on holidays. Seventeen events met these criteria in Water Year 2011. Although the QAPP specifies that the event will have a 75% or greater probability of occurring staff targeted events with as low as 50% probability. Therefore, the number of attempts exceeded 17 at all of the sites. Per field notes, attempts to sample forecasted storms totaled 26 at the LORES site, 30 at the HIRES site and 18 at the COMM site. Not all attempts resulted in samples being pulled and/or analyzed by the laboratory.

Table 2 - QAPP Criteria for Storm Representativeness

Seasonal Period	Wet Season October 1 – April 30 Dry Season May 1 – September 30
Rainfall depth	≥ 0.2 inches
Rainfall duration	No fixed minimum or maximum
Antecedent dry period	≤.02 inches rain in previous 24 hrs. (wet season) ≤.02 inches rain in previous 72 hrs. (dry season)
Inter-event dry period	6 hours
Aliquots	≥ 10
Percent Coverage	≥75% of storm hydrograph if storm <24 hrs, ≥75% of first 24 hrs. of storm hydrograph if storm > 24 hrs.

The permit allows analysis of up to a maximum of three (3) samples that are collected as a result of attempts to sample the eleven (11) required storm events that do not meet the rainfall volume storm event criterion, but do meet the other storm event and sample criteria. None of the samples collected met these conditions.

The rainfall vs. runoff relationship was mapped to help with future estimates of unsampled storm events (where water quality samples are not collected). Results for the three monitoring sites for Water Year 2011 are shown in Figures 7 - 9. This information will be used for future sample pacing and estimations of annual and seasonal loads.

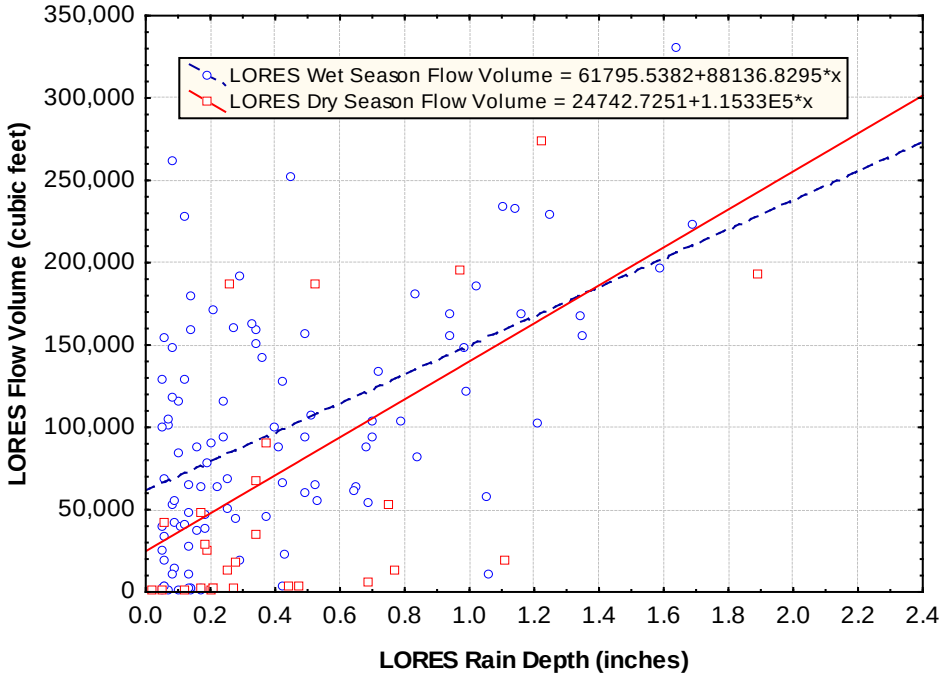

Figure 7 - Rainfall Runoff Relationship for LORES Site

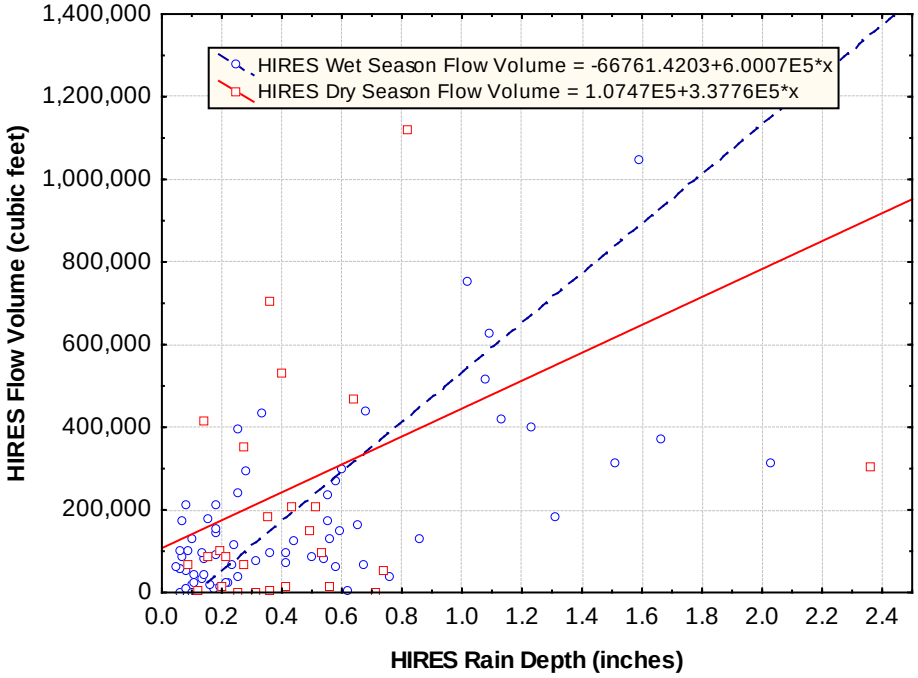

Figure 8 - Rainfall Runoff Relationship for HIRES Site

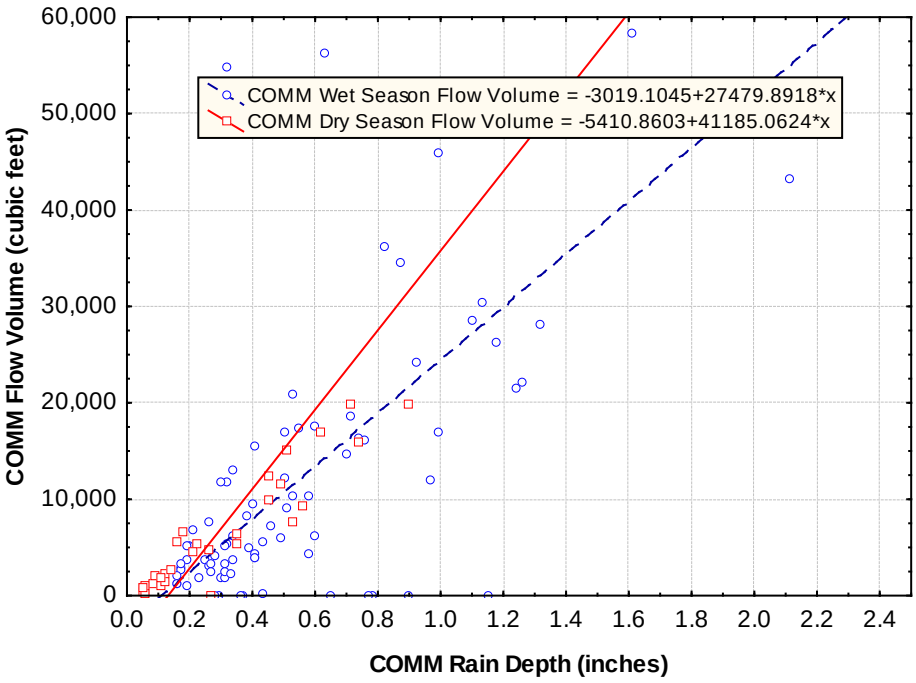

Figure 9 - Rainfall Runoff Relationship for COMM Site

3.2 Storm Sampling

Tables 3 through 5 present the sampling and storm statistics for all sampled events (qualifying and non-qualifying) at the LORES, HIRES, and COMM monitoring locations. The chemistry results from the storms that did meet permit criteria are discussed in more detail in the Stormwater Sampling Section. It is noted on the tables which specific storm criteria outlined in the permit was not met. If criteria were met but the sampled event is still not bolded it means that other issues such as sampling across multiple events, splitting events, or poor hydrograph quality, made the event non-qualifying.

Table 3 - Sampling and Storm Statistics for Each Sampled Event at LORES Site

Event Date	Rain Depth (inches)	Antecedent Dry Period (hours)	Aliquots (count)	Coverage (%)	Criteria Not Met
11/5/2010	**1.06**	**240.5**	**13**	**89**	-
11/9/2010	**0.65**	**37.2**	**22**	**grab**	-
11/17/2010	**0.42**	**36.2**	**8**	**grab**	-
11/29/2010	**1.16**	**70.2**	**82**	**grab**	-
1/12/2011	1.05	113.8	102	35	% coverage
1/20/2011	0.1	56.8	100	20.5	Rain depth, % coverage
1/28/2011	0.07	88	107	33.9	Rain depth, % coverage
2/4/2011	0.08	140.5	8	6.3	Rain depth, aliquots, % coverage
2/12/2011	**0.49**	**110.2**	**110**	**89**	-
2/27/2011	**1.69**	**71.5**	**94**	**82.2**	-
3/8/2011	**0.33**	**79.8**	**113**	**89.3**	-
3/8/2011	**0.33**	**79.8**	**113**	**grab**	-
3/18/2011	0.42	42	45	86.8	Sampling error
3/20/2011	0.09	49	17	38.3	Rain depth, % coverage
4/10/2011	0.22	16.2	39	62.6	Antecedent dry period, % coverage
4/24/2011	0.7	10.2	63	69.8	Antecedent dry period, % coverage
5/2/2011	0.17	100.2	34	73.3	Rain depth, % coverage
5/25/2011	0.34	98.2	28	42.6	% coverage
5/26/2011	0.26	22.5	28	41.3	Antecedent dry period, % coverage
7/25/2011	0.36	183.2	99	17.7	% coverage
8/22/2011	*0.19*	*674.8*	*47*	*93.3*	-
9/25/2011	0.18	70.2	90	97.3	Rain depth, antecedent dry period
9/26/2011	1.22	24.2	11	3.9	Antecedent dry period, % coverage

Bold values are qualifying samples that meet all the storm and sampling criteria.
Italicized values are for the time-weighted toxicity sample

Table 4 - Sampling and Storm Statistics for Each Sampled Event at HIRES Site

Event Date	Rain Depth (inches)	Antecedent Dry Period (hours)	Aliquots (count)	Coverage (%)	Criteria Not Met
11/9/2010	**0.41**	**181.5**	**113**	**83.9**	**-**
11/15/2010	0.25	131.5	113	19.3	% coverage
11/17/2010	**1.02**	**34.2**	**226**	**grab**	**-**
11/29/2010	0.33	261	113	37.2	% coverage
12/7/2010	0.68	126	113	51.7	% coverage
12/8/2010	1.08	10.8	113	65.6	Antecedent dry period, % coverage
12/11/2010	2.93	28	339	24	% coverage
1/12/2011	**1.66**	**40.5**	**113**	**grab**	**-**
1/20/2011	0.65	8.5	67	23.8	Antecedent dry period, % coverage
1/28/2011	0.07	87	113	32	Rain depth, % coverage
2/4/2011	0.08	8	65	98.7	Rain depth, antecedent dry period
2/5/2011	0.1	32.8	5	23.8	Rain depth, aliquots, % coverage
2/12/2011	0.67	116.5	57	87.1	Sampling error
2/13/2011	0.55	16	46	24.6	Antecedent dry period, % coverage
2/27/2011	1.31	85.8	46	91.1	Sampling error
3/8/2011	0.13	80	27	84.7	Rainfall depth, % coverage
3/9/2011	2.03	17.8	31	30.1	Antecedent dry period, % coverage
3/18/2011	**0.44**	**42**	**113**	**93.4**	**-**
4/10/2011	**0.36**	**61.8**	**73**	**90**	**-**
4/25/2011	**0.76**	**8**	**48**	**97.2**	**-**
5/2/2011	**0.2**	**84.8**	**20**	**78.2**	**-**
5/25/2011	**0.27**	**86**	**113**	**88**	**-**

Bold values are qualifying samples that meet all the storm and sampling criteria.

Table 5 - Sampling and Storm Statistics for Each Sampled Event at COMM Site

Event Date	Rain Depth (inches)	Antecedent Dry Period (hours)	Aliquots (count)	Coverage (%)	Criteria Not Met
10/23/2010	2.09	8.2	49	94.4	Antecedent dry period
11/5/2010	**1.4**	**95.8**	**16**	**75**	**-**
11/7/2010	0.11	10	1	0	Rain depth, antecedent dry period, aliquots, % coverage
11/9/2010	**0.5**	**36.2**	**11**	**83.3**	**-**
11/11/2010	0.17	37.8	1	0	Rain depth, aliquots, % coverage
11/13/2010	0.84	14.8	14	74.5	Antecedent dry period, % coverage
11/17/2010	**1.03**	**33.5**	**49**	**93.9**	**-**
11/29/2010	0.4	46	3	51	Aliquots, % coverage
11/30/2010	0.12	8.8	2	75.6	Rain depth, antecedent dry period, aliquots
12/7/2010	0.96	11.2	17	83.9	Antecedent dry period
12/8/2010	1.34	10.5	34	92.2	Antecedent dry period,
12/11/2010	**2.49**	**28**	**49**	**93.8**	**-**
12/13/2010	1.23	12.2	27	82.3	Antecedent dry period
12/15/2010	0.27	13.5	5	74.2	Antecedent dry period, aliquots, % coverage
12/19/2010	0.11	34.8	2	30.1	Rainfall depth, aliquots, % cover
1/12/2011	**2.14**	**40.8**	**60**	**grab**	**-**
1/12/2011	**2.14**	**40.8**	**60**	**92.5**	
1/20/2011	0.75	10	30	94.1	Antecedent dry period
1/23/2011	0.42	61.8	6	50.3	Aliquots, % coverage
1/29/2011	0.66	11.5	18	93.5	Antecedent dry period
2/4/2011	0.06	8.2	1	0	Rainfall depth,

2/6/2011	0.14	8	2	53.6	Antecedent dry period, aliquots, % cover
2/12/2011	0.6	115.8	54	90.6	Sampling problem
2/13/2011	0.06	7.8	3	74.9	Rainfall depth
2/13/2011	0.59	14	3	10.2	Antecedent dry period, aliquots, % coverage
2/27/2011	**1.09**	**74.8**	**58**	**81.3**	**-**
3/8/2011	0.14	16.5	1	0	Rainfall depth, antecedent dry period, aliquots, % cover
3/9/2011	2.49	16	17	10.8	Antecedent dry period, % cover
3/18/2011	**0.52**	**42.2**	**24**	**88.2**	**-**
5/25/2011	0.34	86.2	16	75	Due to sample size, limited to S8.F analysis
5/27/2011	0.71	33	7	9.3	Antecedent dry period, aliquots, % cover
7/25/2011	**0.21**	**186.2**	**39**	**89.4**	**-**
8/22/2011	*0.56*	*679*	*48*	*92*	-
9/17/2011	0.11	12	41	93.1	Antecedent dry period
9/17/2011	0.74	7.8	71	15.6	Antecedent dry period, % cover
9/25/2011	0.45	148.8	83	95.9	Sampling continued into subsequent storm event
9/26/2011	0.45	16.2	28	44.3	Antecedent dry period, % cover

Bold values are qualifying samples that meet all the storm and sampling criteria.
Italicized values are for the time-weighted toxicity sample

Characteristics of all events that met storm event and stormwater sampling criteria are provided in Appendix A.

The permit requires a seasonal distribution of storm collection, with a goal of 60-80% of storms being collected during the wet season and 20-40% of storms being collected during the dry season. At the LORES site, 73% of storms were collected in the wet season and 27% in the dry season. At the COMM site, 80% of storms were collected during the wet season and 20% during the dry season. The distribution goal was met at both of these sites.

At the HIRES site 90% of storms were collected during the wet season and 10% during the dry season. The low percentage captured during the dry season is due to absence of flow at HIRES starting in July and extending until mid-October. Flow was only present for approximately two out of the five months constituting the dry season.

While the permit criteria of collecting eleven valid grab and composite samples per water year was not met, Pierce County made good faith efforts and applied professional judgment to fulfill the permit and QAPP objectives. A portion of the targeted storms were unsuccessful due to antecedent dry period not being met; insufficient rainfall; insufficient number of aliquots; equipment errors (resulting in data gaps or poor hydrograph quality), and human error (failure to collect composite before equipment began sampling into separate storm event, collecting samples before at least 75% of storm event has been sampled).

Pierce County does not have telemetric access to the samplers. To improve storm tracking, the County installed a HOBO U30 Stations GSM at the COMM site. This is a multi-functional data logger that can be accessed on-line to track current weather and rainfall data, check sensor and battery status, and make adjustments to logging intervals. The system is now being used to track the falling limb of the storm and improve sample collection time.

3.3 Stormwater Sampling

For the successful events bolded in Tables 3 through 5, associated flow weighted composite samples and grab samples were analyzed by Spectra Laboratories in Tacoma, Washington. Samples were analyzed for the following constituents.

- Total suspended solids by Standard Method 2540D
- Conductivity by EPA method 120.1
- Turbidity by EPA method 180.1
- Chloride by Standard Method 4500-C
- Biochemical oxygen demand (BOD5) by Standard Method 5210B
- Total Kjeldahl nitrogen (TKN) by Standard Method 4500
- Total phosphorus by EPA method 365.3
- Orthophosphate phosphorus by EPA method 365.3
- Nitrate-nitrite by Standard Method 4500
- Hardness by Standard Method 2340B
- Anionic surfactants as MBAS by Standard Method 5540C
- Total recoverable and dissolved zinc by EPA method 200.8

- Total and dissolved mercury by EPA method 7470A
- Polycyclic aromatic hydrocarbons (PAHs) by EPA method 8270D SIM
- Pentachlorophenol by EPA method 8270D SIM
    - Phthalates by EPA method 8270D

Spectra Laboratories sent project composite samples to subcontracted laboratories to analyze for:

- Total recoverable and dissolved cadmium, copper, and lead by EPA method 200.8 (Columbia Analytical Services)
- Pesticides (chlorpyrifos, diazinon, dichlobenil, malathion, and prometon) by EPA method 8270D (Pacific Agricultural Laboratory)
- Herbicides (2,4-D, MCPP, and triclopyr) by EPA method 8321B (Pacific Agricultural Laboratory)

Results are included in Appendix B, and will be used in loading calculations. No chemicals have been removed from the list of analysis due to two years of non-detect data:

3.4 Sediment Sampling

Annual sediment samples were collected at each of the three monitoring sites as shown in Table 6. Sampling was conducted per the collection and transport standards outlined in the QAPP. All samples were iced and transported to Spectra Laboratories for analysis on the day of collection. Due to the low volume of sediment that had accumulated in the sampling dishes at the HIRES site, the analysis was limited to total organic carbon and metals.

Table 6 -Sediment Sample Collection Dates

Monitoring Site	Collection Date
LORES	10/6/11
HIRES	9/22/11
COMM	10/6/11

Sediment samples were analyzed at Spectra Laboratories for:

- Total solids by Standard Method 2540D
- Total organic carbon by U.S. EPA method 9060
- Total cadmium, copper, lead , and zinc by U.S. EPA method 6020
- Total mercury by U.S. EPA method 7471B
- Polycyclic aromatic hydrocarbons (PAHs) by U.S. EPA method 8270D SIM
- Polychlorinated biphenyls (PCBs) by U.S. EPA method 8082
- Pentachlorophenol by U.S. EPA method 8270D SIM
- Phenolics and phthalates by U.S. EPA method 8270D

Spectra Laboratories sent S8.D project samples to subcontracted laboratories (named in parentheses, below) to analyze for:

- Grain size by ASTM D421 (Analytical Resources, Inc.)
- Pesticides (chlorpyrifos, diazinon, dichlobenil, malathion, and prometon) by EPA method 8270D (Pacific Agricultural Laboratory)
- Herbicides (2,4-D, MCPP, and triclopyr) by EPA method 8321B (Pacific Agricultural Laboratory)

Results are included in Appendix C.

3.5 Baseflow Sampling

The QAPP requires four baseflow grab samples, two in the wet season and two in the dry season, to be analyzed for total suspended solids, total phosphorus, total copper, and pesticides. Note that "baseflow" in this case refers to observable flows between storm events. Flow was not observed at the HIRES or COMM site from approximately June through September between storm events. The LORES site exhibits low flow during the summer months, suggesting groundwater contribution to this system. Baseflow grab samples were collected on the dates shown in Table 7.

Table 7 - Baseflow Sample Collection Dates

Monitoring Site	Collection Date	Season
LORES	4/22/11	Wet
	6/10/11	Dry
	7/27/11	Dry
HIRES	4/22/11	Wet
	6/10/11	Dry
COMM	4/21/11	Wet

Baseflow grab samples were analyzed at Spectra Laboratories for:

- Total suspended solids by Standard Method 2540D
- Total phosphorus by EPA method 365.3
- Total copper by EPA method 200.8

Spectra Laboratories sent project baseflow grab samples to subcontracted laboratories to analyze for:

- Pesticides (chlorpyrifos, diazinon, dichlobenil, malathion, and prometon) by EPA method 8270D (Pacific Agricultural Laboratory)
- Herbicides (2,4-D, MCPP, and triclopyr) by EPA method 8321B (Pacific Agricultural Laboratory)
-

Baseflow information will be used in loading calculations.

3.6 Toxicity Sampling

In accordance with the permit, Pierce County conducted toxicity sampling for the first-flush storm event of Water Year 2011. On 8/23/11, time-weighted composite samples were collected at the COMM and LORES sites. No flow was observed at the HIRES site during this rain event. Fish-embryo toxicity testing was conducted by Nautilus Laboratories, Inc. No significant toxicity was observed in either sample. A summary report from Nautilus is included in Appendix D.

A second attempt was made to collect a storm sample at the HIRES site over the 9/17/11 weekend, again resulting in no observed flow. The two attempts at the HIRES site were determined by the Washington Department of Ecology as having met the "good faith" requirements in S8D.2.a of the permit, thus no further attempts were necessary (personal communication, Vince McGowan,WDOE, 9/19/11).

4 Parameter Concentrations and Annual Pollutant Load Calculations

Parameter concentrations based on qualifying samples collected in Water Year 2011 are included in Appendix B. These include qualifying composite samples, grab samples, and base flow samples. Appendix C includes parameter concentrations for sediment samples.

Insufficient data was collected in Water Year 2011 to calculate parameter concentrations, statistics, trends, or provide calculations of pollutant loads. This information will be included in the 2012 annual report and/or when an adequate number of samples have been collected for analyses.

5 Quality Assurance/Quality Control (QA/QC) Report

Pierce County contracted Herrera Environmental Consultants to conduct a quality assurance/quality control review of hydrologic, water quality, and sediment data associated with this project. Results were evaluated using data quality indicators identified in the project QAPP. This review was conducted to establish the usability of data from these monitoring locations, and to guide corrective actions. This section summarizes Herrera's findings.

5.1 Hydrologic Data Quality Assessment

Hydrologic monitoring at the three outfall characterization sites involved the collection of precipitation, water level and water velocity data; the water level and water velocity data were subsequently converted to a flow time series using the continuity equation. The Measurement Quality Objectives (MQOs) for these measurements are expressed in terms of bias, representativeness, completeness, and comparability, which are discussed below.

5.1.1 Bias

All of the monitoring locations were instrumented with Sigma area-velocity meters. Bias from these meters can be introduced into discharge data by: water level sensor drift over time or displacement

between calibrations; errors in velocity estimation due to non-uniform hydraulics or instrument error; or anomalous data spikes or small data gaps.

Sensor Drift and Displacement

The project QAPP indicated that on a monthly basis field calibrations of the water level sensors were to be conducted during the study to detect sensor drift, operation, or configuration errors. . Sensor drift between calibrations was to be assessed by comparing sensor readings to manually measured depth readings at each monitoring location.

In the first three quarters of Water Year 2011, water levels were manually checked by field personnel, but the manual measurements were not recorded. From 7/1/11 – 9/30/11, water levels were manually checked by field technicians on numerous occasions. However, some of the calibration data were collected when the sensor was not submerged; these data are generally not useful for sensor level checks. The water level checks that did occur when the sensor was submerged are shown in Table 8.; these checks occurred at the LORES and COMM stations. At the LORES station the five usable level checks indicated that the level sensor was accurate for the majority of monitoring period. The one exception was the level check on 7/26/11, which was the first of the calibration checks. This check indicated that the sensor was overestimating the water level by 1.8 inches. The logger was re-calibrated to account for this problem and in subsequent level checks the logger was much more accurate. The two level checks at the COMM station indicated that the sensor was reading water level with <0.5 inch accuracy.

Table 8 - Results (in inches) from Manual Checks of Water Level Sensors

Date	LORES		COMM	
	Manual	Logger	Manual	Logger
7/26/11	3	4.8		
8/12/11	1	0.9	0	0
8/23/11				
8/26/11				
9/15/11	2.5	2.09	0	0
9/17/11				
9/19/11			0.50	0.45
9/20/11	2.3	2.7		
9/23/11			0	0
9/26/11	7.1	7	2.9	2.5

Manual = actual manual depth measurement
Logger = Datalogger reading at time of manual measurement.

Velocity Estimation Error

The project QAPP indicates that six times each year field technicians will check the Sigma velocity readings against the velocity readings from a hand held velocity meter. These checks would help to verify that the velocity data being logged were not biased due to sensor placement, hydraulics, or

calibration. The first three quarters of Water Year 2011, no manual velocity checks were conducted so velocity bias could not be quantified. From 7/1/11 – 9/30/11, five velocity checks were conducted at the LORES station, two at the HIRES station, and one at the COMM station (Table 9). The measurements at the HIRES and COMM stations indicated that the Sigma velocity sensors were accurate to within 0.01 feet per second.

At the LORES station, there was more variability. On 10/10/11, it became apparent that the Sigma velocity sensor was reading too high. It was subsequently replaced and the performance improved.

Table 9 - Results (in Feet Per Second) from Manual Checks of Velocity Sensors

Date	LORES		HIRES		COMM	
	Manual	Logger	Manual	Logger	Manual	Logger
9/26/11	0.87	1.04				
10/6/11	0.26	0.4				
10/10/11	0.42	10.3 [a]			0.75	0.74
10/11/11			0.24	0.21		
10/26/11	0.36	0.34				
10/29/11	0.24	0.60				
11/29/11			1.66	1.67		

[a] New sensor installed after this date

Manual = Velocity measurement taken with hand-held Marsh-McBirney Flow-Mate velocity sensor

Logger = Velocity reading from datalogger (Sigma velocity sensor) at time of manual measurement

Spikes and Data Gaps

Herrera used an Aquarius Data Management System to evaluate the continuous data from each velocity sensor to delete anomalous spikes and fill small data gaps. For the most part, deleted spikes and interpolated gaps were infrequent and of a short duration; therefore bias from these data irregularities was minimal and reduced through data processing

There was one major data gap that, due to its size, could not be interpolated. This occurred at the COMM station from 4/1 – 5/4/11, and is discussed further under the Completeness section below.

5.1.2 Representativeness

The representativeness of the hydrologic data was assessed in terms of precipitation, visual assessment of hydrographs, and assessment of sampling and storm criteria.

Precipitation

In Water Year 2011, precipitation ranged between the 50th – 75th percentile of the historical rainfall for the same period. Therefore, the data are representative of typical climatic conditions for the region.

Visual Assessment of Hydrographs

To assess if monitoring location hydrographs were representative of the associated land use types, visual analyses of the hydrographs were routinely conducted. This analysis was also used to identify anomalies in the flow data which could not be explained by rainfall patterns or other physical processes.

At the LORES monitoring location the hydrograph form was generally good throughout the year; however, during various periods there were spikes in flow that were not correlated with rainfall and occasionally, unexplained sudden decreases in flow. It was hypothesized that errors were potentially caused by debris getting caught on the velocity sensor. On 9/11/11 the area velocity sensor was repositioned to reduce possibly clogging with leaf litter and sediment. Due to continued discrepancies between manual and sensor measurements, the sensor was replaced on 10/26/11 in the off-set position.

The HIRES hydrograph form was characterized by numerous spikes, drops, noisy readings, and occasional hydrograph responses that did not correlate to rainfall. It was hypothesized that these errors were due to sensor placement in this poorly confined natural channel, where sediment repeatedly accumulated in the stilling well next to the sensor. The HIRES station was reconfigured on 8/8/11 and the Sigma area velocity sensor was removed from an earthen channel and placed in an 18-inch pipe. Due to this significant change in site configuration, caution should be used when comparing flow data collected before 8/8/11, with data collected after this date.

The COMM monitoring location exhibited hydrographs that were flashy, had good form, and had predictable response to rainfall.

Assessment of Sampling and Storm Criteria

While the permit criteria of collecting 11 valid grab and composite samples per water year was not met, Pierce County made good faith efforts and applied professional judgment to fulfill the permit and QAPP objectives. A portion of the targeted storms were unsuccessful due to antecedent dry period not being met; insufficient rainfall; insufficient number of aliquots; lack of telemetric access to samplers; equipment errors (resulting in data gaps or poor hydrograph quality), and human error (sampling across back-to-back storm events, collecting multiple samples across a prolonged event).

5.1.3 Completeness

Completeness was assessed based on the occurrence of gaps in the data record for all continuous hydrological data. The MQO for completeness requires that less than 10% of the total data record is missing due to equipment malfunction or other operational problems (i.e., 90% completeness).

During the October 2010 to December 2010 quarter the flow sensors were occasionally turned off between storm events at all three sites to conserve battery power. This resulted in completeness of approximately 47%. Therefore, the completeness MQO was not met during this quarter. This human error was corrected and all flow monitoring equipment was run continuously after this quarter. With the exception of the COMM site, there were no data gaps in the continuous records for the remainder of the water year.

At the COMM station there was a data gap from 4/1 – 5/4/11. Following a heavy rain event in mid-March, a large load of sediment came down through the 18" pipe where the sampling equipment was positioned. Inspection by Pierce County Public Works and Utilities – Roads Division found that a large underground settling vault uphill of (and connected via pipe to) the StormFilter system was filled to capacity and had begun overflowing. The inlet probe and sampling line, and sediment traps at the COMM site were temporarily removed while Roads cleaned out the vault and the StormFilter chambers. Equipment was re-installed on 5/4/11. This resulted in a data gap for this quarter of approximately 35% gap in the April – June quarter. Consequently, the completeness MQO at the COMM station was not met during this quarter.

5.1.4 Comparability

Consistent sampling procedures, laboratory analysis procedures (consistent with the permit and the QAPP), units of measurement, and reporting conventions were used in this study to meet the quality indicator of data comparability.

5.1.5 Hydrologic Data Quality Assessment Summary

The level calibration data indicated that, with the exception of the 7/26/11 calibration at LORES, the sensor level reading were always within approximately 0.5 inches of the actual level. Flow data from the HIRES site collected before 8/8/11, when the sensor was moved, should be used with caution when comparing to data collected after this date. Flow data collected from the LORES site from about 10/10/11 to 10/26/11, when the sensor was replaced, should also be treated with caution. The data from the COMM station can be used without qualification.

Other elements of the data were assessed based on visual inspections of the hydrographs and an assessment of storm and sampling characteristics relative to NPDES criteria for minimum storm depth, antecedent dry period, minimum aliquots, and minimum percent sampling coverage. The results from the assessment produced a subset of the sampled events that were of a high enough quality to be included in the QA of the associated laboratory data.

5.2 Water Quality Data Assessment

The laboratory's performance was reviewed in accordance with the QAPP. This review was based on quality control data summaries submitted by the laboratories; raw data were not reviewed. Data qualifiers (flags) were added to the sample results in the laboratory reports. The following data qualifier definitions are taken from USEPA Contract Laboratory Program National Functional Guidelines for Inorganic Data Review (USEPA 2002).

U	The material was analyzed for, but was not detected above the level of the associated value. The associated value is either the sample quantification limit or the sample detection limit.
J	The associated value is an estimated quantity.
UJ	The material was analyzed for, but was not detected. The associated value is an estimate and may be inaccurate or imprecise.
R	The data are unusable. (Note: analyte may or may not be present.)

Sample collection requirements are shown in Table 10.

Table 10 - Summary of Sample Collection Requirements

Parameter	Method Number [a]	Pre-Filtration Holding Time [b]	Total Holding Time [b]	Field Preservation	Laboratory Preservation
Total suspended solids	EPA 160.2 or SM 2540	7 days	7 days	Maintain ≤ 6°C	Maintain ≤ 6°C
Turbidity	EPA 180.1	NA	48hours	Maintain ≤ 6°C	Maintain ≤ 6°C
Conductivity	SM 2510 or EPA 120.1	NA	28 days	Maintain ≤ 6°C	Maintain ≤ 6°C
Chloride	EPA 300.0 or 325.2	NA	28 days	Maintain ≤ 6°C	Maintain ≤ 6°C
BOD_5	EPA 405.1	NA	48 hours	Maintain ≤ 6°C	Maintain ≤ 6°C
Hardness as $CaCO_3$	EPA 200.7	NA	6 months	Maintain ≤ 6°C	Maintain 4°C, HNO_3 to pH < 2
Methylene Blue Activated Substances (MBAS)	SM 5540C	NA	48 hours	Maintain ≤ 6°C	Maintain ≤ 6°C
Total phosphorus	EPA 365.1	NA	28 days	Maintain ≤ 6°C	Maintain ≤ 6°C, H_2SO_4 to pH < 2
Ortho-phosphate phosphorus	EPA 365.2	24 hours	48 hours	Filter (0.45-micron syringe), Maintain ≤ 6°C	Maintain ≤ 6°C, H_2SO_4 to pH < 2
Total Kjeldahl nitrogen	EPA 351.2	NA	28 days	Maintain ≤ 6°C	Maintain ≤ 6°C, H_2SO_4 to pH < 2
Nitrate + nitrite nitrogen	EPA 353.2	48 hours	28 days	Maintain ≤ 6°C	Maintain ≤ 6°C, H_2SO_4 to pH < 2
Zinc, total	EPA 200.8	24 hours	6 months	Maintain ≤ 6°C	Maintain ≤ 6°C, HNO_3 to pH < 2
Zinc, dissolved		NA		Filter (0.45-micron syringe), Maintain ≤ 6°C	Maintain ≤ 6°C, HNO_3 to pH < 2 after filtration
Lead, total	EPA 200.8	24 hours	6 months	Maintain ≤ 6°C	Maintain ≤ 6°C, HNO_3 to pH < 2
Lead, dissolved		NA		Filter (0.45-micron syringe), Maintain ≤ 6°C	Maintain ≤ 6°C, HNO_3 to pH < 2 after filtration
Copper, total	EPA 200.8	24 hours	6 months	Maintain ≤ 6°C	Maintain ≤ 6°C, HNO_3 to pH < 2
Copper, dissolved		NA		Filter (0.45-micron syringe), Maintain ≤ 6°C	Maintain ≤ 6°C, HNO_3 to pH < 2 after filtration
Cadmium, total	EPA 200.8	24 hours	6 months	Maintain ≤ 6°C	Maintain ≤ 6°C, HNO_3 to pH < 2
Cadmium, dissolved		NA		Filter (0.45-micron syringe), Maintain ≤ 6°C	Maintain ≤ 6°C, HNO_3 to pH < 2 after filtration

Table 10 (continued). Summary of sample collection requirements.

Parameter	Method Number [a]	Pre-Filtration Holding Time [b]	Total Holding Time [b]	Field Preservation (degrees Celcius)	Laboratory Preservation
Mercury, total	EPA 7470	24 hours	28 days	Maintain ≤ 6°C	Maintain ≤ 6°C, HNO_3 to pH < 2
Mercury, dissolved		NA		Filter (0.45-micron syringe), Maintain ≤ 6°C	Maintain ≤ 6°C, HNO_3 to pH < 2 after filtration
Polycyclic aromatic hydrocarbons	EPA 8270D SIM	NA	7 days	Maintain ≤ 6°C	Maintain ≤ 6°C
Phthalates	EPA 8270D	NA	7 days	Maintain ≤ 6°C	Maintain ≤ 6°C
Pentachlorophenol	EPA 8270D SIM	NA	7 days	Maintain ≤ 6°C	Maintain ≤ 6°C
2,4-D and MCPP	EPA 8151	NA	7 days	Maintain ≤ 6°C	Maintain ≤ 6°C
Triclopyr	EPA 8321	NA	7 days	Maintain ≤ 6°C	Maintain ≤ 6°C
Dichlobenil	EPA 8270D SIM	NA	7 days	Maintain ≤ 6°C	Maintain ≤ 6°C
Prometon	EPA 8270D SIM	NA	7 days	Maintain ≤ 6°C	Maintain ≤ 6°C
Chloropyrifos, Diazinon, and Malathion	EPA 8141A	NA	7 days	Maintain ≤ 6°C	Maintain ≤ 6°C

[a] Standard Method numbers are from APHA et al. (1998); EPA method numbers are from U.S. EPA (1983, 1984). The 18th edition of *Standard Methods for the Examination of Water and Wastewater* (APHA et al. 1992) is the current legally adopted version in the *Code of Federal Regulations.*

[b] Holding time specified in U.S. EPA guidance (U.S. EPA 1983, 1984 or referenced in APHWA et al. (1992) for equivalent method.

5.2.1 Custody, Preservation, Holding Times, and Completeness

The samples were properly preserved and sample custody was maintained from sample collection to receipt at the laboratories. With the exceptions noted below, all samples were analyzed within the required holding times (Table 11). The laboratory reports were complete and contained results for all samples and tests requested on the chain-of-custody (COC) forms.

The grab sample collected from the COMM Outfall on 11/17/2010 was analyzed for diesel-range organics one day beyond the holding time. Results from the analyses were qualified as estimated (J or UJ). The BOD analysis of a composite sample collected at COMM on 7/26/2011 was performed 24 hours outside of the 48-hour holding time, due to laboratory oversight. In accordance with the QAPP, the BOD result for the COMM sample collected on 7/26/2011 was rejected (R) due to holding time exceedance.

Table 11 - Summary of Samples Qualified or Rejected Due to Holding Time Exceedances

Sample Date	Parameter	QAPP Station Name	Qualifier
11/17/10	Diesel	COMM	UJ
11/17/10	Oil	COMM	J
07/26/2011	BOD	COMM	R

5.2.2 Laboratory Reporting Limits

The laboratory reporting limits and QAPP specified reporting limits are provided in Table 12. The laboratory reporting limits generally met or were lower than the QAPP specified reporting limits for all analyses. For the few cases where the laboratory reporting limit exceeded the QAPP reporting limit (ortho-phosphorus, total Kjeldahl nitrogen and chloride), the data were reviewed to evaluate whether any measured values were near the reporting limit and should be qualified. No data were qualified based on laboratory reporting limits.

5.2.3 Laboratory Blank Analysis - Method Blanks

Spectra Laboratory analyzed method blanks were analyzed at the required frequency. As noted in Table 13, several method blanks contained levels of target analytes above the laboratory reporting limits.

Sample results associated with method blanks shown in Table 14 were qualified as estimated (J) if the reported value for the sample was less than 5 times the method blank. Sample qualified due to method blank criterion exceedances are shown in Table 14.

Thirteen samples were qualified for Bis(2-ethylhexyl)phthalate due to method blank criterion exceedance. Based on discussions with Spectra Laboratory, it was concluded that Bis(2-ethylhexyl)phthalate is ubiquitous in sampling equipment, laboratory equipment, and possibly reagent grade deionized (blank) water provided by the laboratory. Exceedances are not likely to be due solely to stormwater. These thirteen samples were qualified with a J in the event reports.

In all cases the reported value for conductivity was greater than five times the method blank; those samples were not qualified.

Table 12 - Summary of QAPP and Laboratory Reporting Limits

Parameter	QAPP Reporting Limit	Laboratory Reporting Limit	Units
Total suspended solids	1.0	0.5	mg/L
Turbidity	±0.2	0.1	NTU
Conductivity	±1	0.5	umho/cm
Chloride	0.2	0.5	mg/L
BOD_5	2.0	2	mg/L
Hardness as $CaCO_3$	1.0	1	mg/L
Methylene Blue Activated Substances (MBAS)	0.025	0.02	mg/L
Total phosphorus	0.01	0.01	mg/L
Ortho-phosphate phosphorus	0.005	0.01	mg/L
Total Kjeldahl nitrogen	0.1	0.5	mg/L
Nitrate + nitrite nitrogen	0.01	0.01	mg/L
Zinc, total	5.0	1.0	µg /L
Zinc, dissolved	1.0	1.0	
Lead, total	0.1	0.02	µg /L
Lead, dissolved	0.02	0.02	
Copper, total	0.1	0.1	µg /L
Copper, dissolved	0.1	0.1	
Cadmium, total	0.2	0.02	µg /L
Cadmium, dissolved	0.02	0.02	
Mercury, total	0.1	0.1	µg /L
Mercury, dissolved	0.1	0.1	µg /L
Polycyclic aromatic hydrocarbons	0.1	0.1	µg /L
Phthalates	1.0	1	µg /L
Pentachlorophenol	5.0	0.1	µg /L
2,4-D and MCPP	0.3	0.08	µg /L
Triclopyr	0.3	0.08	µg /L
Dichlobenil	0.3	0.12	µg /L
Prometon	0.3	0.12	µg /L
Chloropyrifos, Diazinon, and Malathion	0.3	0.12	µg /L

Table 13- Summary of Method Blank Criterion Exceedances

Date Analyzed	Parameter	Method Blank Result	Reporting Limit
11/09/2010	Conductivity	0.6 mg/L	umho/cm
11/11/2010	Conductivity	0.6 mg/L	umho/cm
11/14/2010	Bis(2-ethylhexyl)phthalate	1.22 µg/L	1.00 µg/L
11/22/2010	Bis(2-ethylhexyl)phthalate	3.10 µg/L	1.00 µg/L
12/23/2010	Bis(2-ethylhexyl)phthalate	2.01 µg/L	1.00 µg/L
02/02/2011	Conductivity	0.5 mg/L	umho/cm
02/17/2011	Bis(2-ethylhexyl)phthalate	1.09 µg/L	1.00 µg/L
2/17/2011	Bis(2-ethylhexyl)phthalate	9.28 µg/L	1.00 µg/L
03/03/2011	Bis(2-ethylhexyl)phthalate	2.79 µg/L	1.00 µg/L
03/01/2011	Conductivity	0.7 mg/L	0.5 mg/L
03/10/2011	Conductivity	1.2 mg/L	0.5 mg/L
03/17/2011	Bis(2-ethylhexyl)phthalate	2.91 µg/L	1.00 µg/L
03/22/2011	Conductivity	0.7 mg/L	0.5 mg/L
03/31/2011	Bis(2-ethylhexyl)phthalate	4.73 µg/L	1.00 µg/L
04/18/2011	Bis(2-ethylhexyl)phthalate	23.4 µg/L	1.00 µg/L
04/18/2011	Conductivity	0.6 mg/L	0.5 mg/L
04/26/2011	Conductivity	0.8 mg/L	0.5 mg/L
05/05/2011	Bis(2-ethylhexyl)phthalate	1.74 µg/L	1.00 µg/L
05/05/2011	Conductivity	0.8 mg/L	0.5 mg/L
05/13/2011	Bis(2-ethylhexyl)phthalate	1.35 µg/L	1.00 µg/L
05/27/2011	Conductivity	0.8 mg/L	0.5 mg/L
06/02/2011	Bis(2-ethylhexyl)phthalate	4.29 µg/L	1.00 µg/L
07/26/2011	Conductivity	1.0 mg/L	0.5 mg/L
08/01/2011	Bis(2-ethylhexyl)phthalate	2.60 µg/L	1.00 µg/L

Table 14 - Summary of Samples Qualified Due to Method Blank Exceedances

Sample ID	Sample Date	Parameter	Sample Result	Qualifier
LORES	11/08/2010	Bis(2-ethylhexyl)phthalate	1.16 µg/L	J
COMM	12/13/2010	Bis(2-ethylhexyl)phthalate	2.38 µg/L	J
LORES	02/14/2011	Bis(2-ethylhexyl)phthalate	1.51 µg/L	J
COMM	02/28/2011	Bis(2-ethylhexyl)phthalate	6.72 µg/L	J
LORES	02/28/2011	Bis(2-ethylhexyl)phthalate	5.69 µg/L	J
LORES	03/09/2011	Bis(2-ethylhexyl)phthalate	1.71 µg/L	J
COMM	03/21/2011	Bis(2-ethylhexyl)phthalate	4.37 µg/L	J
HIRES	03/21/2011	Bis(2-ethylhexyl)phthalate	2.21 µg/L	J
HIRES	04/11/2011	Bis(2-ethylhexyl)phthalate	2.30 µg/L	J
HIRES	05/05/2011	Bis(2-ethylhexyl)phthalate	1.35 µg/L	J

Sample ID	Sample Date	Parameter	Sample Result	Qualifier
HIRES	05/27/2011	Bis(2-ethylhexyl)phthalate	2.89 µg/L	J
COMM	07/26/2011	Bis(2-ethylhexyl)phthalate	4.64 µg/L	J

5.2.4 Laboratory Blank Analysis - Sampling Equipment Rinsate Blanks

Composite equipment rinsate blanks were collected from the COMM site on 10/27/10 and 1/11/11; from the LORES site on 10/27/10 and 1/11/11; and from the HIRES site on 11/10/10 and 1/3/11. The samples were analyzed for all composite sample parameters in accordance with the QAPP. To meet rinsate blank criterion, reported values for rinsate samples must be less than 2 times the reporting limit. Several parameters were detected above the laboratory reporting limits (Table 15), and included naphthalene, dissolved and total copper, dissolved and total lead, dissolved and total cadmium, dissolved and total copper, total zinc, nitrate/nitrite, and bis(2-ethylhexyl)phthalate.

As discussed above, Bis(2-ethylhexyl)phthalate is a common laboratory contaminant and may also have been introduced by plastic sampling equipment, or was present in the reagent grade deionized (blank) water provided by the laboratory. Results for bis(2-ethylhexyl)phthalate were qualified as not detected (U) or estimated (J) where sample results were less than 5 times the detected blank value.

Metal contaminants may have been introduced by inadequate decontamination of equipment and tubing. Naphthalene was detected in all rinsate blanks and may have been introduced at the sampling locations.

Table 15 - Summary of Detected Parameters in Sampling Equipment Rinsate Blanks

Sample ID	Sample Date	Parameter	Blank Result	Reporting Limit
COMM Blank-	10/27/2010	Naphthalene	**0.261 µg/L**	0.100 µg/L
COMM Blank	10/27/2010	Dissolved cadmium	0.03 µg/L	0.02 µg/L
COMM Blank	10/27/2010	Dissolved copper	**0.3 µg/L**	0.10 µg/L
COMM Blank	10/27/2010	Dissolved lead	**0.05 µg/L**	0.02 µg/L
COMM Blank	10/27/2010	Dissolved zinc	1.0 µg/L	1.0 µg/L
COMM Blank	10/27/2010	Total recoverable cadmium	0.03 µg/L	0.02 µg/L
COMM Blank	10/27/2010	Total recoverable copper	**0.3 µg/L**	0.10 µg/L
COMM Blank	10/27/2010	Total recoverable lead	**0.05 µg/L**	0.02 µg/L
COMM Blank	10/27/2010	Total recoverable zinc	1.1 µg/L	1.0 µg/L
COMM Blank	10/27/2010	Nitrate/nitrite	0.010 mg/L	0.010 mg/L
COMM Blank	10/27/2010	Chloride	0.5 mg/L	0.5 mg/L
COMM Blank	10/27/2010	Bis(2-ethylhexyl)phthalate	**16.4 µg/L**	1.0 µg/L
LORES Blank	10/27/2010	Naphthalene	0.156 µg/L	0.100 µg/L
LORES Blank	10/27/2010	Dissolved cadmium	0.04 µg/L	0.02 µg/L
LORES Blank	10/27/2010	Dissolved lead	0.03 µg/L	0.02 µg/L
LORES Blank	10/27/2010	Total recoverable cadmium	0.04 µg/L	0.02 µg/L
LORES Blank	10/27/2010	Total recoverable lead	0.03 µg/L	0.02 µg/L
LORES Blank	10/27/2010	Total phosphorus	0.01 mg/L	0.01 mg/L
LORES Blank	10/27/2010	Nitrate/nitrite	0.011 mg/L	0.010 mg/L

Sample ID	Sample Date	Parameter	Blank Result	Reporting Limit
LORES Blank	10/27/2010	Chloride	0.5 mg/L	0.5 mg/L
LORES Blank	10/27/2010	Bis(2-ethylhexyl)phthalate	1.32 µg/L	1.0 µg/L
HIRES Blank	11/10/2010	Naphthalene	**0.221 µg/L**	0.100 µg/L
HIRES Blank	11/10/2010	Dissolved cadmium	0.02 µg/L	0.02 µg/L
HIRES Blank	11/10/2010	Dissolved copper	0.2 µg/L	0.10 µg/L
HIRES Blank	11/10/2010	Total recoverable cadmium	0.03 µg/L	0.02 µg/L
HIRES Blank	11/10/2010	Total recoverable copper	0.1 µg/L	0.10 µg/L
HIRES Blank	11/10/2010	Total recoverable zinc	**6.2 µg/L**	1.0 µg/L
HIRES Blank	11/10/2010	Nitrate/nitrite	0.011 mg/L	0.010 mg/L
HIRES Blank	11/10/2010	Chloride	1.0 mg/L	0.5 mg/L
HIRES Blank	11/10/2010	Bis(2-ethylhexyl)phthalate	**6.98 µg/L**	1.0 µg/L
COMM Blank	01/11/2011	Naphthalene	0.162 µg/L	0.100 µg/L
COMM Blank	01/11/2011	Dissolved cadmium	**0.15 µg/L**	0.02 µg/L
COMM Blank	01/11/2011	Dissolved copper	**0.3 µg/L**	0.10 µg/L
COMM Blank	01/11/2011	Dissolved zinc	**4.5 µg/L**	1.0 µg/L
COMM Blank	01/11/2011	Total recoverable cadmium	**0.75 µg/L**	0.02 µg/L
COMM Blank	01/11/2011	Total recoverable copper	**0.3 µg/L**	0.10 µg/L
COMM Blank	01/11/2011	Total recoverable lead	**0.05 µg/L**	0.02 µg/L
COMM Blank	01/11/2011	Total recoverable zinc	**4.6 µg/L**	1.0 µg/L
COMM Blank	01/11/2011	Nitrate/nitrite	**0.030 mg/L**	0.010 mg/L
COMM Blank	01/11/2011	Bis(2-ethylhexyl)phthalate	**7.88 µg/L**	1.0 µg/L
LORES Blank	01/11/2011	Naphthalene	0.132 µg/L	0.100 µg/L
LORES Blank	01/11/2011	Dissolved cadmium	**0.09 µg/L**	0.02 µg/L
LORES Blank	01/11/2011	Dissolved copper	0.2 µg/L	0.10 µg/L
LORES Blank	01/11/2011	Dissolved lead	0.03 µg/L	0.02 µg/L
LORES Blank	01/11/2011	Dissolved zinc	1.4 µg/L	1.0 µg/L
LORES Blank	01/11/2011	Total recoverable cadmium	**0.54 µg/L**	0.02 µg/L
LORES Blank	01/11/2011	Total recoverable copper	0.2 µg/L	0.10 µg/L
LORES Blank	01/11/2011	Total recoverable lead	0.04 µg/L	0.02 µg/L
LORES Blank	01/11/2011	Total recoverable zinc	1.7 µg/L	1.0 µg/L
LORES Blank	01/11/2011	Nitrate/nitrite	0.011 mg/L	0.010 mg/L
LORES Blank	01/11/2011	Bis(2-ethylhexyl)phthalate	**2.07 µg/L**	1.0 µg/L
HIRES Blank	01/03/2011	Naphthalene	**0.202 µg/L**	0.100 µg/L
HIRES Blank	01/03/2011	Dissolved copper	**0.5 µg/L**	0.10 µg/L
HIRES Blank	01/03/2011	Dissolved lead	**0.07 µg/L**	0.02 µg/L
HIRES Blank	01/03/2011	Total recoverable cadmium	**0.27 µg/L**	0.02 µg/L
HIRES Blank	01/03/2011	Total recoverable copper	0.2 µg/L	0.10 µg/L
HIRES Blank	01/03/2011	Nitrate/nitrite	0.012 mg/L	0.010 mg/L
HIRES Blank	01/03/2011	Chloride	0.5 mg/L	0.5 mg/L

Bold blank values exceed QAPP criterion of not to exceed two times the reporting limit.

Since rinsate blank problems are assumed to reflect conditions for all samples collected, all data collected during the associated sampling period were assigned qualifiers based on the rinsate blank results. All Samples that were qualified due to equipment rinsate blank contamination are shown in Table 16.

Table 16 - Summary of Samples Qualified Due to Rinsate Blank Criterion Exceedances

Sample ID	Sample Date	Parameter	Sample Result	Qualifier
COMM	11/08/2010	Bis(2-ethylhexyl)phthalate	24.0 µg/L	J
COMM	11/10/2010	Bis(2-ethylhexyl)phthalate	13.4 µg/L	U
COMM	11/18/2010	Dissolved lead	0.23 µg/L	J
COMM	11/18/2010	Bis(2-ethylhexyl)phthalate	19.6 µg/L	J
COMM	12/13/2010	Bis(2-ethylhexyl)phthalate	2.38 µg/L	U
COMM	12/13/2010	Dissolved lead	0.19 µg/L	J
HIRES	11/10/2010	Naphthalene	0.123 µg/L	U
HIRES	11/10/2010	Total recoverable zinc	5.4 µg/L	U
HIRES	11/10/2010	Bis(2-ethylhexyl)phthalate	27.8 µg/L	J
COMM	02/28/2011	Dissolved cadmium	0.066 µg/L	J
COMM	02/28/2011	Total cadmium	0.237 µg/L	J
COMM	02/28/2011	Nitrate/nitrite	0.085 mg/L	J
COMM	02/28/2011	Bis(2-ethylhexyl)phthalate	6.72 µg/L	J
COMM	03/21/2011	Dissolved cadmium	0.09 µg/L	J
COMM	03/21/2011	Dissolved zinc	5.3 µg/L	J
COMM	03/21/2011	Total cadmium	0.17 µg/L	J
COMM	03/21/2011	Nitrate/nitrite	0.118 mg/L	J
COMM	03/21/2011	Bis(2-ethylhexyl)phthalate	4.37 µg/L	J
LORES	02/14/2011	Dissolved cadmium	0.05 µg/L	J
LORES	02/14/2011	Total cadmium	0.07 µg/L	J
LORES	02/14/2011	Bis(2-ethylhexyl)phthalate	1.51 µg/L	J
LORES	02/28/2011	Dissolved cadmium	0.123 µg/L	J
LORES	02/28/2011	Total cadmium	0.484 µg/L	J
LORES	02/28/2011	Bis(2-ethylhexyl)phthalate	5.67 µg/L	J
LORES	03/09/2011	Dissolved cadmium	0.05 µg/L	J
LORES	03/09/2011	Total cadmium	0.05 µg/L	J
LORES	03/09/2011	Bis(2-ethylhexyl)phthalate	1.71 µg/L	J
HIRES	03/21/2011	Dissolved copper	1.4 µg/L	J
HIRES	03/21/2011	Dissolved lead	0.07 µg/L	J
HIRES	03/21/2011	Total cadmium	0.06 µg/L	J

5.2.5 Laboratory Control Sample Analysis

Laboratory control samples (LCS) were analyzed at the required frequency. The percent recovery values for all sampling events met the QAPP criteria.

5.2.6 Laboratory Matrix Spike Analysis

Matrix spike (MS) samples were analyzed at the required frequency. The percent recovery values for the MS analyses met the control limits established by the QAPP.

5.2.7 Laboratory Duplicate Analysis

Laboratory duplicates were analyzed at the required frequency. The relative percent difference (RPD) was calculated for each analyte where both duplicate values were greater than 5 times the reporting limit (RL). The difference between duplicate values was calculated if the detected compound concentration was less than 5 times the RL in either the sample or the field duplicate. A control limit of less than 20 percent RPD (25 percent for total suspended solids, turbidity, conductivity, BOD_5, MBAS; 35 percent for fecal coliform bacteria and petroleum hydrocarbons) was established in the QAPP and a control limit of 2 times the RL was used to evaluate difference values. With the exceptions shown in Table 17, the relative percent difference (RPD) values met the control limits established by the QAPP, and all difference values were less than two times the RL. All results shown on Table 17 were qualified as estimated (J) due to laboratory duplicate criteria exceedance.

Table 17 - Summary of Laboratory Duplicate Criteria Exceedances

Sample Date	Parameter	Associated Samples	RPD (%)	Qualifier
11/08/2010	Total phosphorus	COMM, LORES	28	J
11/10/2010	Orthophosphate	COMM, HIRES	25	J
02/28/2011	Chloride	COMM	20.8	J
03/09/2011	Chloride	COMM	22	J
04/11/2011	TSS	HIRES	28	J
05/27/2011	TKN	HIRES	40	J
07/26/2011	MBAs	COMM	32	J

5.2.8 Field Duplicate Analysis

Field duplicate samples were collected on 4/25/2011 at the LORES site and on 4/26/2011 at the HIRES site. One field duplicate sample was collected on 7/26/2011 at the LORES site; this was a non-qualifying sample and data were only used to evaluate precision.

The relative percent difference (RPD) was calculated for each analyte where both duplicate values were greater than 5 times the reporting limit (RL). The difference between duplicate values was calculated if the detected compound concentration was less than 5 times the RL, in either the sample or the field duplicate. A control limit of less than 20 percent RPD (25 percent for total suspended solids, turbidity, conductivity,

BOD_5, and MBAS) was established in the QAPP and a control limit of 2 times the RL was used to evaluate difference values. With the exceptions shown in Table 18, the relative percent difference (RPD) values met the control limits established by the QAPP, and all difference values were less than 2 times the RL.

Table 18 - Summary of Field Duplicate Criteria Exceedances

Sample Date	QAPP Station Name	Sample ID	Duplicate ID	Parameter	RPD (%)
04/25/2011	LORES	Lake Sylvia	QA1	Turbidity	52
04/25/2011	LORES	Lake Sylvia	QA1	Dissolved copper	29
04/25/2011	LORES	Lake Sylvia	QA1	Zinc	52
04/26/2011	HIRES	Collinswood	Outfall QA-2	Dissolved copper	38
04/26/2011	HIRES	Collinswood	Outfall QA-2	Zinc	93
04/26/2011	HIRES	Collinswood	Outfall QA-2	Chloride	22

As shown in Table 18, six parameters exceeded field duplicate criteria established in the QAPP. Variability in the stormwater, sample aliquot preparation methods, and analytical methods may have contributed to field duplicate criteria exceedances. No data were qualified for field duplicate exceedances.

5.3 Sediment Data Quality Assessment

The sediment samples were collected at all three sites. The samples were collected using passive sediment samplers that were placed at the base of the outfall pipes; the passive samplers were in place for an entire water year and represent composites over the entire discharge period. Data were evaluated using data quality indicators identified in the QAPP.

Table 19 summarizes sample information related to the sites and samples that were submitted to the laboratories.

Table 19 - Sediment Quality Data Collected and Reviewed

Location	Lab File Number	Sample Collection Date
HIRES	2011090462	09/22/2011
COMM	2011100208	10/06/2011
LORES	2011100663	10/26/2011

The laboratory's performance was reviewed in accordance with quality control criteria outlined in the project QAPP. The performance review was based on quality control data summaries submitted by the laboratories; raw data were not reviewed. Data validation results are summarized below followed by definitions of data qualifiers.

5.3.1 Custody, Preservation, Holding Times, and Completeness

Sample collection requirements are shown in Table 20. The samples were properly preserved, and sample custody was maintained from sample collection to receipt at the laboratories. All samples were analyzed within the required holding times. With the exception noted below, the laboratory reports were complete and contained results for all samples and tests requested on the chain-of-custody (COC) forms.

Table 20 - Summary of Sample Collection Requirements

Parameter	Method Number [a]	Total Holding Time [b]	Field Preservation (degrees Celcius)	Laboratory Preservation
Total solids (% solids)	SM2540	14 days, 6 months if frozen	Refrigerate, 4°C	NA
Total volatile solids	SM2540	6 months	Refrigerate, 4°C	NA
Total phosphorus	EPA 365.3	14 days, 6 months if frozen	Refrigerate, 4°C	NA
Grain-size	ASTM D421	6 months	Refrigerate, 4°C	NA
Total organic carbon	PSEP 1997	14 days, 6 months if frozen	Refrigerate, 4°C	NA
PAHs	EPA 8270D SIM	14 days, 6 months if frozen	Refrigerate, 4°C	NA
Phthalates-BNA	EPA 8270D	14 days, 6 months if frozen	Refrigerate, 4°C	NA
Phenolics-BNA	EPA 8270D	14 days, 6 months if frozen	Refrigerate, 4°C	NA
Pentachlorophenol	EPA 8270D SIM	14 days, 6 months if frozen	Refrigerate, 4°C	NA
PCBs (Total)	EPA 8082B	14 days, 6 months if frozen	Refrigerate, 4°C	NA
2,4-D and MCPP	EPA 8151A	14 days, 6 months if frozen	Refrigerate, 4°C	NA
Triclopyr	EPA 8151A	14 days, 6 months if frozen	Refrigerate, 4°C	NA
Dichlobenil	EPA 8270D SIM	14 days, 6 months if frozen	Refrigerate, 4°C	NA
Prometon	EPA 8270D SIM	14 days, 6 months if frozen	Refrigerate, 4°C	NA
Chloropyrifos, Diazinon, & Malathion	EPA 8141A	14 days, 6 months if frozen	Refrigerate, 4°C	NA
Total copper	EPA 200.8	6 months, 2 years if frozen	Refrigerate, 4°C	NA
Total zinc	EPA 200.8	6 months, 2 years if frozen	Refrigerate, 4°C	NA
Total cadmium	EPA 200.8	6 months, 2 years if frozen	Refrigerate, 4°C	NA
Total lead	EPA 200.8	6 months, 2 years if frozen	Refrigerate, 4°C	NA
Mercury	EPA 7471	28 days	Refrigerate, 4°C	NA
Total Petroleum Hydrocarbons	NWTPH-Dx	14 days	Refrigerate, 4°C	NA

[a] SM method numbers are from APHA et al. (1998); EPA method numbers are from US EPA (1984).
[b] Holding time specified in U.S. EPA guidance (U.S. EPA 1984) or referenced in APHA et al. (1998) for equivalent method.

For the Collinswood sample collected on 9/22/11, less than 2 grams of sediment was submitted to the laboratory. The only analyses performed were total organic carbon and metals, due to the small sample size. No data were qualified.

5.3.2 Laboratory Reporting Limits

The laboratory reporting limits and QAPP-specified reporting limits are provided in Table 21. The laboratory reporting limits generally met or were lower than the permit and QAPP-specified reporting limits for all analyses. Reporting limits for total petroleum hydrocarbons were elevated due to sample dilution and sample matrix interferences. Reporting limits for phenolics and phthalates were elevated for

the sample collected at Canyon Road due to matrix interferences. No data were qualified based on laboratory reporting limits.

Table 21 - Summary of QAPP and Laboratory Reporting Limits

Parameter	QAPP Reporting Limit	Laboratory Reporting Limit	Units
Total solids (% solids)	0.1	0.1	%
Total volatile solids	0.1	0.1	%
Total phosphorus	0.1	0.1	mg/kg
Grain-size	0.1	0.1	%
Total organic carbon	0.1	0.1	%
PAHs	20	1	µg /kg
Phthalates-BNA	20	20 – 100	µg /kg
Phenolics-BNA	20	1 - 10	µg /kg
Pentachlorophenol	100	1	µg /kg
Total PCBs	100	10	µg /kg
2,4-D and MCPP	20	10	µg /kg
Triclopyr	20	10	µg /kg
Dichlobenil	20	6.7	µg /kg
Prometon	20	6.7	µg /kg
Chloropyrifos, Diazinon, and Malathion	20	6.7	µg /kg
Total copper	0.1	0.1	mg/kg
Total zinc	5.0	5.0	mg/kg
Total cadmium	0.1	0.1	mg/kg
Total lead	0.1	0.1	mg/kg
Mercury	0.1	0.05	mg/kg
Total petroleum hydrocarbons	10 (diesel) ; 100 (motor oil)	50 - 110 (diesel) ; 100 - 525 (motor oil)	mg/kg

QAPP = quality assurance project plan
mg/kg = milligrams per kilogram
µg /kg = micrograms per kilogram

5.3.3 Method Blanks

Method blanks were analyzed at the required frequency. Method blanks did not contain levels of target analytes above the laboratory reporting limits. No data were qualified based on laboratory method blank contamination.

5.3.4 Laboratory Control Sample Analysis

Laboratory control samples were analyzed at the required frequency. The percent recovery values for all sampling events met the QAPP criteria.

5.3.5 Matrix Spike Analysis

Matrix spike samples were analyzed at the required frequency. The percent recovery values for the matrix spike analyses met the control limits established by the QAPPs.

5.3.6 Laboratory Duplicate Analysis

Laboratory duplicates were analyzed at the required frequency. The relative percent difference (RPD) was calculated for each analyte where both duplicate values were greater than 5 times the reporting limit (RL). The difference between duplicate values was calculated if the detected compound concentration was less than 5 times the RL in either the sample or the field duplicate. A control limit of less than 35 percent RPD was established in the QAPP and a control limit of 2 times the RL was used to evaluate difference values. The relative percent difference (RPD) values met the control limits established by the QAPP, and all difference values were less than 2 times the RL.

5.3.7 Field Duplicates

No field duplicates of sediment samples were required to be collected in Water Year 2011.

5.3.8 Soil Data Assessment Summary

In general, the data quality for all parameters was found to be acceptable based on holding time, reporting limit, method blank, control standard, matrix spike, and laboratory duplicate criteria. Based on the usability criteria specified in the QAPP, the data are acceptable as qualified. Data that were not qualified were rejected. Efforts will be made in Water Year 2012 to improve equipment decontamination prior to rinsate blank collection.

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Appendices

Appendix A- Individual Storm Reports (Hydrographs and Statistics) of Qualifying Events
Appendix B - Water Quality Data for Qualifying Events
Appendix C - Sediment Data
Appendix D – First Flush Toxicity Report

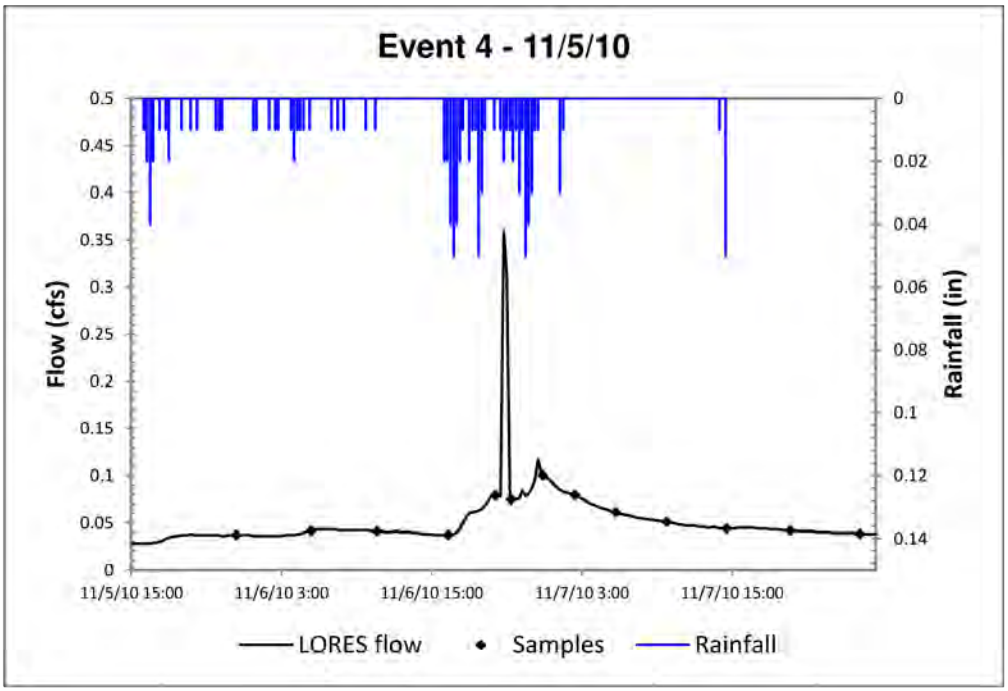
Appendix A

Individual Storm Reports (Hydrographs and Statistics) of Qualifying Events

Pierce County NPDES **LORES: 11/5/10 16:00 Storm Targeted**

Precipitation Statistics	
Start	11/5/10 16:00
Stop	11/7/10 14:30
Duration (hr)	46.5
Depth (in)	1.06
Max Reading (in)	0.05
Max Hourly Intensity (in/hr)	0.17
Average Intensity (in/hr)	0.023
Antecedent Dry Period (hr)	240.5
% Complete	100

Flow Statistics	
	LORES FM
Start	11/5/10 16:00
Stop	11/8/10 02:30
Duration (hours)	58.75
Peak Flow (cfs)	0.359
Average Flow (cfs)	0.051
Volume (cf)	10760.39
% Complete	100

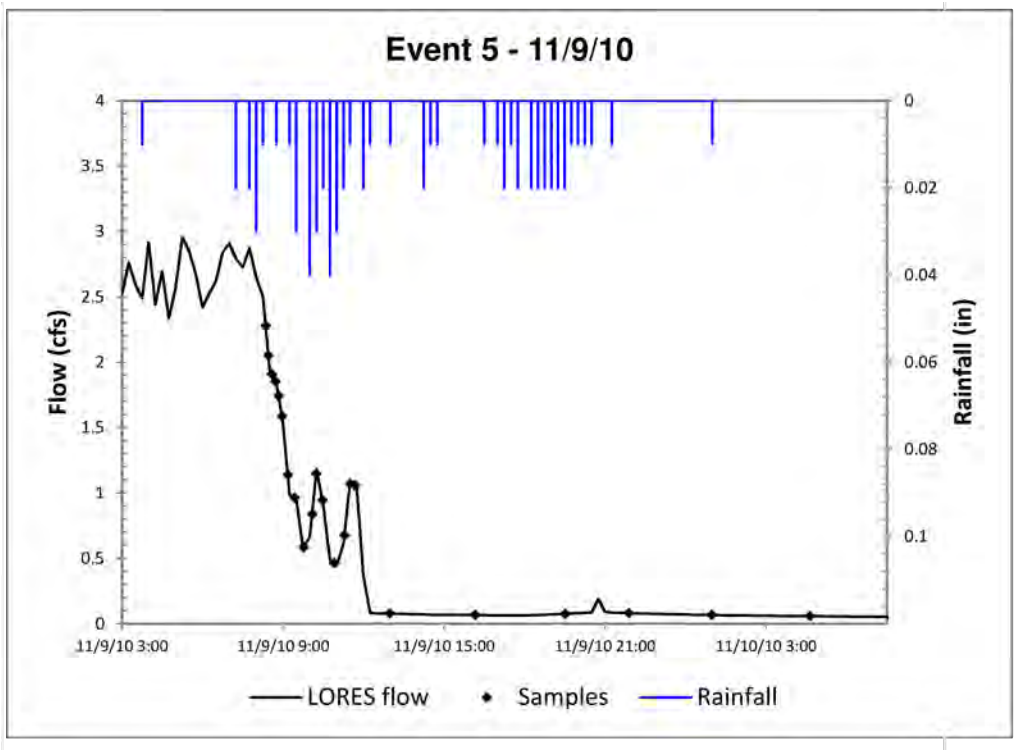


Sample Statistics	
	LORES FM
Aliquots	13
Volume Sampled (cf)	1601.2
% Coverage	48.3
First Aliquot Time	11/5/10 23:23
Last Aliquot Time	11/8/10 01:13
Sampling Duration (hours)	49.8

Pierce County NPDES **LORES: 11/9/10 03:45 Storm Targeted**

Precipitation Statistics	
Start	11/9/10 03:45
Stop	11/10/10 01:00
Duration (hr)	21.25
Depth (in)	0.65
Max Reading (in)	0.04
Max Hourly Intensity (in/hr)	0.16
Average Intensity (in/hr)	0.031
Antecedent Dry Period (hr)	37.2
% Complete	100

Flow Statistics	
	LORES FM
Start	11/9/10 03:45
Stop	11/10/10 07:30
Duration (hours)	28
Peak Flow (cfs)	2.953
Average Flow (cfs)	0.632
Volume (cf)	63701.12
% Complete	100

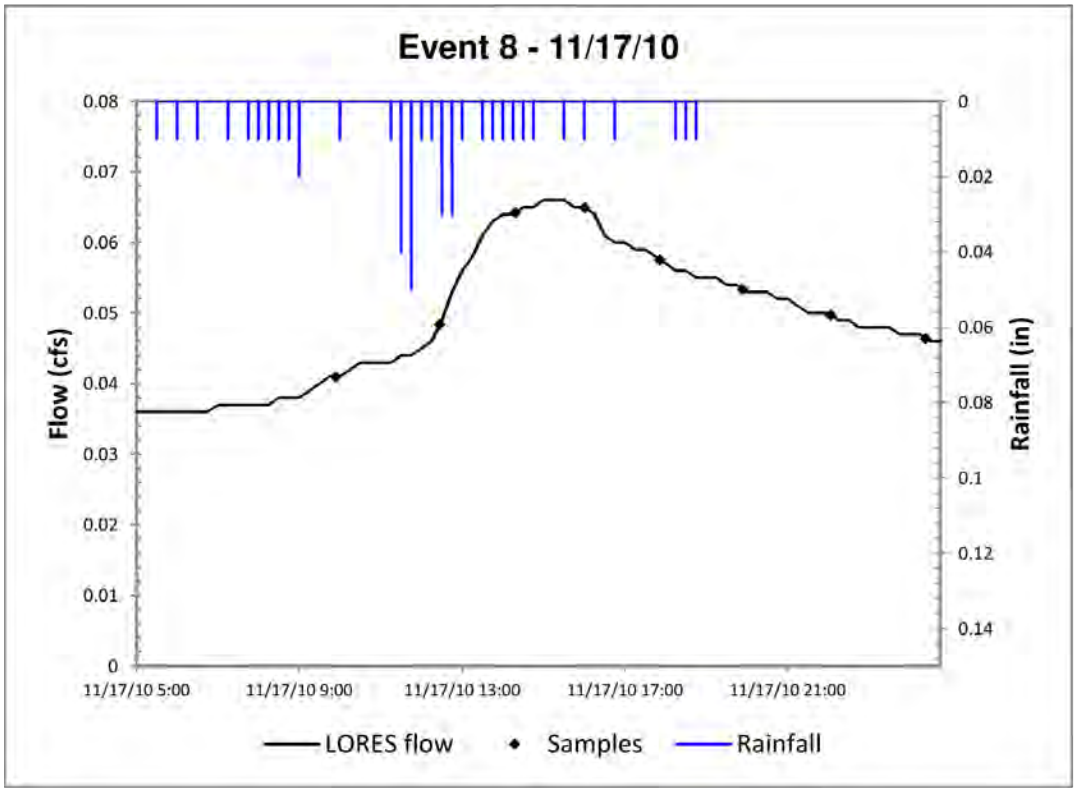


Sample Statistics	
	LORES FM
Aliquots	22
Volume Sampled (cf)	15953.6
% Coverage	25.3
First Aliquot Time	11/9/10 08:21
Last Aliquot Time	11/10/10 04:38
Sampling Duration (hours)	20.3

Pierce County NPDES **LORES: 11/17/10 05:30 Storm Targeted**

Precipitation Statistics	
Start	11/17/10 05:30
Stop	11/17/10 18:45
Duration (hr)	13.25
Depth (in)	0.42
Max Reading (in)	0.05
Max Hourly Intensity (in/hr)	0.14
Average Intensity (in/hr)	0.032
Antecedent Dry Period (hr)	36.2
% Complete	100

Flow Statistics	
	LORES FM
Start	11/17/10 05:30
Stop	11/18/10 00:45
Duration (hours)	19.5
Peak Flow (cfs)	0.066
Average Flow (cfs)	0.05
Volume (cf)	3493.8
% Complete	100

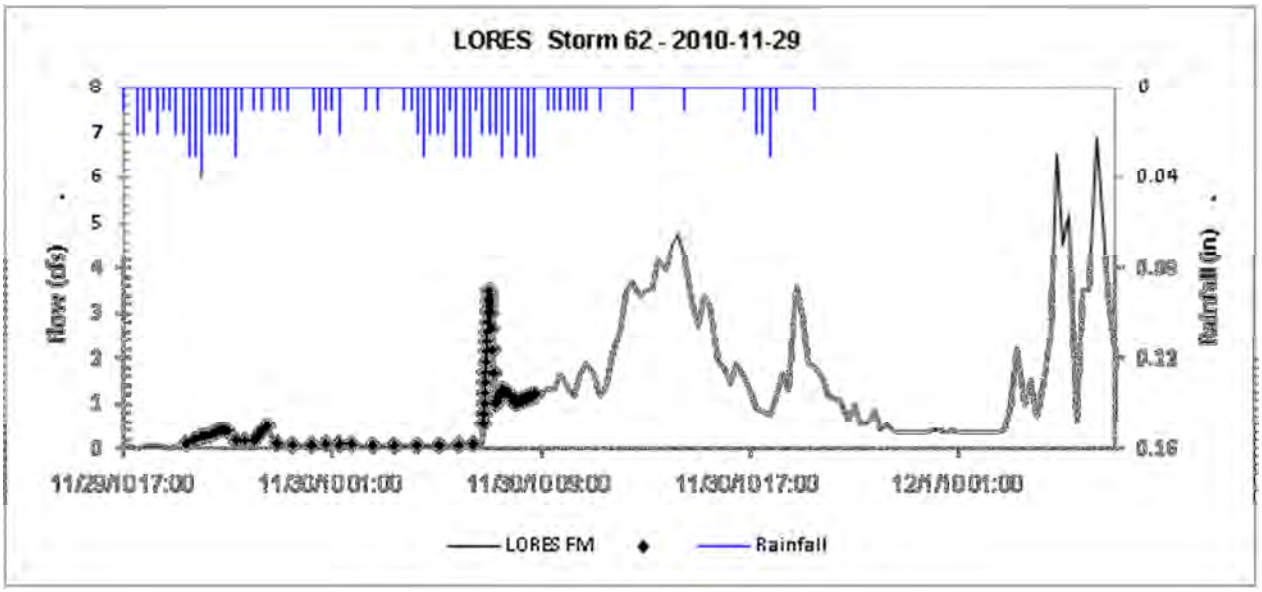


Sample Statistics	
	LORES FM
Aliquots	8
Volume Sampled (cf)	2808.9
% Coverage	80.4
First Aliquot Time	11/17/10 09:54
Last Aliquot Time	11/18/10 00:24
Sampling Duration (hours)	14.5

Pierce County NPDES **LORES: 11/29/10 17:00 Storm Targeted**

Precipitation Statistics	
Start	11/29/10 17:00
Stop	11/30/10 19:30
Duration (hr)	26.5
Depth (in)	1.16
Max Reading (in)	0.04
Max Hourly Intensity (in/hr)	0.14
Average Intensity (in/hr)	0.044
Antecedent Dry Period (hr)	70.2
% Complete	100

Flow Statistics	
	LORES FM
Start	11/29/10 17:00
Stop	12/1/10 07:30
Duration (hours)	33.25
Peak Flow (cfs)	6.907
Average Flow (cfs)	1.413
Volume (cf)	169127.1
% Complete	100

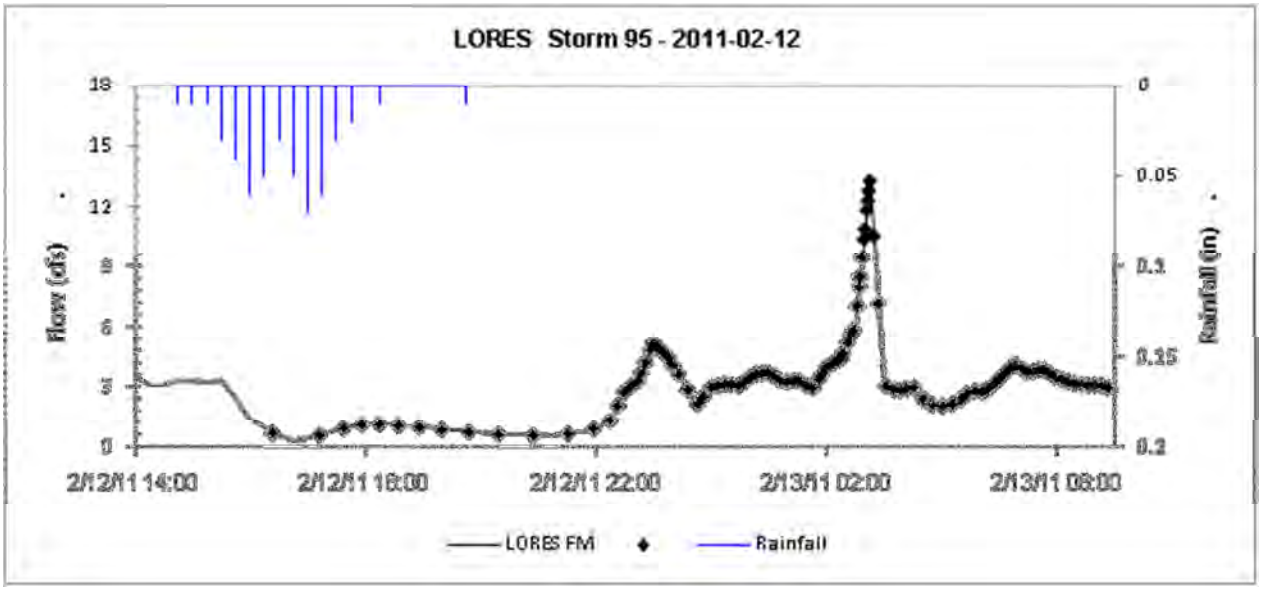


Sample Statistics	
	LORES FM
Aliquots	82
Volume Sampled (cf)	16231.4
% Coverage	17.9
First Aliquot Time	11/29/10 19:21
Last Aliquot Time	11/30/10 08:47
Sampling Duration (hours)	13.4

Pierce County NPDES **LORES: 2/12/11 14:45 Storm Targeted**

Precipitation Statistics	
Start	2/12/11 14:45
Stop	2/12/11 19:45
Duration (hr)	5
Depth (in)	0.49
Max Reading (in)	0.07
Max Hourly Intensity (in/hr)	0.26
Average Intensity (in/hr)	0.098
Antecedent Dry Period (hr)	110.2
% Complete	100

Flow Statistics	
	LORES FM
Start	2/12/11 14:45
Stop	2/13/11 07:45
Duration (hours)	17.25
Peak Flow (cfs)	13.292
Average Flow (cfs)	2.537
Volume (cf)	157542.3
% Complete	100

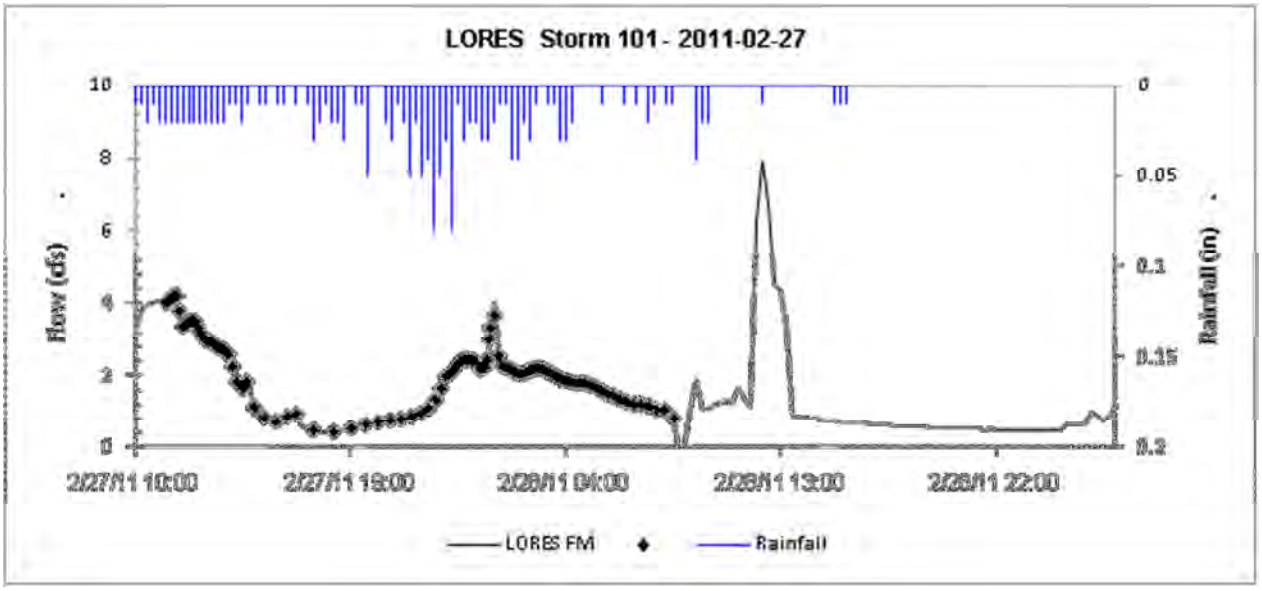


Sample Statistics	
	LORES FM
Aliquots	110
Volume Sampled (cf)	140169.4
% Coverage	89
First Aliquot Time	2/12/11 16:24
Last Aliquot Time	2/13/11 07:41
Sampling Duration (hours)	15.3

Pierce County NPDES **LORES: 2/27/11 10:00 Storm Targeted**

Precipitation Statistics	
Start	2/27/11 10:00
Stop	2/28/11 15:45
Duration (hr)	29.75
Depth (in)	1.69
Max Reading (in)	0.08
Max Hourly Intensity (in/hr)	0.28
Average Intensity (in/hr)	0.057
Antecedent Dry Period (hr)	71.5
% Complete	100

Flow Statistics	
	LORES FM
Start	2/27/11 10:00
Stop	3/1/11 03:45
Duration (hours)	41.5
Peak Flow (cfs)	7.958
Average Flow (cfs)	1.495
Volume (cf)	223393.5
% Complete	100

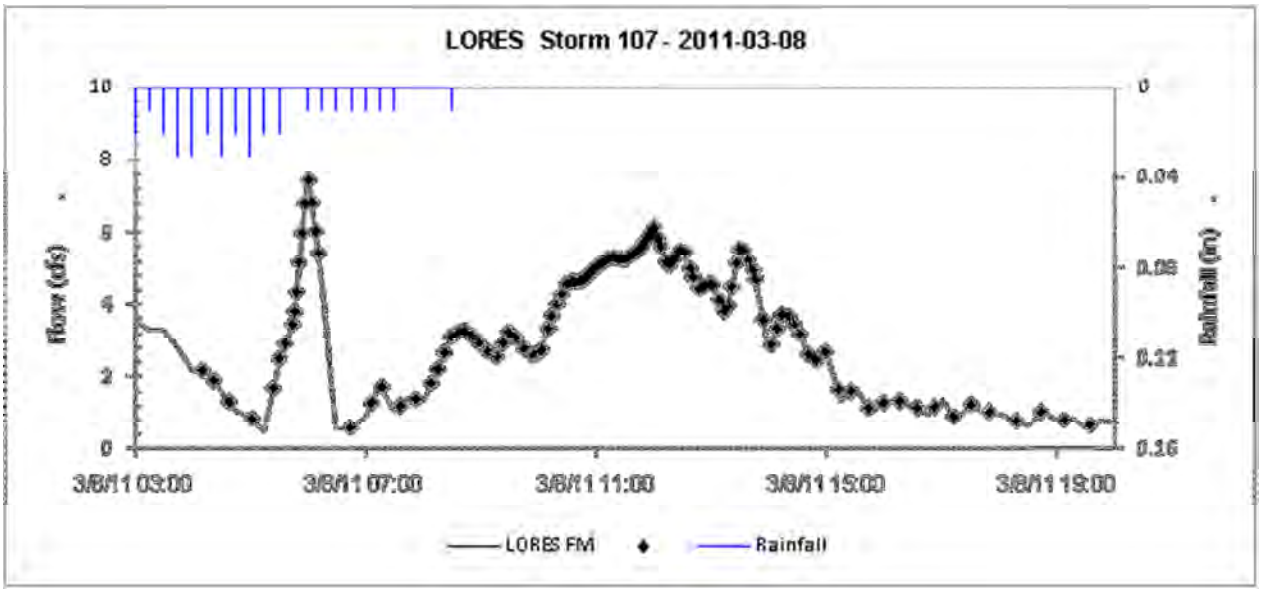


Sample Statistics	
	LORES FM
Aliquots	94
Volume Sampled (cf)	123620.7
% Coverage	82.2
First Aliquot Time	2/27/11 11:20
Last Aliquot Time	2/28/11 08:33
Sampling Duration (hours)	21.2

Pierce County NPDES **LORES: 3/8/11 03:00 Storm Targeted**

Precipitation Statistics	
Start	3/8/11 03:00
Stop	3/8/11 08:30
Duration (hr)	5.5
Depth (in)	0.33
Max Reading (in)	0.03
Max Hourly Intensity (in/hr)	0.13
Average Intensity (in/hr)	0.06
Antecedent Dry Period (hr)	79.8
% Complete	100

Flow Statistics	
	LORES FM
Start	3/8/11 03:00
Stop	3/8/11 20:30
Duration (hours)	17.75
Peak Flow (cfs)	7.603
Average Flow (cfs)	2.546
Volume (cf)	162684.9
% Complete	100

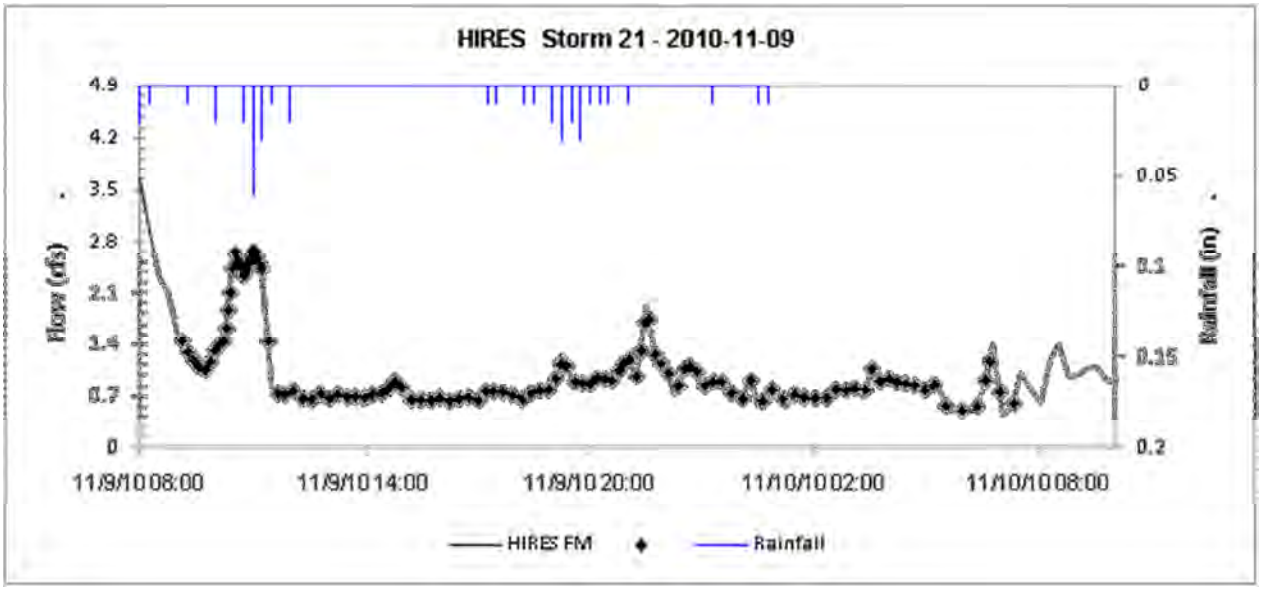


Sample Statistics	
	LORES FM
Aliquots	113
Volume Sampled (cf)	145246.8
% Coverage	89.3
First Aliquot Time	3/8/11 04:10
Last Aliquot Time	3/8/11 19:34
Sampling Duration (hours)	15.4

Pierce County NPDES **HIRES: 11/9/10 08:00 Storm Targeted**

Precipitation Statistics	
Start	11/9/10 08:00
Stop	11/10/10 00:45
Duration (hr)	16.75
Depth (in)	0.41
Max Reading (in)	0.06
Max Hourly Intensity (in/hr)	0.12
Average Intensity (in/hr)	0.024
Antecedent Dry Period (hr)	181.5
% Complete	100

Flow Statistics	
	HIRES FM
Start	11/9/10 08:00
Stop	11/10/10 10:30
Duration (hours)	26.75
Peak Flow (cfs)	3.641
Average Flow (cfs)	0.994
Volume (cf)	95741.12
% Complete	100

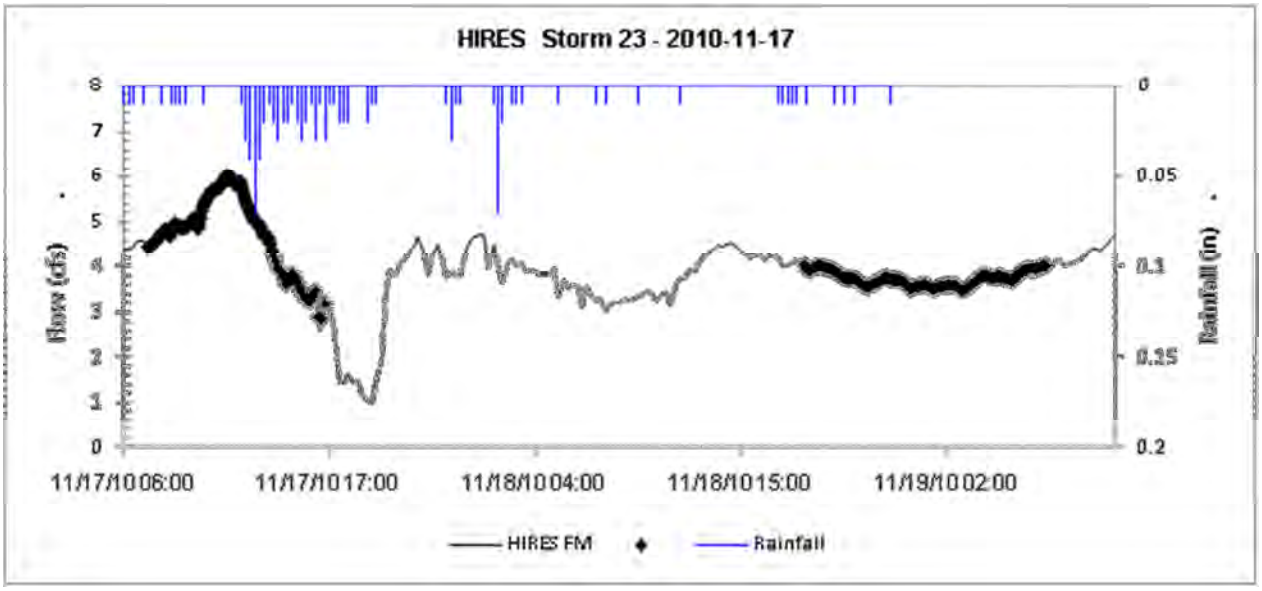


Sample Statistics	
	HIRES FM
Aliquots	113
Volume Sampled (cf)	72253.7
% Coverage	83.9
First Aliquot Time	11/9/10 09:06
Last Aliquot Time	11/10/10 07:17
Sampling Duration (hours)	22.2

Pierce County NPDES **HIRES: 11/17/10 06:00 Storm Targeted**

Precipitation Statistics	
Start	11/17/10 06:00
Stop	11/18/10 23:00
Duration (hr)	41
Depth (in)	1.02
Max Reading (in)	0.07
Max Hourly Intensity (in/hr)	0.2
Average Intensity (in/hr)	0.025
Antecedent Dry Period (hr)	34.2
% Complete	100

Flow Statistics	
	HIRES FM
Start	11/17/10 06:00
Stop	11/19/10 11:00
Duration (hours)	53.25
Peak Flow (cfs)	6.019
Average Flow (cfs)	3.92
Volume (cf)	751509.8
% Complete	100

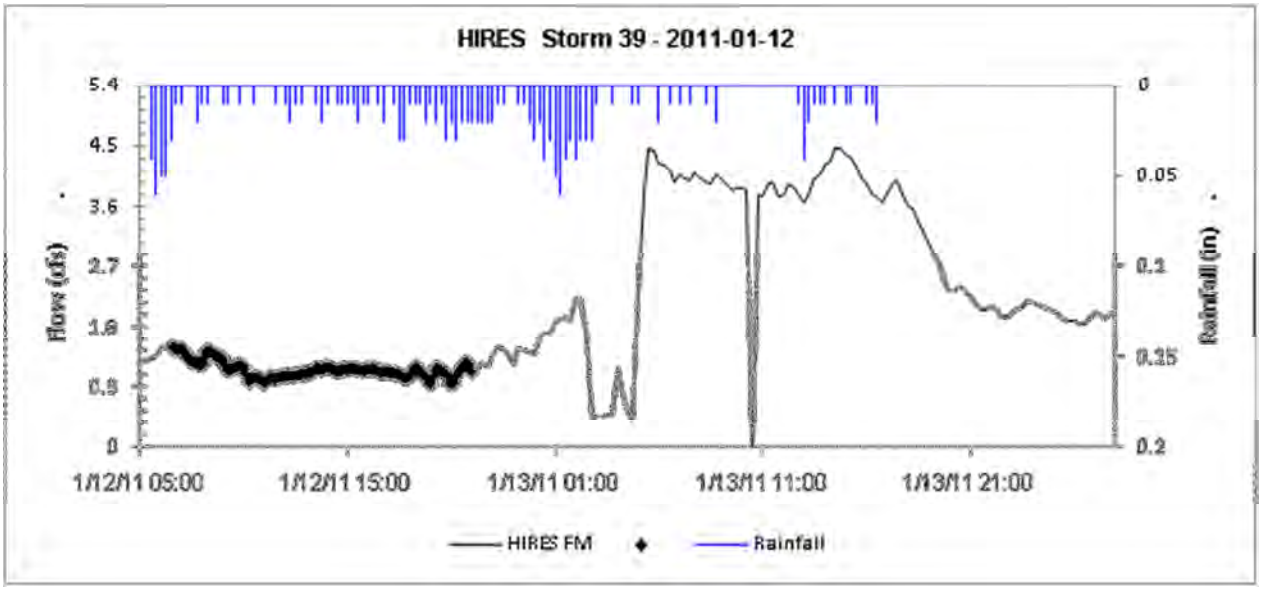


Sample Statistics	
	HIRES FM
Aliquots	226
Volume Sampled (cf)	157988.2
% Coverage	45
First Aliquot Time	11/17/10 07:20
Last Aliquot Time	11/19/10 07:16
Sampling Duration (hours)	47.9

Pierce County NPDES **HIRES: 1/12/11 5:30 Storm Targeted**

Precipitation Statistics	
Start	1/12/11 05:30
Stop	1/13/11 16:30
Duration (hr)	35
Depth (in)	1.66
Max Reading (in)	0.06
Max Hourly Intensity (in/hr)	0.23
Average Intensity (in/hr)	0.047
Antecedent Dry Period (hr)	40.5
% Complete	100

Flow Statistics	
	HIRES FM
Start	1/12/11 05:30
Stop	1/14/11 04:15
Duration (hours)	46.75
Peak Flow (cfs)	4.475
Average Flow (cfs)	2.208
Volume (cf)	371557.8
% Complete	100

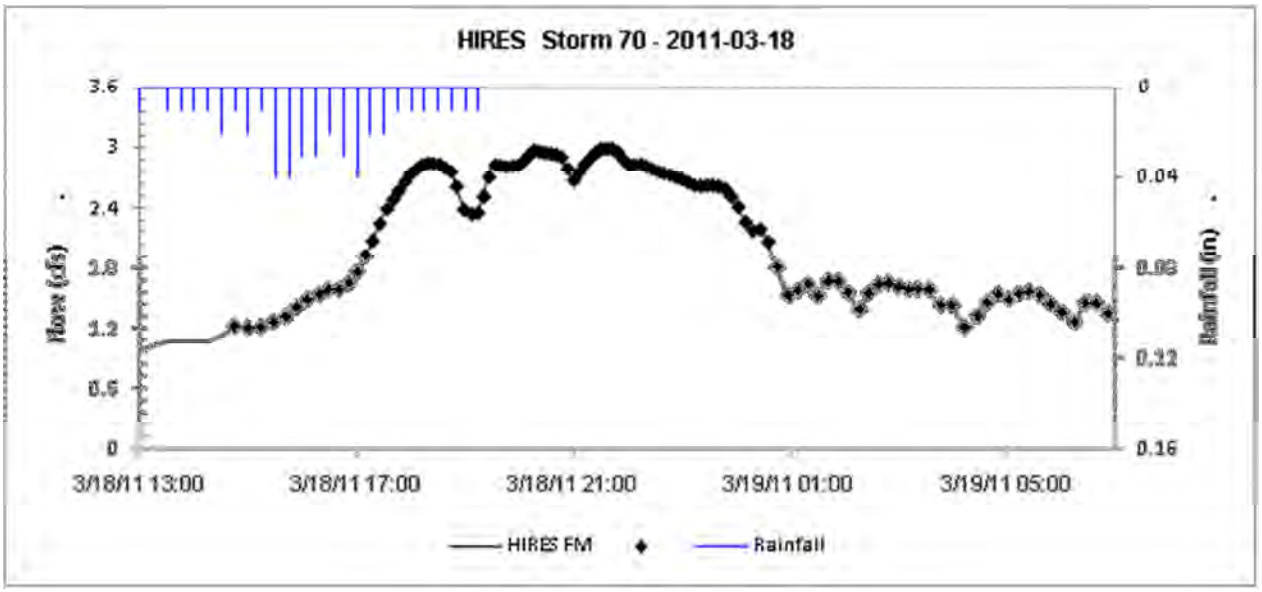


Sample Statistics	
	HIRES FM
Aliquots	113
Volume Sampled (cf)	59771.3
% Coverage	52.9
First Aliquot Time	1/12/11 06:35
Last Aliquot Time	1/12/11 20:57
Sampling Duration (hours)	14.4

Pierce County NPDES **HIRES: 3/18/11 13:00 Storm Targeted**

Precipitation Statistics	
Start	3/18/11 13:00
Stop	3/18/11 19:15
Duration (hr)	6.25
Depth (in)	0.44
Max Reading (in)	0.04
Max Hourly Intensity (in/hr)	0.16
Average Intensity (in/hr)	0.07
Antecedent Dry Period (hr)	42
% Complete	100

Flow Statistics	
	HIRES FM
Start	3/18/11 13:00
Stop	3/19/11 07:15
Duration (hours)	18.5
Peak Flow (cfs)	2.99
Average Flow (cfs)	1.921
Volume (cf)	127924.2
% Complete	100

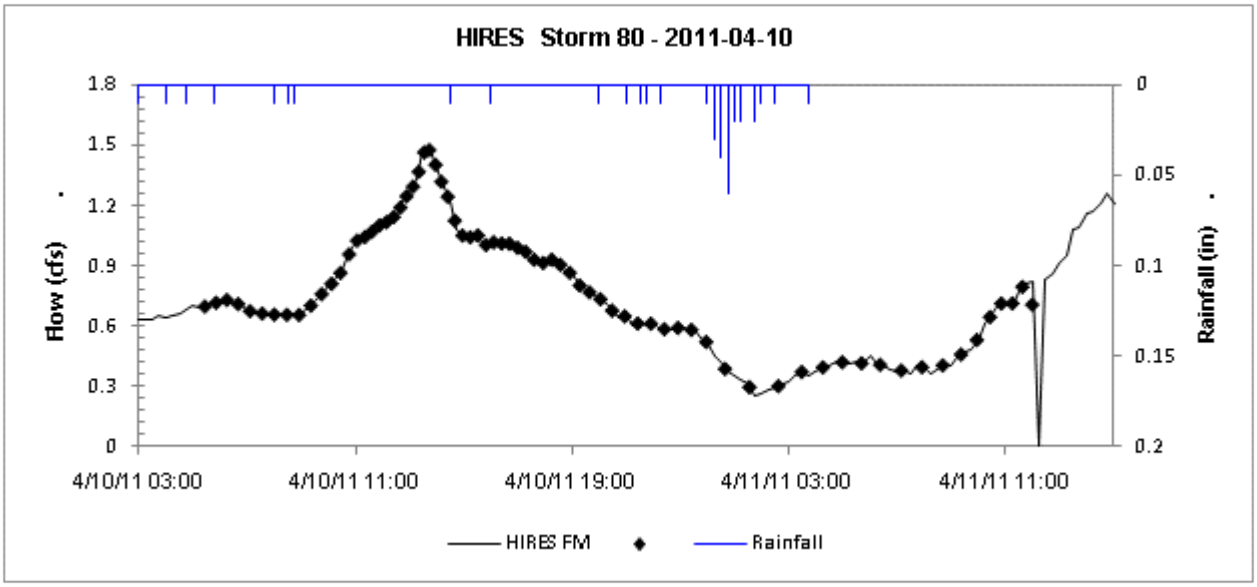


Sample Statistics	
	HIRES FM
Aliquots	113
Volume Sampled (cf)	119458.9
% Coverage	93.4
First Aliquot Time	3/18/11 14:44
Last Aliquot Time	3/19/11 07:05
Sampling Duration (hours)	16.4

Pierce County NPDES HIRES: 4/10/11 03:00 Storm Targeted

Precipitation Statistics	
Start	4/10/11 03:00
Stop	4/11/11 03:45
Duration (hr)	24.75
Depth (in)	0.36
Max Reading (in)	0.06
Max Hourly Intensity (in/hr)	0.17
Average Intensity (in/hr)	0.015
Antecedent Dry Period (hr)	61.8
% Complete	100

Flow Statistics	
	HIRES FM
Start	4/10/11 03:00
Stop	4/11/11 15:45
Duration (hours)	36.75
Peak Flow (cfs)	1.484
Average Flow (cfs)	0.734
Volume (cf)	97093.79
% Complete	100

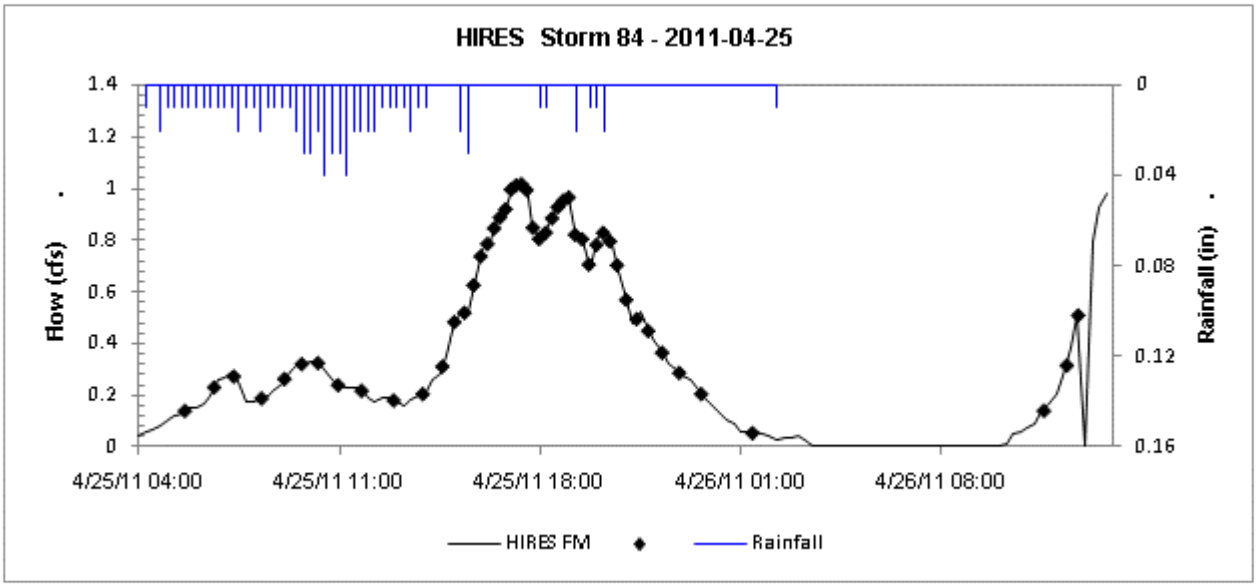


Sample Statistics	
	HIRES FM
Aliquots	73
Volume Sampled (cf)	60953.8
% Coverage	90
First Aliquot Time	4/10/11 05:25
Last Aliquot Time	4/11/11 12:02
Sampling Duration (hours)	30.6

Pierce County NPDES HIRES: 4/25/11 04:15 Storm Targeted

Precipitation Statistics	
Start	4/25/11 04:15
Stop	4/26/11 02:15
Duration (hr)	22
Depth (in)	0.76
Max Reading (in)	0.04
Max Hourly Intensity (in/hr)	0.16
Average Intensity (in/hr)	0.035
Antecedent Dry Period (hr)	8
% Complete	100

Flow Statistics	
	HIRES FM
Start	4/25/11 04:15
Stop	4/26/11 14:15
Duration (hours)	27
Peak Flow (cfs)	1.122
Average Flow (cfs)	0.375
Volume (cf)	36458.1
% Complete	100

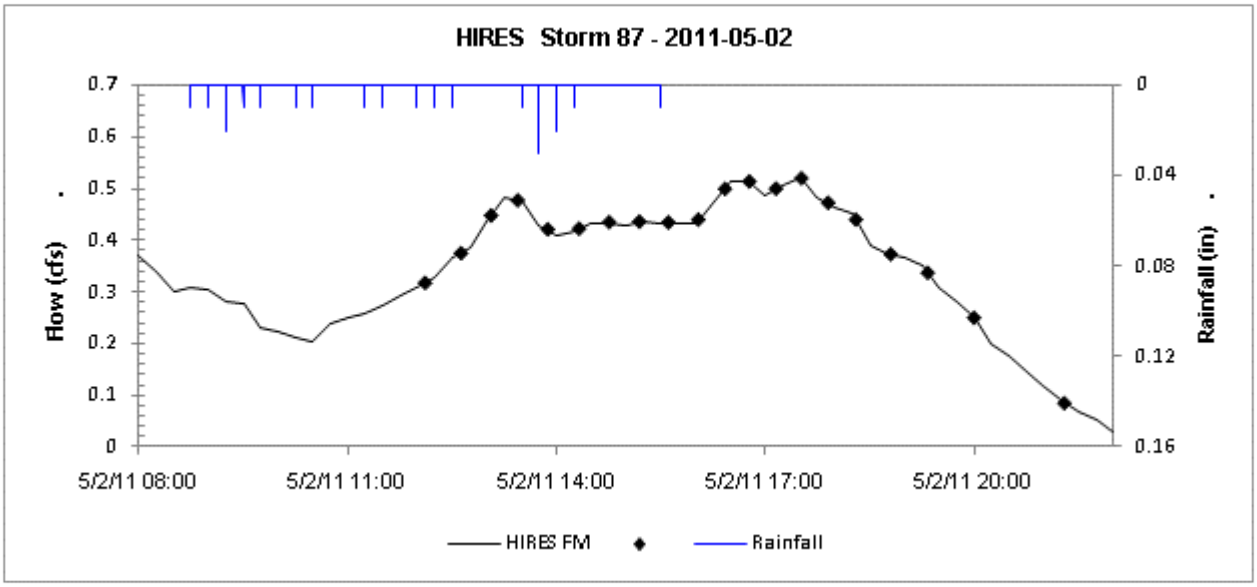


Sample Statistics	
	HIRES FM
Aliquots	48
Volume Sampled (cf)	29490.7
% Coverage	97.2
First Aliquot Time	4/25/11 05:38
Last Aliquot Time	4/26/11 12:45
Sampling Duration (hours)	31.1

Pierce County NPDES HIRES: 5/2/11 08:45 Storm Targeted

Precipitation Statistics	
Start	5/2/11 08:45
Stop	5/2/11 15:30
Duration (hr)	6.75
Depth (in)	0.2
Max Reading (in)	0.03
Max Hourly Intensity (in/hr)	0.07
Average Intensity (in/hr)	0.03
Antecedent Dry Period (hr)	84.8
% Complete	100

Flow Statistics	
	HIRES FM
Start	5/2/11 08:45
Stop	5/2/11 22:00
Duration (hours)	13.5
Peak Flow (cfs)	0.521
Average Flow (cfs)	0.334
Volume (cf)	16226.1
% Complete	100

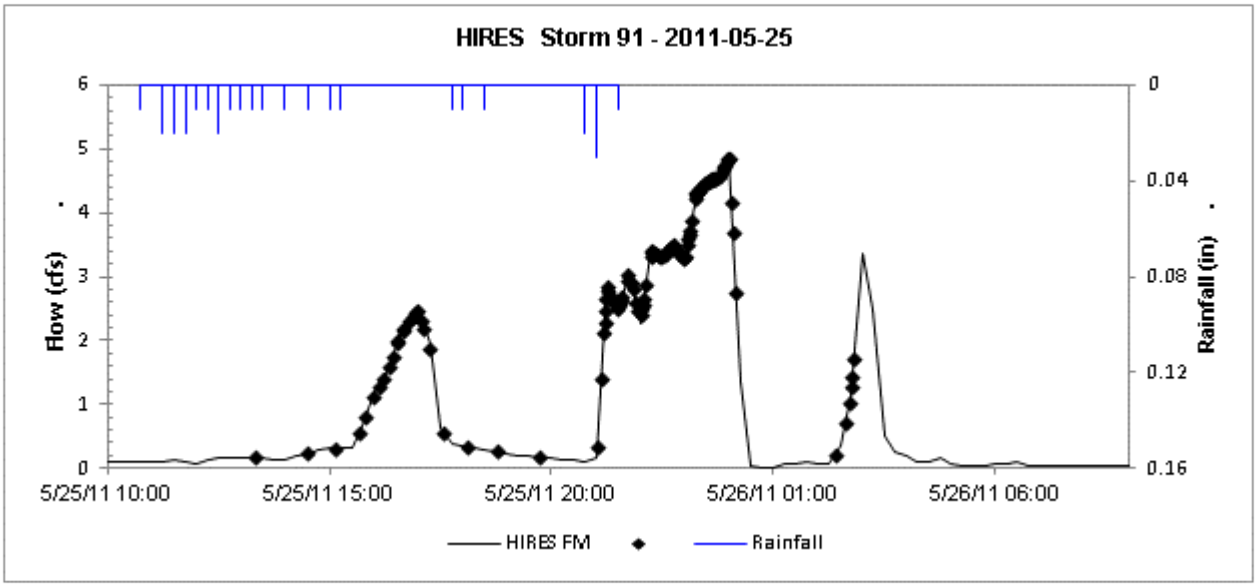


Sample Statistics	
	HIRES FM
Aliquots	20
Volume Sampled (cf)	12687.8
% Coverage	78.2
First Aliquot Time	5/2/11 12:06
Last Aliquot Time	5/2/11 21:18
Sampling Duration (hours)	9.2

Pierce County NPDES HIRES: 5/25/11 10:45 Storm Targeted

Precipitation Statistics	
Start	5/25/11 10:45
Stop	5/25/11 21:30
Duration (hr)	10.75
Depth (in)	0.27
Max Reading (in)	0.03
Max Hourly Intensity (in/hr)	0.08
Average Intensity (in/hr)	0.025
Antecedent Dry Period (hr)	86
% Complete	100

Flow Statistics	
	HIRES FM
Start	5/25/11 10:45
Stop	5/26/11 09:30
Duration (hours)	19.25
Peak Flow (cfs)	4.858
Average Flow (cfs)	0.946
Volume (cf)	65576.7
% Complete	100

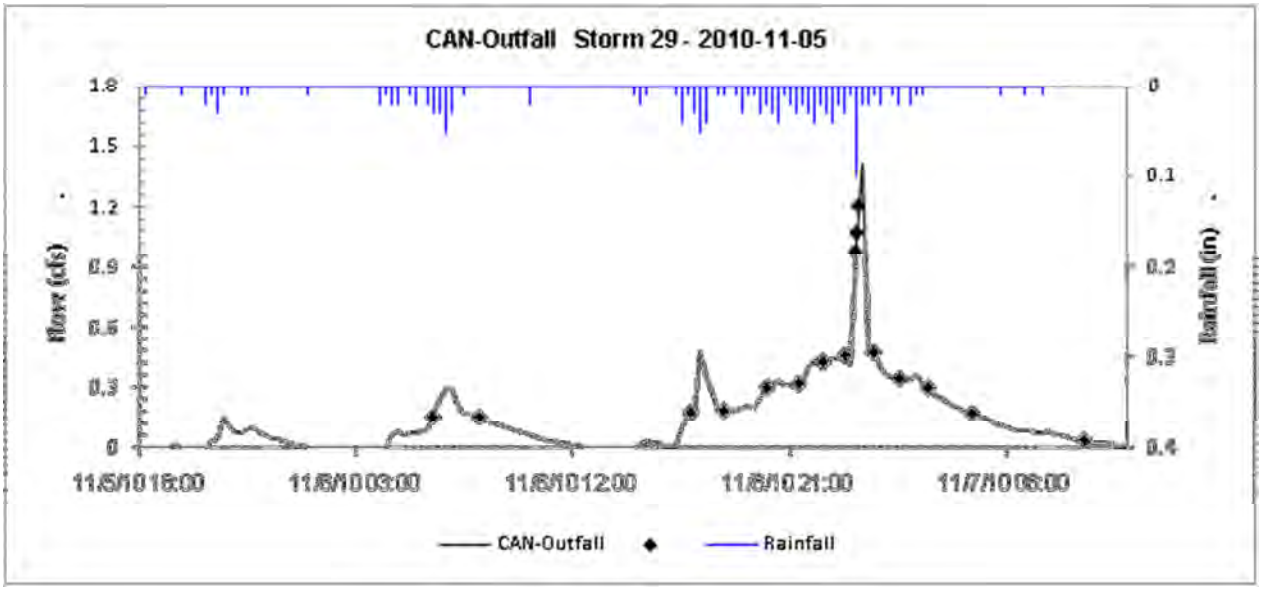


Sample Statistics	
	HIRES FM
Aliquots	113
Volume Sampled (cf)	57725.8
% Coverage	88
First Aliquot Time	5/25/11 13:20
Last Aliquot Time	5/26/11 02:49
Sampling Duration (hours)	13.5

Pierce County NPDES CAN - Outfall: 11/5/10 18:15 Storm Targeted

Precipitation Statistics	
Start	11/5/10 18:15
Stop	11/7/10 07:30
Duration (hr)	37.25
Depth (in)	1.4
Max Reading (in)	0.1
Max Hourly Intensity (in/hr)	0.2
Average Intensity (in/hr)	0.038
Antecedent Dry Period (hr)	95.8
% Complete	100

Flow Statistics	
	CAN-Outfall
Start	11/5/10 19:30
Stop	11/7/10 11:00
Duration (hours)	31.5
Peak Flow (cfs)	1.416
Average Flow (cfs)	0.186
Volume (cf)	21060
% Complete	100

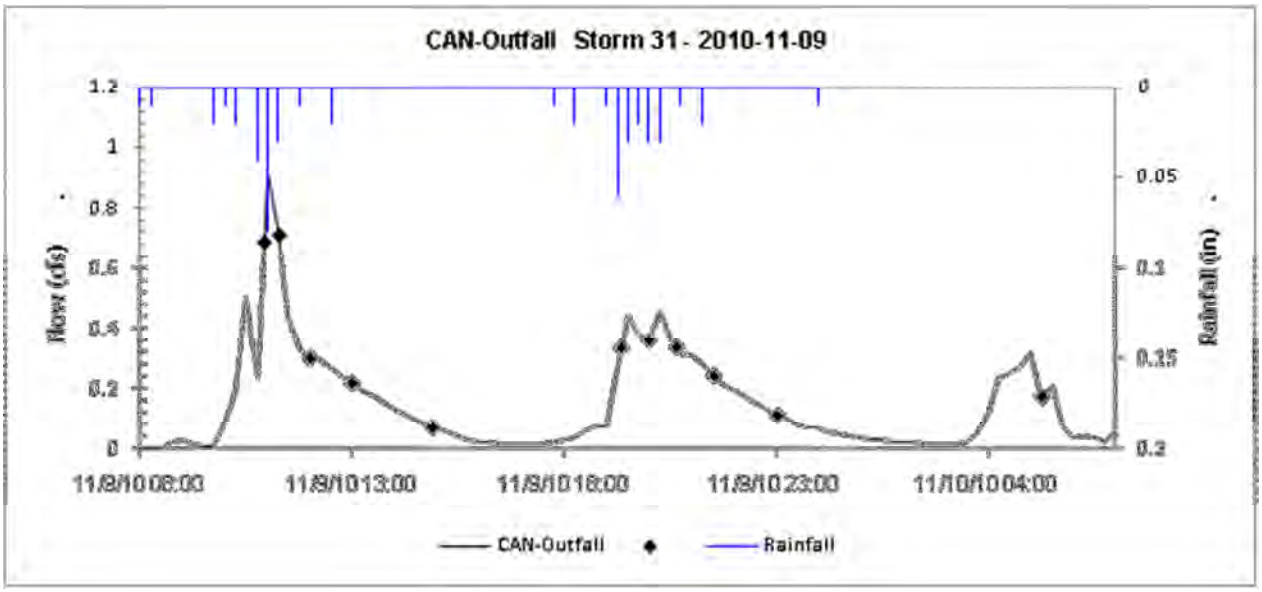


Sample Statistics	
	CAN-Outfall
Aliquots	16
Volume Sampled (cf)	3004.3
% Coverage	50.9
First Aliquot Time	11/6/10 06:10
Last Aliquot Time	11/7/10 09:15
Sampling Duration (hours)	27.1

Pierce County NPDES CAN - Outfall: 11/9/10 08:00 Storm Targeted

Precipitation Statistics	
Start	11/9/10 08:00
Stop	11/10/10 00:00
Duration (hr)	16
Depth (in)	0.5
Max Reading (in)	0.08
Max Hourly Intensity (in/hr)	0.17
Average Intensity (in/hr)	0.031
Antecedent Dry Period (hr)	36.2
% Complete	100

Flow Statistics	
	CAN-Outfall
Start	11/9/10 08:30
Stop	11/10/10 07:45
Duration (hours)	23.5
Peak Flow (cfs)	0.916
Average Flow (cfs)	0.145
Volume (cf)	12262.5
% Complete	100

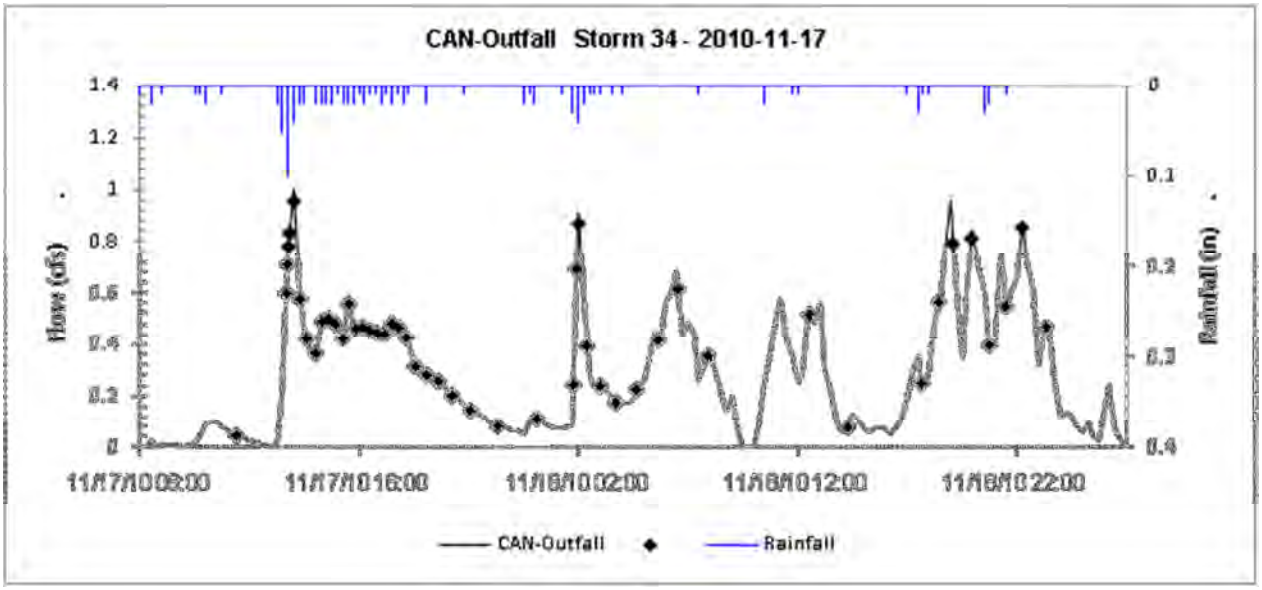


Sample Statistics	
	CAN-Outfall
Aliquots	11
Volume Sampled (cf)	10216.2
% Coverage	83.3
First Aliquot Time	11/9/10 10:55
Last Aliquot Time	11/10/10 05:15
Sampling Duration (hours)	18.3

Pierce County NPDES CAN - Outfall: 11/17/10 06:00 Storm Targeted

Precipitation Statistics	
Start	11/17/10 06:00
Stop	11/18/10 21:30
Duration (hr)	39.5
Depth (in)	1.03
Max Reading (in)	0.1
Max Hourly Intensity (in/hr)	0.23
Average Intensity (in/hr)	0.026
Antecedent Dry Period (hr)	33.5
% Complete	100

Flow Statistics	
	CAN-Outfall
Start	11/17/10 06:30
Stop	11/19/10 03:45
Duration (hours)	42.75
Peak Flow (cfs)	1.008
Average Flow (cfs)	0.29
Volume (cf)	44572.48
% Complete	100

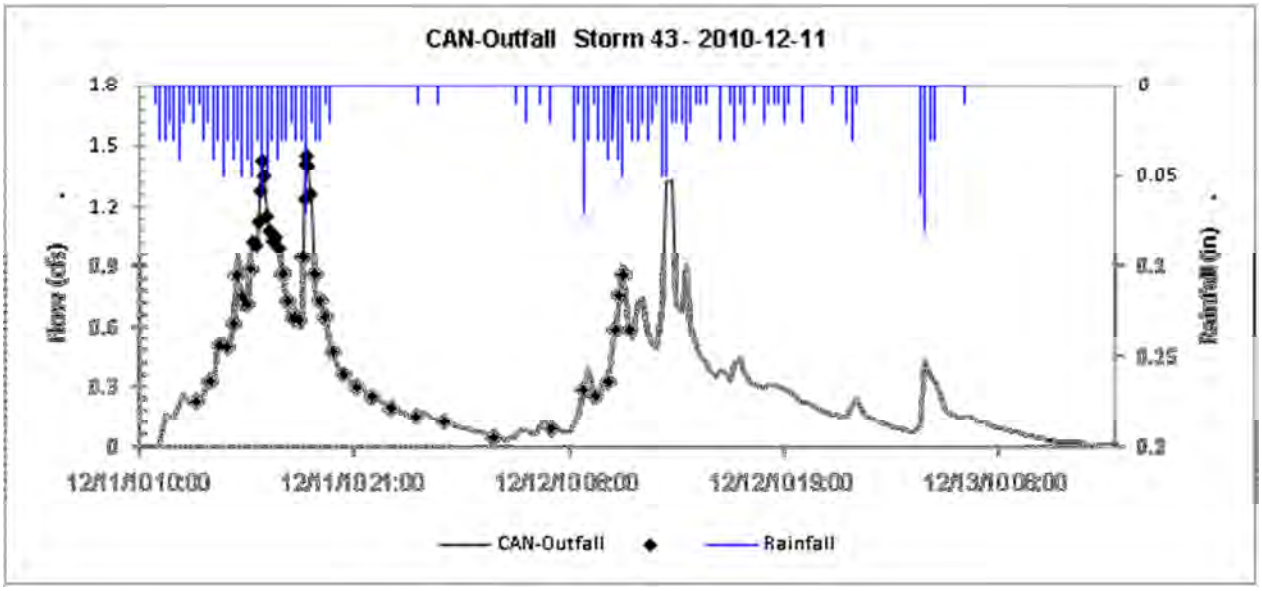


Sample Statistics	
	CAN-Outfall
Aliquots	49
Volume Sampled (cf)	19058.1
% Coverage	93.9
First Aliquot Time	11/17/10 10:24
Last Aliquot Time	11/18/10 23:23
Sampling Duration (hours)	37

Pierce County NPDES CAN - Outfall: 12/11/10 10:45 Storm Targeted

Precipitation Statistics	
Start	12/11/10 10:45
Stop	12/13/10 04:15
Duration (hr)	41.5
Depth (in)	2.49
Max Reading (in)	0.08
Max Hourly Intensity (in/hr)	0.23
Average Intensity (in/hr)	0.06
Antecedent Dry Period (hr)	28
% Complete	100

Flow Statistics	
	CAN-Outfall
Start	12/11/10 10:45
Stop	12/13/10 12:30
Duration (hours)	49.5
Peak Flow (cfs)	1.478
Average Flow (cfs)	0.305
Volume (cf)	54355.48
% Complete	100

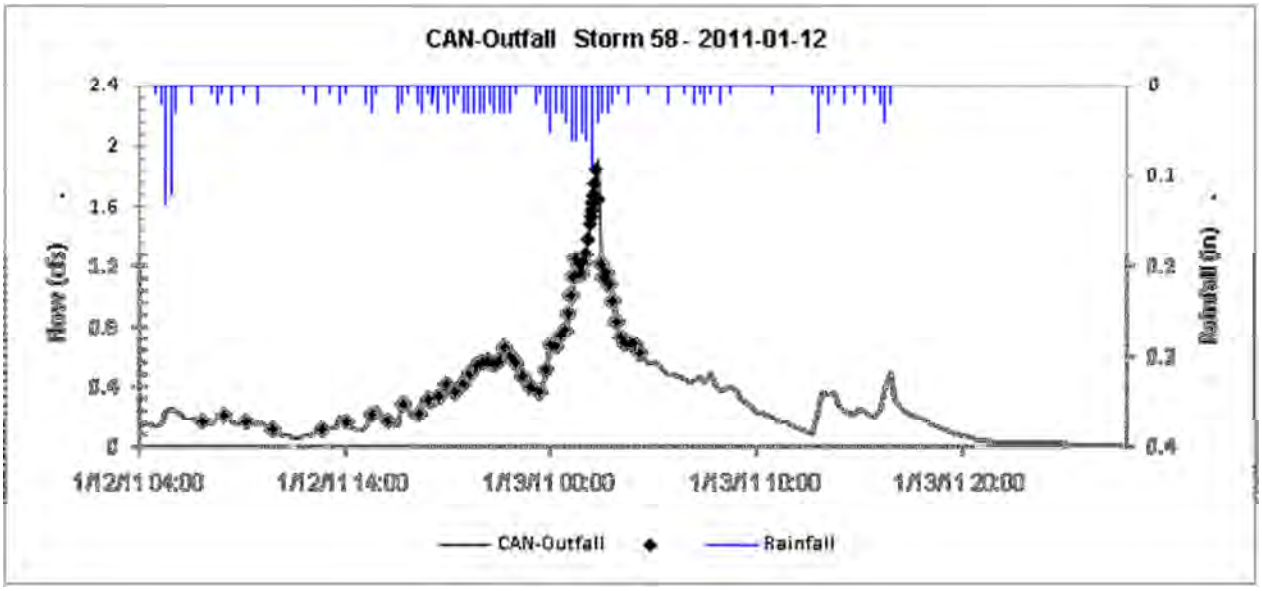


Sample Statistics	
	CAN-Outfall
Aliquots	49
Volume Sampled (cf)	29039.5
% Coverage	93.8
First Aliquot Time	12/11/10 12:51
Last Aliquot Time	12/12/10 11:06
Sampling Duration (hours)	22.3

Pierce County NPDES CAN - Outfall: 1/12/11 04:45 Storm Targeted

Precipitation Statistics	
Start	1/12/11 04:45
Stop	1/13/11 16:30
Duration (hr)	35.75
Depth (in)	2.14
Max Reading (in)	0.13
Max Hourly Intensity (in/hr)	0.32
Average Intensity (in/hr)	0.06
Antecedent Dry Period (hr)	40.8
% Complete	100

Flow Statistics	
	CAN-Outfall
Start	1/12/11 04:45
Stop	1/14/11 04:15
Duration (hours)	47.75
Peak Flow (cfs)	1.927
Average Flow (cfs)	0.3
Volume (cf)	51575.14
% Complete	100

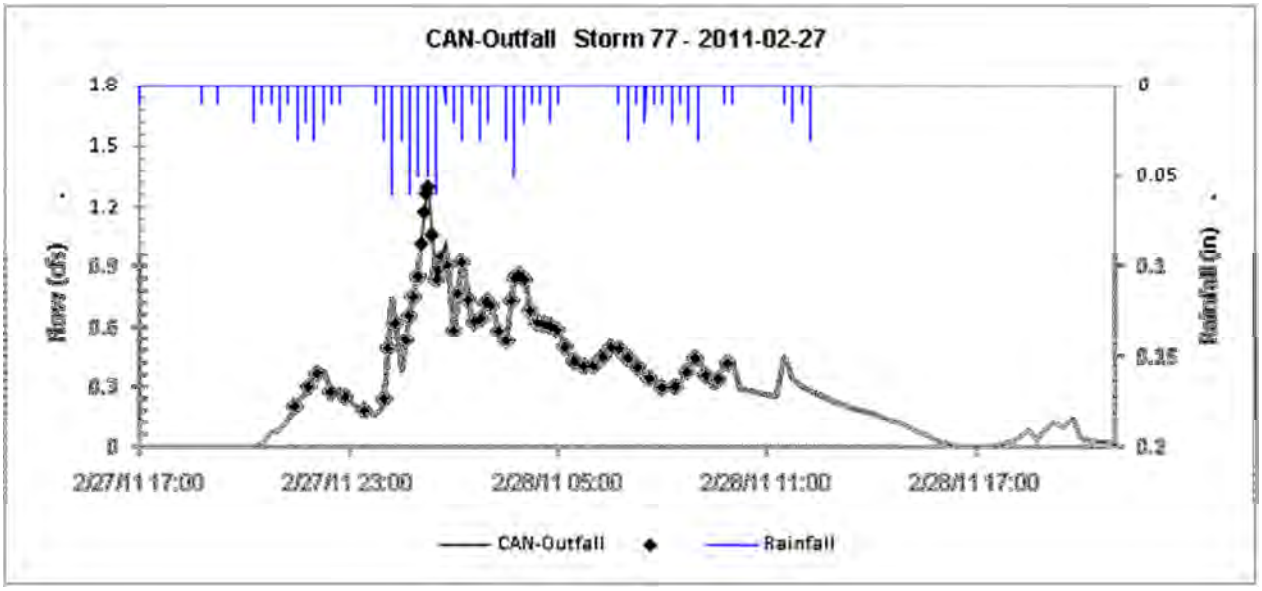


Sample Statistics	
	CAN-Outfall
Aliquots	60
Volume Sampled (cf)	32296
% Coverage	92.5
First Aliquot Time	1/12/11 07:00
Last Aliquot Time	1/13/11 04:21
Sampling Duration (hours)	21.4

Pierce County NPDES CAN - Outfall: 2/27/11 17:00 Storm Targeted

Precipitation Statistics	
Start	2/27/11 17:00
Stop	2/28/11 12:15
Duration (hr)	19.25
Depth (in)	1.09
Max Reading (in)	0.06
Max Hourly Intensity (in/hr)	0.25
Average Intensity (in/hr)	0.057
Antecedent Dry Period (hr)	74.8
% Complete	100

Flow Statistics	
	CAN-Outfall
Start	2/27/11 19:15
Stop	2/28/11 21:15
Duration (hours)	23.5
Peak Flow (cfs)	1.33
Average Flow (cfs)	0.336
Volume (cf)	28464.75
% Complete	100

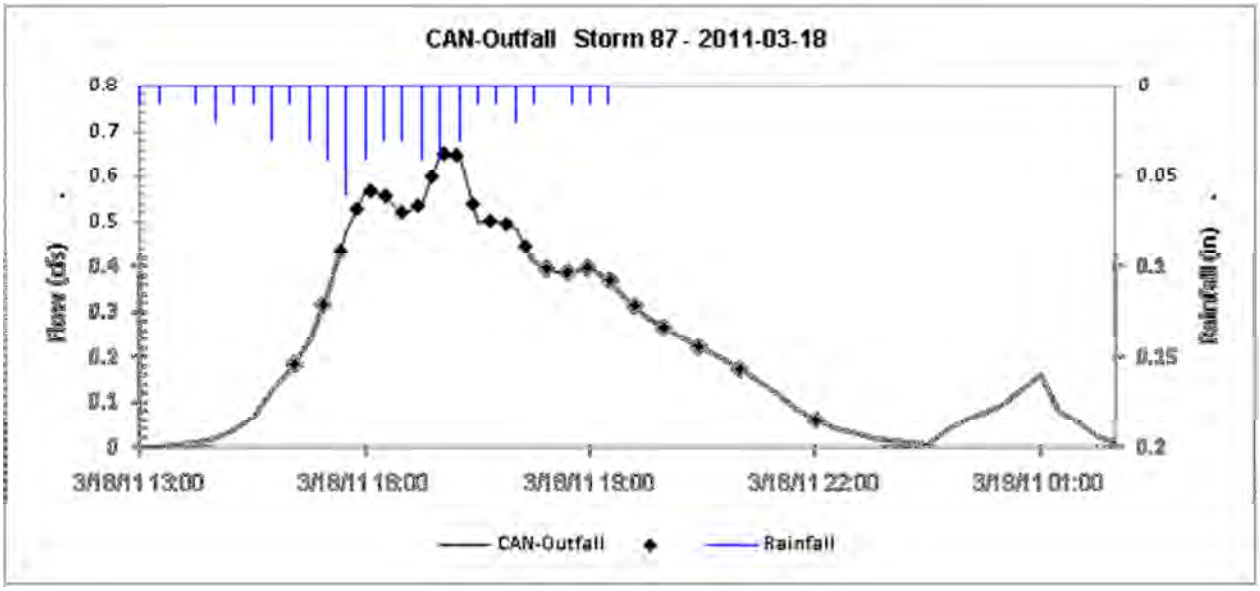


Sample Statistics	
	CAN-Outfall
Aliquots	58
Volume Sampled (cf)	22552.3
% Coverage	81.3
First Aliquot Time	2/27/11 21:27
Last Aliquot Time	2/28/11 09:52
Sampling Duration (hours)	12.4

Pierce County NPDES CAN - Outfall: 3/18/11 13:00 Storm Targeted

Precipitation Statistics	
Start	3/18/11 13:00
Stop	3/18/11 19:15
Duration (hr)	6.25
Depth (in)	0.52
Max Reading (in)	0.06
Max Hourly Intensity (in/hr)	0.2
Average Intensity (in/hr)	0.083
Antecedent Dry Period (hr)	42.2
% Complete	100

Flow Statistics	
	CAN-Outfall
Start	3/18/11 13:30
Stop	3/19/11 02:00
Duration (hours)	12.75
Peak Flow (cfs)	0.651
Average Flow (cfs)	0.224
Volume (cf)	10265.4
% Complete	100

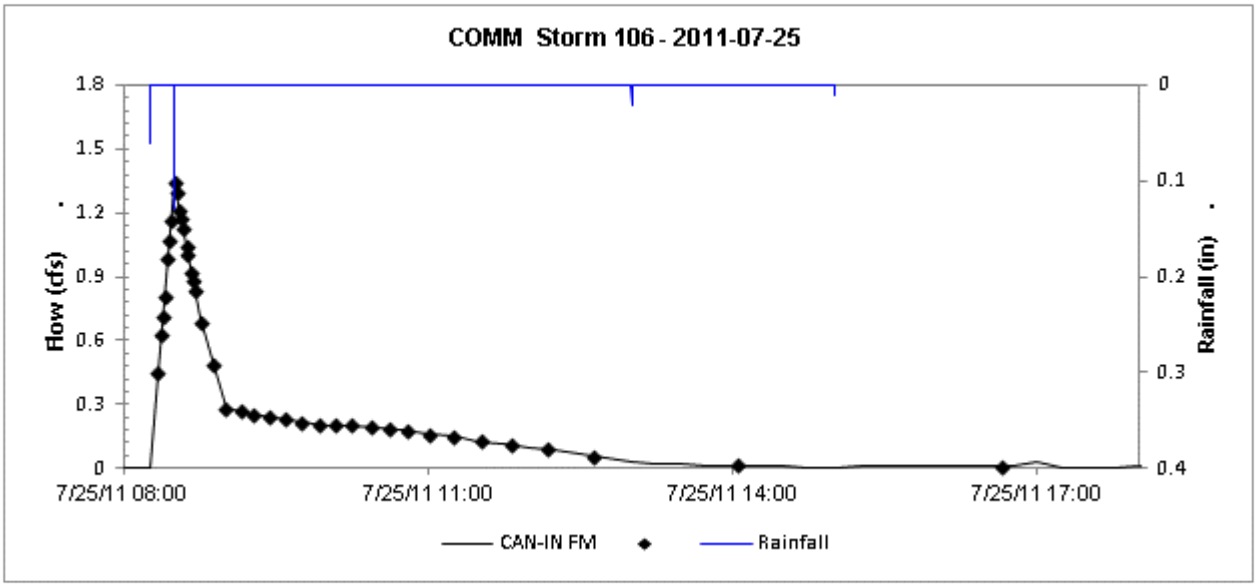


Sample Statistics	
	CAN-Outfall
Aliquots	24
Volume Sampled (cf)	9055.6
% Coverage	88.2
First Aliquot Time	3/18/11 15:03
Last Aliquot Time	3/18/11 22:01
Sampling Duration (hours)	7

Pierce County NPDES COMM: 7/25/11 08:15 Storm Targeted

Precipitation Statistics	
Start	7/25/11 08:15
Stop	7/25/11 15:00
Duration (hr)	6.75
Depth (in)	0.16
Max Reading (in)	0.13
Max Hourly Intensity (in/hr)	0.19
Average Intensity (in/hr)	0.024
Antecedent Dry Period (hr)	183.5
% Complete	100

Flow Statistics	
	CAN-IN FM
Start	7/25/11 08:15
Stop	7/25/11 18:45
Duration (hours)	7.5
Peak Flow (cfs)	1.339
Average Flow (cfs)	0.166
Volume (cf)	4474.799
% Complete	



Sample Statistics	
	CAN-IN FM
Aliquots	39
Volume Sampled (cf)	4001.3
% Coverage	89.4
First Aliquot Time	7/25/11 08:20
Last Aliquot Time	7/25/11 16:39
Sampling Duration (hours)	8.3

Appendix B

Water Quality Data for Qualifying Events

Pierce County NPDES Outfall Monitoring data for Commercial, Low Density Residential and High Density Residential stations.

| Collection Date | Storm Start Date | Spectra File # | Polycyclic Aromatic Hydrocarbons (PAHs) | | |

Pierce County NPDES Outfall Monitoring data for Commercial, Low Density Residential and High Density Residential stations.

			Phthalates																											
Collection Date	Storm Start Date	Spectra File #	bis(2-Ethylhexyl)Phthalate (µg/L)		Butylbenzyl phthalate (µg/L)		Diethylphthalate (µg/L)		Dimethyl Phthalate (µg/L)		Di-n-Butylphthalate (µg/L)		Di-n-Octyl Phthalate (µg/L)		Dissolved Cadmium (µg/L)		Dissolved Copper (µg/L)		Dissolved Lead (µg/L)		Dissolved Zinc (µg/L)		Total Recoverable Cadmium (µg/L)		Total Recoverable Copper (µg/L)		Total Recoverable Lead (µg/L)		Total Recoverable Zinc (µg/L)		Dissolved Mercury (µg/L)		Mercury (µg/L)	
Commercial-Canyon Road																														
11/8/2010	11/5/2010 18:15	20101101 88	24	J	1.00	U	1.00	U	1.00	U	1.00	U	1.00	U	0.11		3.7		0.29		24.2		0.13		8.7		6.80		61.7		0.1	U	0.1	U
11/9/2010	11/9/2010 8:00	2010110202																												
11/10/2010	11/9/2010 8:00	2010110247	13.4	J	1.00	U	1.00	U	1.00	U	1.00	U	1.00	U	0.06		5.1		0.35		40.3		0.11		10.8		4.47		42.8		0.1	U	0.1	U
11/17/2010	11/17/2010 6:00	2010110415																												
11/18/2010	11/17/2010 6:00	2010110454	19.6	J	1.00	U	1.00	U	1.00	U	1.00	U	1.00	U	0.04		3.9		0.23	J	6.4		0.14		17.1		13.10		8.6		0.1	U	0.1	U
12/13/2010	12/11/2011 10:45	2010120305	2.38	J	1.00	U	1.00	U	1.00	U	1.00	U	1.00	U	0.05		2.8		0.19	J	14.8		0.10		8.6		5.24		40.8		0.1	U	0.1	U
1/12/2011	1/12/2011 4:45	2011010200																												
1/13/2011	1/12/2011 4:45	2011010232	3.93	J	1.00	U	1.00	U	1.00	U	1.00	U	1.00	U	0.09	J	3.0		0.19		34.9		0.09	J	6.6		3.08		99.1		0.1	U	0.1	U
2/28/2011	2/27/2011 17:00	2011020558	6.72	J	1.18		1.00	U	1.00	U	1.83		1.00	U	0.066	J	2.6		0.09		51.5		0.237	J	18.8		16.40		99.7		0.1	U	0.1	U
3/21/2011	3/18/2011 13:00	2011030418	4.37	J	1.00	U	1.00	U	1.00	U	1.00	U	1.00	U	0.09	J	3.7		0.14		5.3	J	0.17	J	10.2		5.36		25.5		0.1	U	0.1	U
4/21/2011	NA	2011040452																							2.6									
7/26/2011	7/25/2011 8:15	2011040453	4.64	J	1.00	U	1.00	U	1.00	U	1.34		1.00	U	0.08		11.7		1.60		62.3		0.38		47.0		64.10		262		0.1	U	0.1	U
Median			5.680		1.000		1.000		1.000		1.000		1.000		0.073		3.685		0.210		29.550		0.135		10.200		6.080		52.250		0.100		0.100	
Mean			9.880		1.023		1.000		1.000		1.146		1.000		0.073		4.559		0.385		29.963		0.170		14.489		14.819		80.025		0.100		0.100	
Min			2.380		1.000		1.000		1.000		1.000		1.000		0.040		2.600		0.090		5.300		0.090		2.600		3.080		8.600		0.100		0.100	
Max			24.000		1.180		1.000		1.000		1.830		1.000		0.110		11.700		1.600		62.300		0.380		47.000		64.100		262.000		0.100		0.100	
Low Density Residential-Lake Sylvia																														
10/28/2010	NA	2010100682																												
11/8/2010	11/5/2010 16:00:00 PM	2010110186	1.16	J	1.000	U	1.000	U	1.000	U	1.000	U	1.00	U	0.02		1.2		0.18		4.1		0.03		1.3		0.37		7.4		0.1	U	0.1	U
11/9/2010	11/9/2010 3:45	2010110202																												
11/17/2010	11/17/2010 5:30	2010110413																												
11/30/2010	11/29/2010 17:00	2010110580																												
2/14/2011	2/12/2011 14:45	2011020333	1.51	J	1.000	U	0.024	U	0.024	U	0.026		1.00	U	0.05	J	0.9		0.12		0.5	U	0.07	J	1.3		0.73		4.7		0.1	U	0.1	U
2/28/2011	2/27/2011 10:00	2011020560	5.69	J	1.000	U	1.000	U	1.000	U	1.730		1.00	U	0.123	J	1.0		0.11		0.8		0.484	J	2.5		1.83		6.1		0.1	U	0.1	U
3/9/2011	3/8/2011 3:00	2011030203	1.71	J	1.000	U	1.000	U	1.000	U	1.000	U	1.00	U	0.05	J	1.3		0.10		0.5	J	0.06	U	1.0		0.41		2.8		0.1	U	0.1	U
4/21/2011	NA	2011040452																							0.5	U								
6/10/2011	NA	2011060232																							0.5	U								
7/27/2011	NA	2011070474																							0.5	U								
Median			1.61		1.000		1.000		1.000		1.000		1.00		0.05		1.1		0.11		0.7		0.07		1.0		0.57		5.4		0.1		0.1	
Mean			2.52		1.000		0.756		0.756		0.939		1.00		0.06		1.1		0.13		1.5		0.16		1.1		0.84		5.3		0.1		0.1	
Min			1.16		1.000		0.024		0.024		0.026		1.00		0.02		0.9		0.10		0.5		0.03		0.5		0.37		2.8		0.1		0.1	
Max			5.69		1.000		1.000		1.000		1.730		1.00		0.12		1.3		0.18		4.1		0.48		2.5		1.83		7.4		0.1		0.1	
High Density Residential-Collinswood																														
11/9/2010	11/9/2010 8:00	2010110202																												
11/10/2010	11/9/2010 8:00	2010110245	27.8	J	1.000	U	1.000	U	1.000	U	1.000	U	1.00	U	0.06		2.3		0.13		4.2		0.11		2.6		0.27		5.4	J	0.1	U	0.1	U
11/17/2010	11/17/2010 6:00	2010110414																												
1/12/2011	1/12/2011 5:30	2011010200																												
2/10/2011		2011020241																												
3/21/2011	3/18/2011 13:00	2011030419	2.21	J	1.000	U	1.000	U	1.000	U	1.000	U	1.00	U	0.03		1.4	J	0.07	J	0.9		0.06	J	2.0		0.46		4.9		0.1	U	0.1	U
4/11/2011	4/10/2011 3:00	2011040209	2.30	J	1.000	U	1.000	U	1.000	U	1.000	U	1.00	U	0.02		1.4		0.07		1.6		0.13		1.6		0.22		3.7		0.1	U	0.1	U
4/21/2011	NA	2011040452																							1.0									
4/26/2011	4/25/2011 14:15	2011040526	11		1.000	U	1.000	U	1.000	U	1.000	U	1.00	U	0.08		2.2		0.07		1.2		0.05		2.9		0.73		14.0		0.1	U	0.1	U
5/5/2011	5/2/2011 8:45	2011050136	1.35	J	1.000	U	1.000	U	1.000	U	1.000	U	1.00	U	0.05		1.2		0.05		0.6		0.04		1.3		0.14		1.4		0.1	U	0.1	U
5/27/2011	5/25/2011 10:45	2011050624	2.89	J	1.000	U	1.000	U	1.000	U	1.000	U	1.00	U	0.13		1.9		0.08		1.6		0.61		3.3		0.45		9.3		0.1	U	0.1	U
6/10/2011	NA	2011060232																							0.5									
Median			2.595		1.000		1.000		1.000		1.000		1.00		0.06		1.7		0.07		1.4		0.09		2.0		0.36		5.2		0.1		0.1	
Mean			7.925		1.000		1.000		1.000		1.000		1.00		0.06		1.7		0.08		1.7		0.17		2.1		0.38		6.5		0.1		0.1	
Min			1.35		1.000		1.000		1.000		1.000		1.00		0.02		1.2		0.05		0.6		0.04		1.0		0.14		1.4		0.1		0.1	
Max			27.8		1.000		1.000		1.000		1.000		1.00		0.13		2.3		0.13		4.2		0.61		3.3		0.73		14.0		0.1		0.1	

Pierce County NPDES Outfall Monitoring data for Commercial, Low Density Residential and High Density Residential stations.

Collection Date	Storm Start Date	Spectra File #	Conductivity (umhos/cm)	Turbidity (NTU)	Orthophospha te, as P (mg/L)		Total Phosphorus (mg/L)		Total Hardness (mg/L)		Total Suspended Solids (mg/L)	Chloride (mg/L)		TKN (mg/L)		Nitrate/Nitrite (mg/L)		BOD (mg/L)		Anionic Surf. as MBAS (mg/L)	
Commercial-Canyon Road																					
11/8/2010	11/5/2010 18:15	2010110188	26	10	0.010		0.04	J	9.6		19	2.0		0.5	U	0.157		3.5		0.06	
11/9/2010	11/9/2010 8:00	2010110202																			
11/10/2010	11/9/2010 8:00	2010110247	34	25	0.030	J	0.13		10.3		46	2.0		0.5		1.020		2.7		0.08	
11/17/2010	11/17/2010 6:00	2010110415																			
11/18/2010	11/17/2010 6:00	2010110454	36	22	0.010	U	0.12		8.0		64	2.5		0.5	U	0.095		4.0		0.10	
12/13/2010	12/11/2011 10:45	2010120305	38	12	0.020		0.04		7.8		30	5.0		0.5	U	0.088		2.0	U	0.08	
1/12/2011	1/12/2011 4:45	2011010200																			
1/13/2011	1/12/2011 4:45	2011010232	87	55	0.048		0.20		10.8		210	39		0.7		0.134	J	2.5		0.10	
2/28/2011	2/27/2011 17:00	2011020558	270	61	0.044		0.12		10.8		90	63	J	0.5	U	0.085	J	3.4		0.08	
3/21/2011	3/18/2011 13:00	2011030418	46	22	0.026		0.05		10.0	U	44	5.8		0.8		0.118	J	4.0		0.08	
4/21/2011	NA	2011040452					0.02				1.6										
7/26/2011	7/25/2011 8:15	2011040453	85	75	0.055		0.51		28.0		200	7.50		1.1		0.740			R	0.54	J
Median			42.000	23.500	0.028		0.120		10.150		46.000	5.400		0.500		0.126		3.400		0.080	
Mean			77.750	35.250	0.030		0.137		11.913		78.289	15.850		0.638		0.305		3.157		0.140	
Min			26.000	10.000	0.010		0.020		7.800		1.600	2.000		0.500		0.085		2.000		0.060	
Max			270.000	75.000	0.055		0.510		28.000		210.000	63.000		1.100		1.020		4.000		0.540	
Low Density Residential-Lake Sylvia																					
10/28/2010	NA	2010100682																			
11/8/2010	11/5/2010 16:00:00 PM	2010110186	100	3.6	0.020		0.02	J	34.5		2.8	8.5		0.6		0.213		3.500		0.04	
11/9/2010	11/9/2010 3:45	2010110202																			
11/17/2010	11/17/2010 5:30	2010110413																			
11/30/2010	11/29/2010 17:00	2010110580																			
2/14/2011	2/12/2011 14:45	2011020333	96	12.0	0.021		0.04		16.7		18	7.5		0.5	U	0.350		2.10		0.09	
2/28/2011	2/27/2011 10:00	2011020560	95	32.0	0.035		0.08		33.2		61	9.5		0.5	U	0.418		2.00	U	0.04	
3/9/2011	3/8/2011 3:00	2011030203	96	4.7	0.041		0.05		33.8		8.5	7.5	J	0.5	U	0.302		2.00	UJ	0.15	
4/21/2011	NA	2011040452					0.02				4.4										
6/10/2011	NA	2011060232					0.01	U			6										
7/27/2011	NA	2011070474					0.02				7										
Median			96	8.4	0.028		0.02		33.5		7	8.0		0.5		0.326		2.05		0.065	
Mean			97	13.1	0.029		0.03		29.6		15	8.3		0.5		0.321		2.40		0.08	
Min			95	3.6	0.020		0.01		16.7		2.8	7.5		0.5		0.213		2.00		0.04	
Max			100	32.0	0.041		0.08		34.5		61	9.5		0.6		0.418		3.50		0.15	
High Density Residential-Collinswood																					
11/9/2010	11/9/2010 8:00	2010110202																			
11/10/2010	11/9/2010 8:00	2010110245	130	3.4	0.030	J	0.05		38.3		2.4	5.5		0.5	U	0.657		3.300		0.08	
11/17/2010	11/17/2010 6:00	2010110414																			
1/12/2011	1/12/2011 5:30	2011010200																			
2/10/2011		2011020241																			
3/21/2011	3/18/2011 13:00	2011030419	99	6.1	0.016		0.05		18.1		20	7.0		0.5		0.300		2.00	U	0.11	
4/11/2011	4/10/2011 3:00	2011040209	130	1.6	0.011		0.02		33.1		3.6	8.0	J	0.5	U	0.441		2.50		0.09	
4/21/2011	NA	2011040452					0.03				8.8										
4/26/2011	4/25/2011 14:15	2011040526	118	5.7	0.031		0.05		44.2		18	6.0		0.5	U	0.265		2.00	U	0.13	
5/5/2011	5/2/2011 8:45	2011050136	148	1.4	0.009		0.01		51.4		1.2	8.0		0.5	U	0.451		2.00	U	0.08	
5/27/2011	5/25/2011 10:45	2011050624	180	2.1	0.012		0.04		65.9		2.6	10.0		0.5	UJ	0.848		2.00	U	0.1	
6/10/2011	NA	2011060232					0.01	U			4.8										
Median			130	2.8	0.014		0.04		41.3		3.6	7.5		0.50		0.446		2.00		0.10	
Mean			134	3.4	0.018		0.04		41.8		8	7.4		0.50		0.494		2.30		0.10	
Min			99	1.4	0.009		0.01		18.1		1.2	5.5		0.50		0.265		2.00		0.08	
Max			180	6.1	0.031		0.05		65.9		20	10.0		0.50		0.848		3.30		0.13	

Pierce County NPDES Outfall Monitoring data for Commercial, Low Density Residential and High Density Residential stations.

			Herbicides						Pesticides																	
Collection Date	Storm Start Date	Spectra File #	2,4-D (µg/L)		MCPP (µg/L)		Triclopyr (µg/L)		Chlorpyrifos (µg/L)		Diazinon (µg/L)		Dichlobenil (µg/L)		Malathion (µg/L)		Prometon (µg/L)		Diesel (mg/L)		Oil (mg/L)		Gasoline (mg/L)		Fecal Coliform by MPN in Water (MPN/100ml)	
Commercial-Canyon Road																										
11/8/2010	11/5/2010 18:15	2010110188	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.041		0.024	U	0.024	U								
11/9/2010	11/9/2010 8:00	2010110202																	0.10	U	0.50	U	0.05	U	17	
11/10/2010	11/9/2010 8:00	2010110247	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.049		0.024	U	0.024	U								
11/17/2010	11/17/2010 6:00	2010110415																	0.11	UJ	2.05	J	0.05	U	1300	
11/18/2010	11/17/2010 6:00	2010110454	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.057		0.024	U	0.024	U								
12/13/2010	12/11/2011 10:45	2010120305	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.024	U	0.024	U	0.024	U								
1/12/2011	1/12/2011 4:45	2011010200																	0.10	U	3.98		0.05	U	900	
1/13/2011	1/12/2011 4:45	2011010232	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.024	U	0.024	U	0.024	U								
2/28/2011	2/27/2011 17:00	2011020558	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.670		0.024	U	0.024	U								
3/21/2011	3/18/2011 13:00	2011030418	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.220		0.024	U	0.024	U								
4/21/2011	NA	2011040452	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.024	U	0.024	U	0.024	U								
7/26/2011	7/25/2011 8:15	2011040453	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.066		0.024	U	0.024	U								
Median			0.080		0.080		0.080		0.024		0.024		0.049		0.024		0.024		0.100		2.050		0.050		900.000	
Mean			0.080		0.080		0.080		0.024		0.024		0.131		0.024		0.024		0.103		2.177		0.050		739.000	
Min			0.080		0.080		0.080		0.024		0.024		0.024		0.024		0.024		0.100		0.500		0.050		17.000	
Max			0.080		0.080		0.080		0.024		0.024		0.670		0.024		0.024		0.110		3.980		0.050		1300.000	
Low Density Residential-Lake Sylvia																										
10/28/2010	NA	2010100682																	0.1	U	0.5	U	0.05	U	60	
11/8/2010	11/5/2010 16:00:00 PM	2010110186	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.024	U	0.024	U	0.024	U								
11/9/2010	11/9/2010 3:45	2010110202																	0.10	U	0.50	U	0.05	U	2	U
11/17/2010	11/17/2010 5:30	2010110413																	0.10	U	0.50	U	0.05	U	27	
11/30/2010	11/29/2010 17:00	2010110580																	0.10	U	0.50	U	0.05	U	130	
2/14/2011	2/12/2011 14:45	2011020333	0.080	U	0.080	U	0.080	U	1.000	U	1.000	U	1.000	U	0.024	U	0.024	U								
2/28/2011	2/27/2011 10:00	2011020560	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.037		0.024	U	0.024	U								
3/9/2011	3/8/2011 3:00	2011030203	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.024	U	0.024	U	0.024	U								
4/21/2011	NA	2011040452	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.024	U	0.024	U	0.024	U								
6/10/2011	NA	2011060232	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.024	U	0.024	U	0.024	U								
7/27/2011	NA	2011070474	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.024	U	0.024	U	0.024	U								
Median			0.080		0.080		0.080		0.024		0.024		0.024		0.024		0.024		0.1		0.5		0.05		43.5	
Mean			0.080		0.080		0.080		0.163		0.163		0.165		0.024		0.024		0.1		0.5		0.05		54.75	
Min			0.080		0.080		0.080		0.024		0.024		0.024		0.024		0.024		0.1		0.5		0.05		2	
Max			0.080		0.080		0.080		1.000		1.000		1.000		0.024		0.024		0.1		0.5		0.05		130	
High Density Residential-Collinswood																										
11/9/2010	11/9/2010 8:00	2010110202																	0.10	U	2.54		0.05	U	2	U
11/10/2010	11/9/2010 8:00	2010110245	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.170		0.024	U	0.024	U								
11/17/2010	11/17/2010 6:00	2010110414																	0.10	U	0.50	U	0.05	U	300	
1/12/2011	1/12/2011 5:30	2011010200																	0.10	U	0.50	U	0.05	U	50	
2/10/2011		2011020241																	0.10	U	0.50	U	0.05	U	16	
3/21/2011	3/18/2011 13:00	2011030419	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.051		0.024	U	0.024	U								
4/11/2011	4/10/2011 3:00	2011040209	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.024	U	0.080	U	0.024	U								
4/21/2011	NA	2011040452	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.024	U	0.024	U	0.024	U								
4/26/2011	4/25/2011 14:15	2011040526	0.300		0.110		0.080	U	0.024	U	0.024	U	0.067		0.024	U	0.024	U								
5/5/2011	5/2/2011 8:45	2011050136	0.080	U	0.080	U	0.080	U	0.024	U	0.024	U	0.040		0.024	U	0.024	U								
5/27/2011	5/25/2011 10:45	2011050624	0.150		0.110		0.080	U	0.024	U	0.024	U	0.041		0.024	U	0.024	U								
6/10/2011	NA	2011060232	0.080	U	0.080	U	0.080	U	0.026	U	0.026	U	0.026	U	0.026	U	0.026	U								
Median			0.080		0.080		0.080		0.024		0.024		0.041		0.024		0.024		0.1		0.5		0.05		33	
Mean			0.116		0.088		0.080		0.024		0.024		0.055		0.031		0.024		0.1		1.01		0.05		92	
Min			0.080		0.080		0.080		0.024		0.024		0.024		0.024		0.024		0.1		0.5		0.05		2	
Max			0.300		0.110		0.080		0.026		0.026		0.170		0.080		0.026		0.1		2.54		0.05		300	

Appendix C

Sediment Data

Pierce County NPDES S8.D Monitoring Sediment Data.

Collection Date	Spectra File #	Total organic carbon (%)		Total Solids (%)		Total Cadmium (mg/kg)		Total Copper (mg/kg)		Total Lead (mg/kg)		Total Zinc (mg/kg)		Total Mercury (mg/kg)		TPH-Diesel (mg/kg)		TPH-Oil (mg/kg)	
Commercial-Canyon Road																			
10/6/2011	2011100208	46.8		28.3		4.9		452		523		2460		0.05	U	NA		NA	
Low Density Residential-Lake Sylvia																			
10/26/2011	2011100663	12.2		63.6		0.1		2.4		2.3		18.7		0.05	U	NA		NA	
High Density Residential-Collinswood																			
9/22/2011	2011090462	34.2		NA		0.1	U	20.0		10.0		128		0.05	U	NA		NA	

Pierce County NPDES S8.D Monitoring Sediment

		Polycyclic Aromatic Hydrocarbons and Pentachlorophenol																																					
Collection Date	Spectra File #	1-Methylnaphthalene--SIM (µg/kg)		2-Methylnaphthalene--SIM (µg/kg)		Acenaphthene--SIM (µg/kg)		Acenaphthylene--SIM (µg/kg)		Anthracene--SIM (µg/kg)		Benzo(a)Anthracene--SIM (µg/kg)		Benzo(a)Pyrene--SIM (µg/kg)		Benzo(b)Fluoranthene--SIM (µg/kg)		Benzo(ghi)Perylene--SIM (µg/kg)		Benzo(k)Fluoranthene--SIM (µg/kg)		Chrysene--SIM (µg/kg)		Dibenz(a,h)Anthracene--SIM (µg/kg)		Dibenzofuran--SIM (µg/kg)		Fluoranthene--SIM (µg/kg)		Fluorene--SIM (µg/kg)		Indeno(1,2,3-cd)Pyrene--SIM (µg/kg)		Naphthalene--SIM (µg/kg)		Pentachlorophenol--SIM (µg/kg)		Phenanthrene--SIM (µg/kg)		Pyrene--SIM (µg/kg)	
Commercial-Canyon Road																																							
10/6/2011	2011100208	11.1		16.2		17.3		10	U	31.0		108		134		143		152		131		278		27.4		NA		362		31.8		111		21.0		183		163		728	
Low Density Residential-Lake Sylvia																																							
10/26/2011	2011100663	1.00	U	1.00	U	1.00	U	1.00	U	1.00	U	1.00	U	1.00	U	1.07		1.00	U	1.00	U	1.00	U	1.00	U	1.00	U	1.02		1.00	U	1.00	U	1.56		1.00	U	2.16		1.37	
High Density Residential-Collinswood																																							
9/22/2011	2011090462	NA		NA		NA		NA		NA		NA		NA		NA		NA		NA		NA		NA		NA		NA		NA		NA		NA		NA		NA		NA	

Pierce County NPDES S8.D Monitoring Sediment Data.

		Pesticides						Phthalates - BNA																					
Collection Date	Spectra File #	2,4-D (µg/kg)		MCPP (µg/kg)		Triclopyr (µg/kg)		bis(2-Ethylhexyl)Phthalate (µg/kg)		Butylbenzyl phthalate (µg/kg)		Diethylphthalate (µg/kg)		Dimethyl Phthalate (µg/kg)		Di-n-Butylphthalate (µg/kg)		Di-n-Octyl Phthalate (µg/kg)		Chlorpyrifos (mg/kg)		Diazinon (mg/kg)		Dichlobenil (mg/kg)		Malathion (mg/kg)		Prometon (mg/kg)	
Commercial-Canyon Road																													
10/6/2011	2011100208	0.010	U	0.010	U	0.010	U	8220		740		100	U	100	U	100	U	100	U	0.0067	U	0.0067	U	0.026		0.0067	U	0.0067	U
Low Density Residential-Lake Sylvia																													
10/26/2011	2011100663	0.010	U	0.010	U	0.010	U	20	U	20	U	20	U	20	U	20	U	20	U	0.0067	U	0.0067	U	0.0067	U	0.0067	U	0.0067	U
High Density Residential-Collinswood																													
9/22/2011	2011090462	NA		NA		NA		NA		NA		NA		NA		NA		NA		NA		NA		NA		NA		NA	

Pierce County NPDES S8.D Monitoring Sediment Data.

		Phenolics - BNA																											
Collection Date	Spectra File #	2,4,5-Trichlorophenol (µg/kg)		2,4,6-Trichlorophenol (µg/kg)		2,4-Dichlorophenol (µg/kg)		2,4-Dimethylphenol (µg/kg)		2,4-Dinitrophenol (µg/kg)		2-Chlorophenol (µg/kg)		2-Methylphenol (µg/kg)		2-Nitrophenol (µg/kg)		4,6-Dinitro-2-methylphenol (µg/kg)		4-Chloro-3-methylphenol (µg/kg)		4-Methylphenol (µg/kg)		4-Nitrophenol (µg/kg)		Phenol (µg/kg)	
Commercial-Canyon Road																													
10/6/2011	2011100208	10	U	10	U	10	U	17.4		10	U	10	U	40.2		10	U	10	U	10	U	NA	U	10	U	92.8			
Low Density Residential-Lake Sylvia																													
10/26/2011	2011100663	1.00	U	1.00	U	1.00	U	1.00	U	1.00	U	1.00	U	1.00	U	1.00	U	1.00	U	1.00	U	31.3		4.64		4.52			
High Density Residential-Collinswood																													
9/22/2011	2011090462	NA		NA		NA		NA		NA		NA		NA		NA		NA		NA		NA		NA		NA			

Pierce County NPDES S8.D Monitoring Sediment Data.

				Grain Size - Retained in each size fraction (percent)										
Collection Date	**Spectra File #**	**Total PCB (mg/kg)**		**Coarse Gravel (0.75 - 3")**	**Gravel (4750 microns - 0.75")**	**Coarse Sand (2000-4750 microns)**	**Medium Sand (425-2000 microns)**	**Fine Sand (75-425 microns)**	**Very Coarse Silt (32-75 microns)**	**Coarse Silt (22-32 microns)**	**Medium Silt (13-22 microns)**	**Fine Silt (7-13 microns)**	**Very Fine Silt (3.2-7 microns)**	**Clay (<3.2 microns)**
Commercial-Canyon Road														
10/6/2011	2011100208	0.03		0.0	0.3	1.4	13.2	53.5	12.0	4.9	3.4	6.4	2.0	3.0
Low Density Residential-Lake Sylvia														
10/26/2011	2011100663	0.01	U	0.0	0.0	0.2	23.4	70.0	0.4	1.4	0.5	1.0	2.3	0.9
High Density Residential-Collinswood														
9/22/2011	2011090462	NA		NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

Appendix D

First Flush Toxicity Report

Nautilus Environmental, LLC

NPDES Sublethal Toxicity Testing: Pierce County

August 2011

Report date: September 9, 2011

Submitted to:

Pierce County Public Works and Utilities
Surface Water Mgt.
2702 South 42nd Street, Suite 201
Tacoma, WA 98409-7322

Washington Laboratory
5009 Pacific Hwy East
Suite 2
Tacoma, WA 98424

1.0 INTRODUCTION

Rainbow trout embryo development tests were performed on grab samples collected from Pierce County during a first flush storm event, as part of the Phase I Municipal Stormwater Permit. Two samples were collected from the following stations on August 23, 2011: COMM FIRST FLUSH (COMM) and LORES FIRST FLUSH (LORES). Testing was conducted at the Nautilus Environmental Washington Laboratory located in Tacoma, Washington.

2.0 METHODS

2.1 Sample Collection and Transport

Composite samples of stormwater effluent were collected in low-density polyethylene (LDPE) cubitainers by Pierce County personnel. Samples were packed in coolers containing ice and transported to Nautilus Environmental by Pierce County personnel. Appropriate chain-of-custody (COC) procedures were employed during collection and transport.

2.2 Sample Receipt

Upon arrival at the laboratory, the coolers were opened, samples inspected and the contents verified against information provided on the COC form. Receipt temperatures were measured in the samples and recorded on the COC. Standard water quality parameters were measured and recorded on a sample check-in sheet (Appendix E). The samples were stored at 4°C in the dark until used for testing.

2.3 Test Methods

Toxicity tests were conducted using rainbow trout gametes according to procedures presented by Environment Canada (1998) with modifications from Canaria et al. (1999). This test involved a seven-day exposure of fertilized eggs to the test solutions. Test methods are summarized in Table 1.

Under WDOE guidelines there were two separate endpoints evaluated: survival and development. Survival is defined as the number of survivors (live eggs) divided by the start count. Development is defined as the number of normal embryos divided by the number of survivors. Both endpoints are presented in the results section.

Table 1. Summary of procedures for the 7-day rainbow trout embryo development test.

Test initiation date and time	COMM and LORES: 8/24/2011; 1400h
Test termination date and time	COMM and LORES: 8/31/2011; 1315h
Test organism	Rainbow trout (*Oncorhynchus mykiss*)
Test organism source	Troutlodge; Sumner, WA
Test organism age	Within 30 minutes of fertilization
Test duration	7 days
Feeding	None
Test chamber	1-liter plastic cup
Test solution volume	500 mL
Test temperature	14 ± 1°C
Dilution water	Moderately Hard Synthetic Water
Test concentrations (% sample)	100, 50, 25, 12.5, 6.25, control
Number of organisms/chamber	30
Number of replicates	4
Photoperiod	24-hr dark; low intensity light used during solution renewals
Aeration	Continuous gentle aeration
Test protocol	Environment Canada (1998); modifications Canaria et al. (1999)
Test acceptability criterion for controls	≥ 70% survivors, ≥ 70% of which are normally developed
Reference toxicant	SDS

3.0 RESULTS

Details of water quality parameters measured in the stormwater samples upon receipt are provided in Table 2.

Table 2. Sample information.

Sample ID	LORES FIRST FLUSH	COMM FIRST FLUSH
Nautilus Log-in No.	11-237	11-238
Collection date and time	8/23/2011; 1120h	8/23/2011; 1255h
Receipt date and time	8/23/2011; 1410h	8/23/2011; 1410h
Receipt temp (°C)	8.7	8.2
Dissolved oxygen (mg/L)	11.0	9.2
pH	7.06	6.92
Conductivity (µS/cm)	104	86
Hardness (mg/L $CaCO_3$)	56	44
Alkalinity (mg/L $CaCO_3$)	48	32
Total chlorine (mg/L)	<0.03	<0.03
Total ammonia (mg/L)	<1.0	<1.0

Results of the toxicity tests are summarized in Table 3. No significant toxicity was observed in the tests with the samples collected from LORES and COMM. The no observed effects concentration (NOEC) was 100 percent sample for the LORES sample for either endpoint tested. There was a slight statistical significance in the COMM 100 percent sample for the development endpoint, however, when alpha was lowered to 0.01 for the small difference in response, this significance was removed. The 50 percent effective concentration (EC_{50}) was greater than 100 percent for both samples.

Statistical summaries for the tests, copies of the laboratory bench sheets, reference toxicant test results, control QC plots, sample check-in sheet, and chain-of-custody form, are provided in Appendices A through F.

Table 3. Results of rainbow trout embryo toxicity tests conducted with stormwater samples from Pierce County.

	Sample ID	LORES	COMM
Survival	NOEC[1] (%)	100	100
	LOEC[2] (%)	>100	>100
	EC_{50}[3] (%)	>100	>100
	EC_{25}[4] (%)	>100	>100
	Mean control	100	100
	(95% C.I.)	(100-100)	(100-100)
	Mean 100% sample	97.5	99.2
	(95% C.I.)	(96.3-98.7)	(98.5-99.8)
Development	NOEC[1] (%)	100	50[a], 100[b]
	LOEC[2] (%)	>100	100[a], >100[b]
	EC_{50}[3] (%)	>100	>100
	EC_{25}[4] (%)	>100	>100
	Mean control	92.5	97.5
	(95% C.I.)	(89.7-95.3)	(96.3-98.7)
	Mean 100% sample	94.1	84.1
	(95% C.I.)	(92.5-95.6)	(81.3-86.8)

[a]Alpha = 0.05, [b]Alpha = 0.01, [1]No Observed Effect Concentration; [2]Lowest Observed Effect Concentration, [3]Effective Concentration for 50% of test organisms; [4]Effective Concentration for 25% of test organisms

4.0 QA/QC

Samples were received in good condition and within the temperature range specified by WDOE (2008). The toxicity tests met all acceptability criteria for performance of control organisms.

There were no deviations from the protocols and all water quality parameters remained within the ranges specified in the corresponding test methods throughout the tests.

Results for the reference toxicant test conducted concurrently with the samples are summarized in Table 4 and provided in Appendix D. The results for the reference toxicant test fell within the acceptable range of mean ± two standard deviations of historical test results. The coefficient of variation (CV) for the test is also shown in the table. Dilution water control mean and control CV for development for all tests were also within two standard deviations of the historical mean (Appendix C). Based on the reference toxicant and control results, test organisms appeared to be of an appropriate degree of sensitivity.

Table 4. Reference toxicant test results.

Date initiated	Endpoint	EC_{50} (mg/L SDS)	Acceptable Range	CV (%)
8/24/2011	Combined Development	1.8	0.46 – 2.67	35.2

5.0 REFERENCES

Canaria, E.C., Elphick, J.R. and Bailey, H.C. 1999. A simplified procedure for conducting small scale short-term embryo toxicity tests with salmonids. Environ. Toxicol. 14:301-307.

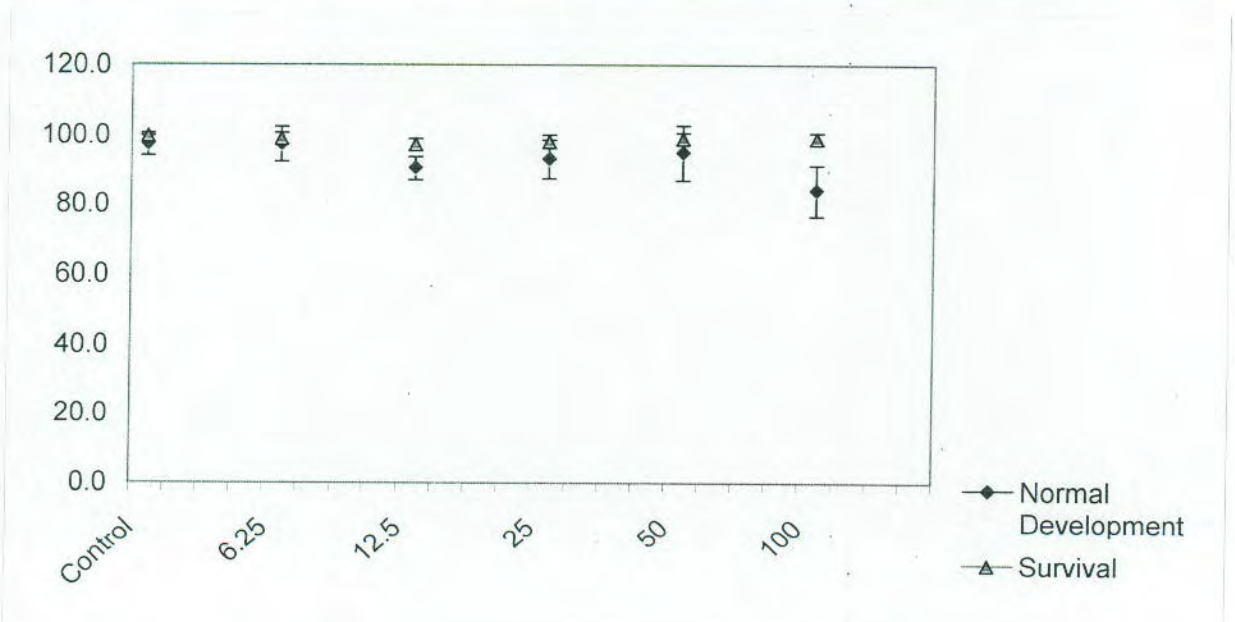
Environment Canada. 1998. Biological Test Method: Toxicity Tests Using Early Life Stages of Salmonid Fish (Rainbow Trout). Second Edition. EPS/1/RM/28, July 1998.

WDOE. 2008. Laboratory Guidance and Whole Effluent Toxicity Test Review Criteria. Washington State Department of Ecology. Water Quality Program. Publication number: WQ-R-95-80, Revised December 2008.

Appendix A
Summaries of Test Data

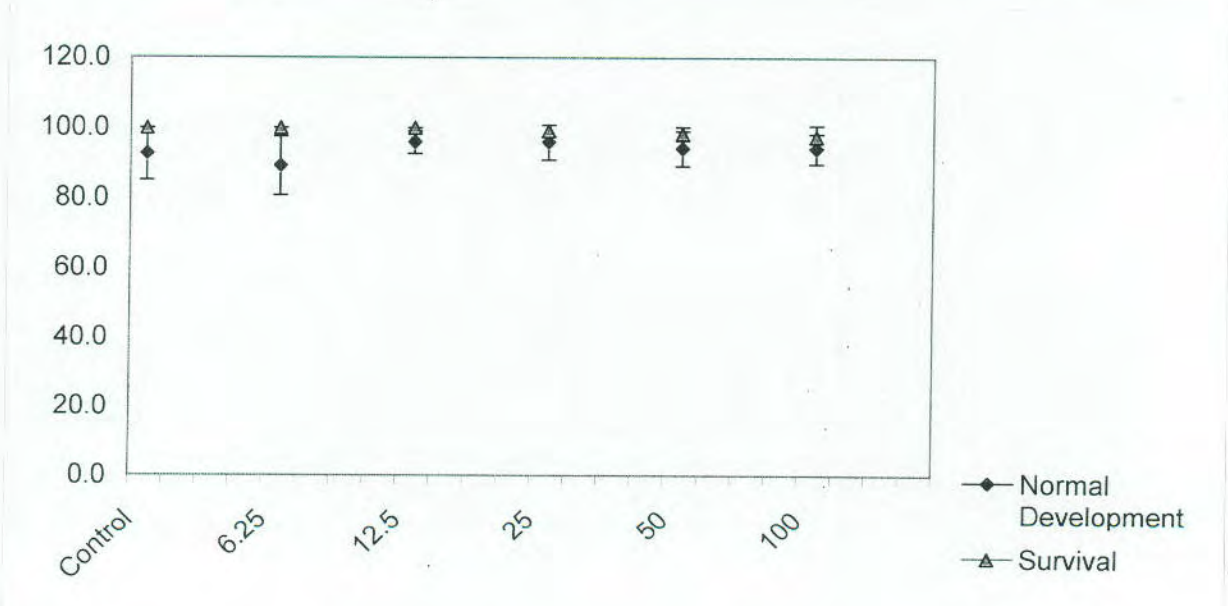
Pierce County
Municioal Stormwater Sublethal Toxicity Tests
Sample ID: COMM FIRST FLUSH
August 24, 2011

Concentration %	Rep	Initial Egg count	Survival # Survived	Survival % Survived	Survival Mean % Survived	Survival St. Dev. % Survival	Normal Development # Normal	Normal Development % Normal	Normal Development Mean % Normal	Normal Development St. Dev. % Normal
Control	1	30	30	100	100.0	0.0	29	97	97.5	3.2
	2	30	30	100			30	100		
	3	30	30	100			30	100		
	4	30	30	100			28	93		
6.25	1	30	30	100	99.2	1.7	30	100	97.5	5.0
	2	30	30	100			27	90		
	3	30	30	100			30	100		
	4	30	29	97			29	100		
12.5	1	30	29	97	97.5	1.7	26	90	90.6	3.4
	2	30	29	97			25	86		
	3	30	30	100			28	93		
	4	30	29	97			27	93		
25	1	30	30	100	98.3	1.9	28	93	93.2	5.6
	2	30	29	97			27	93		
	3	30	30	100			30	100		
	4	30	29	97			25	86		
50	1	30	29	97	99.2	1.7	28	97	95.0	7.9
	2	30	30	100			25	83		
	3	30	30	100			30	100		
	4	30	30	100			30	100		
100	1	30	30	100	99.2	1.7	28	93	84.1	7.3
	2	30	30	100			23	77		
	3	30	30	100			24	80		
	4	30	29	97			25	86		



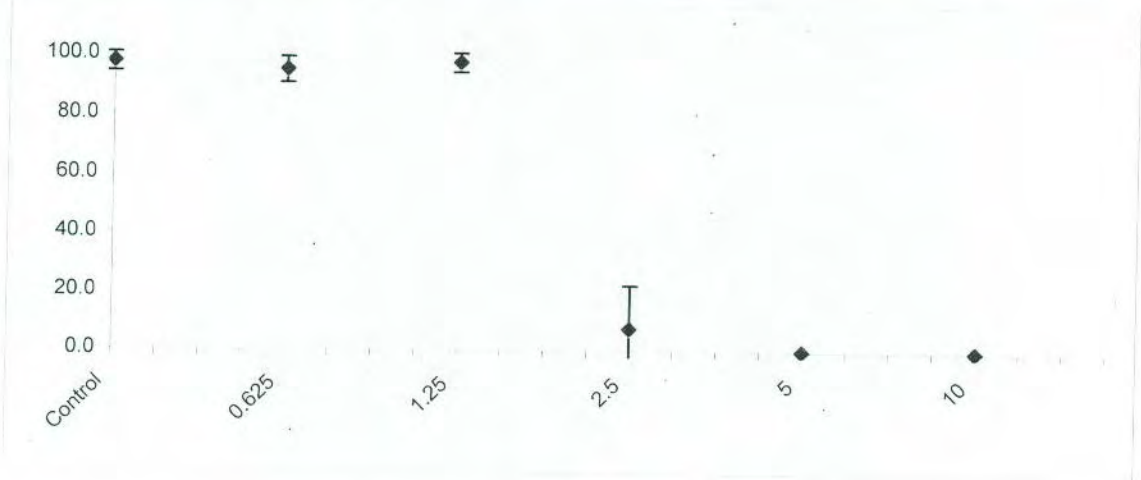
Pierce County
Municioal Stormwater Sublethal Toxicity Tests
Sample ID: LORES FIRST FLUSH
August 24, 2011

Concentration %	Rep	Initial Egg count	Survival # Survived	Survival % Survived	Survival Mean % Survived	Survival St. Dev. % Survival	Normal Development # Normal	Normal Development % Normal	Normal Development Mean % Normal	Normal Development St. Dev. % Normal
Control	1	30	30	100	100.0	0.0	25	83	92.5	7.4
	2	30	30	100			30	100		
	3	30	30	100			29	97		
	4	30	30	100			27	90		
6.25	1	30	30	100	100.0	0.0	24	80	89.2	8.3
	2	30	30	100			30	100		
	3	30	30	100			26	87		
	4	30	30	100			27	90		
12.5	1	30	30	100	100.0	0.0	29	97	95.8	3.2
	2	30	30	100			30	100		
	3	30	30	100			28	93		
	4	30	30	100			28	93		
25	1	30	30	100	99.2	1.7	30	100	95.8	5.0
	2	30	29	97			29	100		
	3	30	30	100			28	93		
	4	30	30	100			27	90		
50	1	30	29	97	98.3	1.9	28	97	94.1	5.0
	2	30	29	97			28	97		
	3	30	30	100			26	87		
	4	30	30	100			29	97		
100	1	30	29	97	97.5	3.2	29	100	94.0	4.2
	2	30	28	93			26	93		
	3	30	30	100			28	93		
	4	30	30	100			27	90		



Reference Toxicant Test
Sample ID: 10 mg/L SDS
August 24, 2011

Concentration mg/L SDS	Replicate	Initial Egg count	# Survived	# Normal	% Combined Development (CD)	Mean % CD	St. Dev. % CD
Control	1	30	28	28	93	97.5	3
	2	30	30	29	97		
	3	30	30	30	100		
	4	30	30	30	100		
0.625	1	30	30	29	97	95.0	4
	2	30	29	28	93		
	3	30	30	27	90		
	4	30	30	30	100		
1.25	1	30	30	30	100	97.5	3
	2	30	29	29	97		
	3	30	30	30	100		
	4	30	29	28	93		
2.5	1	30	1	0	0	7.5	15
	2	30	0	0	0		
	3	30	10	9	30		
	4	30	1	0	0		
5	1	30	0	0	0	0	0
	2	30	0	0	0		
	3	30	0	0	0		
	4	30	0	0	0		
10	1	30	0	0	0	0	0
	2	30	0	0	0		
	3	30	0	0	0		
	4	30	0	0	0		



Appendix B
Rainbow Trout Embryo Development
Raw Data and Statistical Summaries

LORES

CETIS Summary Report

Salmonid Embryo Survival and Development Test — **Nautilus Environmental WA**

Batch ID:	19-1373-2074	Test Type:	Survival-Development	Analyst:	Cat Curran
Start Date:	24 Aug-11 14:00	Protocol:	EC/EPS 1/RM/28	Diluent:	Mod-Hard Synthetic Water
Ending Date:	31 Aug-11 13:15	Species:	Oncorhynchus mykiss	Brine:	
Duration:	6d 23h	Source:	Trout Lodge Fish Farm	Age:	

Sample ID:	18-0839-0980	Code:	11-237	Client:	Pierce County - SW
Sample Date:	23 Aug-11 11:20	Material:	Stormwater	Project:	
Receive Date:	23 Aug-11 14:10	Source:	Pierce County Municipal Stormwater (WAR		
Sample Age:	27h (8.7 °C)	Station:	LORES		

Comparison Summary

Analysis ID	Endpoint	NOEL	LOEL	TOEL	PMSD	TU	Method
17-8883-3720	Development Rate	100	>100	N/A	13.0%	1	Dunnett's Multiple Comparison Test
13-5797-6267	Survival Rate	100	>100	N/A	2.75%	1	Steel Many-One Rank Test

Point Estimate Summary

Analysis ID	Endpoint	Level	%	95% LCL	95% UCL	TU	Method
15-7403-8753	Development Rate	EC25	>100	N/A	N/A	<1	Linear Interpolation (ICPIN)
		EC50	>100	N/A	N/A	<1	
05-4744-5736	Survival Rate	EC25	>100	N/A	N/A	<1	Linear Interpolation (ICPIN)
		EC50	>100	N/A	N/A	<1	

Development Rate Summary

Conc-%	Control Type	Count	Mean	95% LCL	95% UCL	Min	Max	Std Err	Std Dev	CV%	%Effect
0	Dilution Water	4	0.925	0.8974	0.9526	0.8333	1	0.03696	0.07391	7.99%	0.0%
6.25		4	0.8917	0.8605	0.9228	0.8	1	0.04167	0.08333	9.35%	3.6%
12.5		4	0.9583	0.9464	0.9703	0.9333	1	0.01596	0.03191	3.33%	-3.6%
25		4	0.9583	0.9397	0.977	0.9	1	0.025	0.05	5.22%	-3.6%
50		4	0.9411	0.9226	0.9596	0.8667	0.9667	0.02481	0.04962	5.27%	-1.74%
100		4	0.9405	0.9247	0.9563	0.9	1	0.02116	0.04232	4.5%	-1.67%

Survival Rate Summary

Conc-%	Control Type	Count	Mean	95% LCL	95% UCL	Min	Max	Std Err	Std Dev	CV%	%Effect
0	Dilution Water	4	1	1	1	1	1	0	0	0.0%	0.0%
6.25		4	1	1	1	1	1	0	0	0.0%	0.0%
12.5		4	1	1	1	1	1	0	0	0.0%	0.0%
25		4	0.9917	0.9854	0.9979	0.9667	1	0.008333	0.01667	1.68%	0.83%
50		4	0.9833	0.9761	0.9905	0.9667	1	0.009623	0.01925	1.96%	1.67%
100		4	0.975	0.9631	0.9869	0.9333	1	0.01596	0.03191	3.27%	2.5%

Development Rate Detail

Conc-%	Control Type	Rep 1	Rep 2	Rep 3	Rep 4
0	Dilution Water	0.8333	1	0.9667	0.9
6.25		0.8	1	0.8667	0.9
12.5		0.9667	1	0.9333	0.9333
25		1	1	0.9333	0.9
50		0.9655	0.9655	0.8667	0.9667
100		1	0.9286	0.9333	0.9

Survival Rate Detail

Conc-%	Control Type	Rep 1	Rep 2	Rep 3	Rep 4
0	Dilution Water	1	1	1	1
6.25		1	1	1	1
12.5		1	1	1	1
25		1	0.9667	1	1
50		0.9667	0.9667	1	1
100		0.9667	0.9333	1	1

Nautilus Environmental
Washington Laboratory
Client: Pierce County
Sample ID. LORES FIRST FLUSH
Test No: 1108-T034
Log-In#: 11-237

Initial and Final Chemistries
Seven Day Chronic Freshwater Bioassay
Start Date & Time: 8/24/11 1400
Stop Date & Time: 8/31/11 1315
Test Species: Oncorhynchus mykiss

Conc. or % CON	Days	0		1		2		3		4		5		6	
		init.	final	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final
	pH	7.13	7.31	7.19	7.21	7.41	7.47	7.42	7.27	7.31	7.40	7.34	7.31	7.40	7.35
	DO (mg/l)	10.1	10.5	10.5	10.1	10.1	10.4	10.4	10.4	10.4	10.1	10.3	10.4	9.6	9.5
	Cond. (µmhos-cm)	217	225	221	224	227	224	221	225	231	228	229	226	252	247
	Temperature (°C)	14.0	13.9	13.9	14.8	14.9	14.3	13.8	14.0	14.9	14.6	14.3	14.2	14.5	14.3
6.25	Days	0		1		2		3		4		5		6	
		init.	final	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final
	pH	7.17	7.32	7.21	7.25	7.40	7.53	7.46	7.32	7.34	7.40	7.39	7.32	7.57	7.36
	DO (mg/l)	10.6	10.5	10.5	10.2	10.1	10.5	10.5	10.4	10.5	10.2	10.3	10.2	9.8	9.7
	Cond. (µmhos-cm)	211	220	214	214	218	213	218	220	225	219	220	217	248	236
	Temperature (°C)	14.2	13.8	13.8	14.3	14.8	14.0	13.9	14.2	14.9	14.5	14.2	14.1	14.7	14.3
12.5	Days	0		1		2		3		4		5		6	
		init.	final	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final
	pH	7.19	7.37	7.24	7.30	7.39	7.54	7.47	7.35	7.34	7.39	7.41	7.33	7.56	7.36
	DO (mg/l)	10.5	10.5	10.5	10.3	10.1	10.4	10.5	10.5	10.5	9.9	10.2	10.1	10.0	9.8
	Cond. (µmhos-cm)	204	212	208	210	213	209	210	214	218	213	214	223	240	230
	Temperature (°C)	14.1	13.8	13.9	14.3	14.8	14.2	14.1	14.3	14.9	14.2	14.3	14.1	14.6	14.2
25	Days	0		1		2		3		4		5		6	
		init.	final	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final
	pH	7.20	7.35	7.28	7.31	7.37	7.54	7.46	7.37	7.34	7.37	7.41	7.40	7.53	7.36
	DO (mg/l)	10.6	10.3	10.5	10.0	9.9	10.5	10.5	10.4	10.5	10.0	10.2	10.1	10.1	9.8
	Cond. (µmhos-cm)	190	199	174	195	201	193	197	201	205	199	200	198	223	215
	Temperature (°C)	14.2	13.9	14.0	14.4	14.7	14.2	14.2	14.3	14.8	14.2	14.3	14.1	14.7	14.1
50	Days	0		1		2		3		4		5		6	
		init.	final	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final
	pH	7.19	7.35	7.25	7.32	7.32	7.52	7.42	7.38	7.32	7.39	7.39	7.38	7.47	7.35
	DO (mg/l)	10.7	10.4	10.5	10.2	9.9	10.5	10.5	10.5	10.5	10.2	10.1	10.1	10.3	9.8
	Cond. (µmhos-cm)	164	176	170	170	176	169	170	171	176	171	173	170	189	183
	Temperature (°C)	14.2	14.0	14.0	14.3	14.8	14.3	14.5	14.5	14.8	14.2	14.6	14.1	14.8	14.0
100	Days	0		1		2		3		4		5		6	
		init.	final	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final
	pH	7.09	7.30	7.23	7.28	7.19	7.49	7.30	7.35	7.25	7.39	7.29	7.35	7.31	7.33
	DO (mg/l)	10.6	10.5	10.5	10.3	10.3	10.4	10.5	10.5	10.5	10.3	10.2	10.2	10.9	10.6
	Cond. (µmhos-cm)	109	114	111	113	118	111	113	113	115	111	114	112	116	114
	Temperature (°C)	14.2	14.1	14.1	14.3	14.8	14.5	14.9	14.6	14.7	14.2	14.8	14.1	14.7	14.0
	Tech. Initials	JS	JS	JS	RT	RT	BP	BP	BP	BP	RT	RT	RT	JS	MF

Dilution Water Batch #: 063 MHSW
Test Chamber: Env Rm C

QA Check: (N)

Sample Description:
Animal Source: Trout Lodge
Comments:

Date Received: 8/24/11 Date of Hatch: —

Nautilus Environmental
Washington Laboratory
5009 Pacific Hwy. E., Suite 2
Tacoma, WA 98424

Raw Data Sheet
Rainbow Trout
(Oncorhynchus mykiss)
Trout Embryo Test

Client Name: Pierce County **Test No.:** 1108-T034

Sample ID: LORES FIRST FLUSH

Embryos/Container

Conc.	Cont.	Rep.	Days 0	1	2	3	4	5	6	7	# Normal	# Abnormal	Mean % Viable
CON	201	1	30	30	30	30	30	30	30	30	25	5	
	202	2	30	30	30	30	30	30	30	30	30	0	
	203	3	30	30	30	30	30	30	30	30	29	1	
	204	4	30	30	30	30	30	30	30	30	27	3	
6.25	205	1	30	30	30	30	30	30	30	30	24	6	
	206	2	30	30	30	30	30	30	30	30	30	0	
	207	3	30	30	30	30	30	30	30	30	26	4	
	208	4	30	30	30	30	30	30	30	30	27	3	
12.5	209	1	30	30	30	30	30	30	30	30	29	1	
	210	2	30	30	30	30	30	30	30	30	30	0	
	211	3	30	30	30	30	30	30	30	30	28	2	
	212	4	30	30	30	30	30	30	30	30	28	2	
25	213	1	30	30	30	30	30	30	30	30	30	0	
	214	2	30	30	30	30	30	30	30	29	29	0	
	215	3	30	30	30	30	30	30	30	30	28	2	
	216	4	30	30	30	30	30	30	30	30	27	3	
50	217	1	30	30	30	30	30	30	30	29	28	1	
	218	2	30	30	30	30	30	30	29	29	28	1	
	219	3	30	30	30	30	30	30	30	30	26	4	
	220	4	30	30	30	30	30	30	30	30	29	1	
100	221	1	30	30	30	30	30	30	29	29	29	0	
	222	2	30	30	30	30	30	30	30	28	26	2	
	223	3	30	30	30	30	30	30	30	30	28	2	
	224	4	30	30	30	30	30	30	30	30	27	3	
		1											
		2											
		3											
		4											
		1											
		2											
		3											
		4											
Tech Initials			MF	JS	RT	JSP	JSP	RT	RT	MF	MF	MF	

QA Check: (M)

Comments: _____

COMM

CETIS Summary Report

Salmonid Embryo Survival and Development Test — **Nautilus Environmental WA**

Batch ID:	13-9148-9913	**Test Type:**	Survival-Development	**Analyst:**	Cat Curran
Start Date:	24 Aug-11 14:00	**Protocol:**	EC/EPS 1/RM/28	**Diluent:**	Mod-Hard Synthetic Water
Ending Date:	31 Aug-11 14:00	**Species:**	Oncorhynchus mykiss	**Brine:**	
Duration:	7d 0h	**Source:**	Trout Lodge Fish Farm	**Age:**	

Sample ID:	14-7976-6925	**Code:**	11-238	**Client:**	Pierce County - SW
Sample Date:	23 Aug-11 12:55	**Material:**	Stormwater	**Project:**	
Receive Date:	23 Aug-11 14:10	**Source:**	Pierce County Municipal Stormwater (WAR		
Sample Age:	25h (8.2 °C)	**Station:**	COMM		

Comparison Summary

Analysis ID	Endpoint	NOEL	LOEL	TOEL	PMSD	TU	Method
00-5819-5395	Development Rate $\alpha = 0.05$	50	100	70.71	9.14%	2	Dunnett's Multiple Comparison Test
15-1826-2812	$\alpha = 0.01$	100	>100	N/A	13.5%	1	Dunnett's Multiple Comparison Test
05-2720-0417	Survival Rate	100	>100	N/A	2.71%	1	Dunnett's Multiple Comparison Test

Point Estimate Summary

Analysis ID	Endpoint	Level	%	95% LCL	95% UCL	TU	Method
01-9748-0667	Development Rate	EC25	>100	N/A	N/A	<1	Linear Interpolation (ICPIN)
		EC50	>100	N/A	N/A	<1	
13-0564-9937	Survival Rate	EC25	>100	N/A	N/A	<1	Linear Interpolation (ICPIN)
		EC50	>100	N/A	N/A	<1	

Development Rate Summary

Conc-%	Control Type	Count	Mean	95% LCL	95% UCL	Min	Max	Std Err	Std Dev	CV%	%Effect
0	Dilution Water	4	0.975	0.9631	0.9869	0.9333	1	0.01596	0.03191	3.27%	0.0%
6.25		4	0.975	0.9563	0.9937	0.9	1	0.025	0.05	5.13%	0.0%
12.5		4	0.9057	0.8932	0.9183	0.8621	0.9333	0.01681	0.03363	3.71%	7.1%
25		4	0.9316	0.9106	0.9526	0.8621	1	0.02816	0.05632	6.05%	4.45%
50		4	0.9497	0.9201	0.9793	0.8333	1	0.03964	0.07927	8.35%	2.59%
100		4	0.8405	0.8131	0.8679	0.7667	0.9333	0.03671	0.07343	8.74%	13.79%

Survival Rate Summary

Conc-%	Control Type	Count	Mean	95% LCL	95% UCL	Min	Max	Std Err	Std Dev	CV%	%Effect
0	Dilution Water	4	1	1	1	1	1	0	0	0.0%	0.0%
6.25		4	0.9917	0.9854	0.9979	0.9667	1	0.008333	0.01667	1.68%	0.83%
12.5		4	0.975	0.9688	0.9812	0.9667	1	0.008333	0.01667	1.71%	2.5%
25		4	0.9833	0.9761	0.9905	0.9667	1	0.009623	0.01925	1.96%	1.67%
50		4	0.9917	0.9854	0.9979	0.9667	1	0.008333	0.01667	1.68%	0.83%
100		4	0.9917	0.9854	0.9979	0.9667	1	0.008333	0.01667	1.68%	0.83%

Salmonid Embryo Survival and Development Test — Nautilus Environmental WA

Development Rate Detail

Conc-%	Control Type	Rep 1	Rep 2	Rep 3	Rep 4
0	Dilution Water	0.9667	1	1	0.9333
6.25		1	0.9	1	1
12.5		0.8966	0.8621	0.9333	0.931
25		0.9333	0.931	1	0.8621
50		0.9655	0.8333	1	1
100		0.9333	0.7667	0.8	0.8621

Survival Rate Detail

Conc-%	Control Type	Rep 1	Rep 2	Rep 3	Rep 4
0	Dilution Water	1	1	1	1
6.25		1	1	1	0.9667
12.5		0.9667	0.9667	1	0.9667
25		1	0.9667	1	0.9667
50		0.9667	1	1	1
100		1	1	1	0.9667

Nautilus Environmental
Washington Laboratory

Client: Pierce County
Sample ID. COMM FIRST FLUSH
Test No: 1108-T033
Log-In#: 11-236

Initial and Final Chemistries
Seven Day Chronic Freshwater Bioassay
Start Date & Time: 8/24/11 1400
Stop Date & Time: 8/31/11 1400
Test Species: Oncorhynchus mykiss

Conc. or % CON	Days													
	0		1		2		3		4		5		6	
	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final
pH	7.34	7.31	7.32	7.23	7.32	7.40	7.35	7.47	7.44	7.42	7.45	7.36	7.63	7.44
DO (mg/l)	10.3	10.5	10.5	10.2	10.1	10.4	10.4	10.5	10.5	10.2	10.0	10.5	9.5	9.8
Cond. (µmhos-cm)	215	230	219	223	225	218	225	223	231	232	233	228	252	248
Temperature (°C)	13.6	13.9	14.0	14.5	14.8	13.8	14.7	13.5	14.2	14.5	14.2	14.3	13.9	14.4

6.25	Days													
	0		1		2		3		4		5		6	
	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final
pH	7.37	7.32	7.34	7.30	7.33	7.40	7.38	7.49	7.45	7.44	7.46	7.37	7.60	7.47
DO (mg/l)	10.3	10.5	10.5	10.4	10.1	10.5	10.5	10.5	10.5	10.2	10.0	10.5	9.7	10.1
Cond. (µmhos-cm)	205	215	211	213	218	209	220	214	218	219	220	218	241	238
Temperature (°C)	13.5	13.7	14.0	14.1	14.8	13.9	14.8	13.6	14.3	14.1	14.1	14.0	14.0	14.3

12.5	Days													
	0		1		2		3		4		5		6	
	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final
pH	7.36	7.34	7.33	7.31	7.31	7.40	7.34	7.48	7.42	7.44	7.44	7.38	7.52	7.50
DO (mg/l)	10.5	10.5	10.5	10.1	10.0	10.4	10.3	10.5	10.4	10.3	9.9	10.5	9.4	10.3
Cond. (µmhos-cm)	197	209	204	206	212	202	213	207	209	211	212	215	231	228
Temperature (°C)	13.5	13.8	14.2	14.1	14.8	13.8	14.9	13.6	14.0	14.0	14.1	13.9	14.1	14.2

25	Days													
	0		1		2		3		4		5		6	
	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final
pH	7.32	7.35	7.29	7.32	7.26	7.38	7.26	7.42	7.32	7.43	7.39	7.37	7.45	7.51
DO (mg/l)	10.4	10.5	10.4	10.2	9.8	10.3	10.2	10.5	10.2	10.3	9.6	10.6	9.5	10.2
Cond. (µmhos-cm)	179	190	189	189	197	188	196	191	194	193	194	193	213	209
Temperature (°C)	13.3	13.7	14.3	14.2	14.7	13.9	14.9	13.7	14.1	14.0	14.1	14.0	14.1	14.2

50	Days													
	0		1		2		3		4		5		6	
	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final
pH	7.14	7.35	7.25	7.30	7.12	7.34	7.07	7.40	7.07	7.39	7.17	7.34	7.10	7.50
DO (mg/l)	10.2	10.5	10.1	10.3	9.3	10.3	9.8	10.4	9.6	10.0	9.0	10.4	9.2	10.2
Cond. (µmhos-cm)	148	161	158	161	164	154	162	156	161	160	161	158	174	173
Temperature (°C)	13.3	13.8	14.3	14.2	14.7	14.0	14.9	13.6	14.3	14.2	14.3	14.0	14.0	14.1

100	Days													
	0		1		2		3		4		5		6	
	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final
pH	6.74	7.10	6.70	7.24	6.67	7.08	6.68	6.83	6.64	7.27	6.70	7.20	6.50	7.47
DO (mg/l)	9.0	10.1	8.5	10.1	8.2	10.4	7.4	8.4	6.5	10.1	6.0	9.9	6.9	10.0
Cond. (µmhos-cm)	85	91	89	97	94	93	92	89	89	92	89	90	91	99
Temperature (°C)	13.5	13.9	14.5	14.2	14.6	14.1	15.0	13.6	14.5	14.0	14.5	14.3	15.0	14.2
Tech. Initials	JJ	JJ	JJ	RT	RT	BP	BP	BP	BP	MW	MW	MW	JJ	AF

Dilution Water Batch #: 063 MHSW
Test Chamber: Env Rm C

QA Check: MW

Sample Description:
Animal Source: Trout Lodge
Comments:

Date Received: 8/24/11 Date of Hatch: —

Nautilus Environmental
Washington Laboratory
5009 Pacific Hwy. E., Suite 2
Tacoma, WA 98424

Raw Data Sheet
Rainbow Trout
(Oncorhynchus mykiss)
Trout Embryo Test

Client Name: Pierce County **Test No.:** 1108-T033

Sample ID: COMM FIRST FLUSH

Embryos/Container

Conc.	Cont.	Rep.	Days 0	1	2	3	4	5	6	7	# Normal	# Abnormal	Mean % Viable
CON	301	1	30	30	30	30	30	30	30	30	29	1	
	302	2	30	30	30	30	30	30	30	30	30	0	
	303	3	30	30	30	30	30	30	30	30	30	0	
	304	4	30	30	30	30	30	30	30	30	28	2	
6.25	305	1	30	30	30	30	30	30	30	30	30	2	
	306	2	30	30	30	30	30	30	30	30	27	3	
	307	3	30	30	30	30	30	30	30	30	30	0	
	308	4	30	30	30	30	30	29	29	29	29	0	
12.5	309	1	30	30	30	30	30	30	29	29	26	3	
	310	2	30	30	30	30	30	30	30	29	25	4	
	311	3	30	30	30	30	30	30	30	30	28	2	
	312	4	30	29	29	29	29	29	29	29	27	2	
25	313	1	30	30	30	30	30	30	30	30	28	2	
	314	2	30	30	30	30	29	29	29	29	27	2	
	315	3	30	30	30	30	30	30	30	30	30	0	
	316	4	30	30	30	30	29	29	29	29	25	4	
50	317	1	30	30	30	30	30	30	30	29	28	1	
	318	2	30	30	30	30	30	30	30	30	25	5	
	319	3	30	30	30	30	30	30	30	30	30	0	
	320	4	30	30	30	30	30	30	30	30	30	0	
100	321	1	30	30	30	30	30	30	30	30	28	2	
	322	2	30	30	30	30	30	30	30	30	23	7	
	323	3	30	30	30	30	30	30	30	30	24	6	
	324	4	30	30	30	30	30	30	30	29	25	4	
		1											
		2											
		3											
		4											
		1											
		2											
		3											
		4											
Tech Initials			ME	JS	CT	BSP	BSP	(M)	(M)	ME	ME	ME	

QA Check: (M)

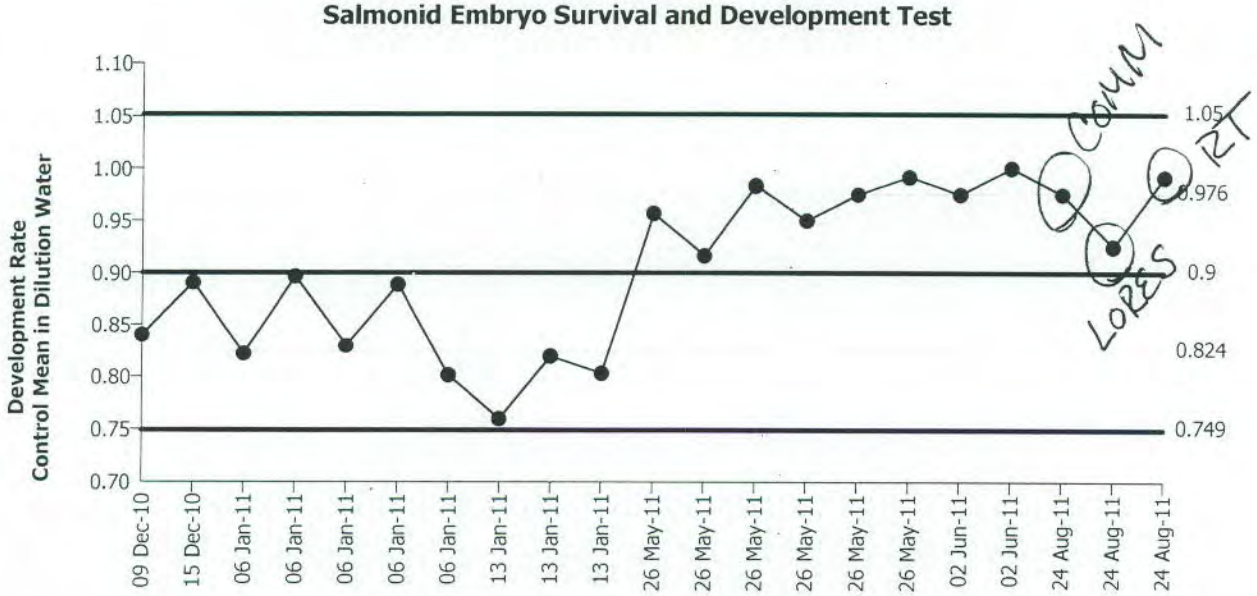
Comments:

Appendix C
Control QC Plot

Salmonid Embryo Survival and Development Test — Nautilus Environmental WA

Test Type:	Survival-Development	**Organism:**	Oncorhynchus mykiss (Rainbow Tro	**Material:**	All Materials
Protocol:	EC/EPS 1/RM/28	**Endpoint:**	Development Rate	**Source:**	All SampleID Sources

Salmonid Embryo Survival and Development Test



Mean:	0.9001	**Count:**	20	**-1s Warning Limit:**	0.8244	**-2s Action Limit:**	0.7487
Sigma:	0.07572	**CV:**	8.41%	**+1s Warning Limit:**	0.9758	**+2s Action Limit:**	1.052

Quality Control Data

Point	Year	Month	Day	QC Data	Delta	Sigma	Warning	Action	Test ID	Analysis ID
1	2010	Dec	9	0.8402	-0.0599	-0.7911			11-6961-0538	
2			15	0.8909	-0.0092	-0.1215			19-0590-0808	
3	2011	Jan	6	0.8222	-0.0779	-1.029	(-)		19-6411-6735	
4			6	0.8968	-0.0033	-0.04358			13-9924-3103	
5			6	0.8298	-0.0703	-0.9284			13-6888-4572	
6			6	0.8891	-0.011	-0.1453			03-2374-4587	
7			6	0.8012	-0.0989	-1.306	(-)		00-2805-9161	
8			13	0.7598	-0.1403	-1.853	(-)		03-0330-2750	
9			13	0.8196	-0.0805	-1.063	(-)		21-0624-5132	
10			13	0.803	-0.0971	-1.282	(-)		05-9744-8253	
11		May	26	0.9569	0.0568	0.7501			01-4238-5878	
12			26	0.9167	0.0166	0.2192			06-7406-7916	
13			26	0.9833	0.0832	1.099	(+)		14-0794-4631	
14			26	0.95	0.0499	0.659			08-6056-6483	
15			26	0.975	0.0749	0.9892			02-9870-0536	
16			26	0.9917	0.0916	1.21	(+)		02-6305-8372	
17		Jun	2	0.975	0.0749	0.9892			11-6590-4701	
18			2	1	0.0999	1.319	(+)		05-1781-0260	
19		Aug	24	0.975	0.0749	0.9892			08-7317-1377	
20			24	0.925	0.0249	0.3288			17-9413-5277	
21			24	0.9917	0.0916	1.21	(+)		09-4243-1992	

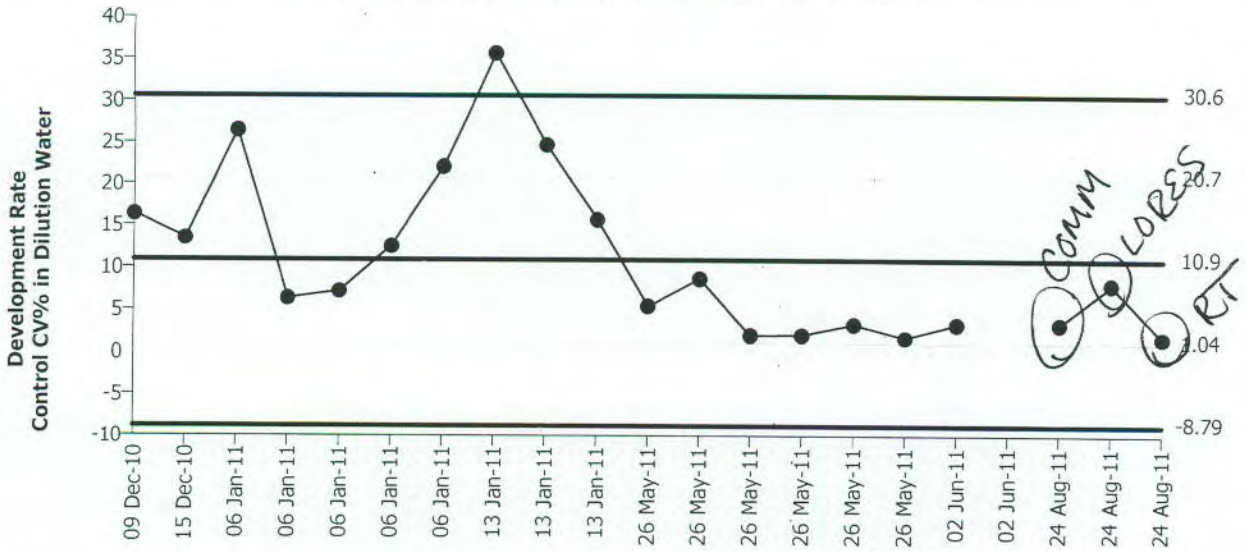
Analyst: _____ QA: _____

Salmonid Embryo Survival and Development Test

Nautilus Environmental WA

Test Type:	Survival-Development	**Organism:**	Oncorhynchus mykiss (Rainbow Tro	**Material:**	All Materials
Protocol:	EC/EPS 1/RM/28	**Endpoint:**	Development Rate	**Source:**	All SampleID Sources

Salmonid Embryo Survival and Development Test



Mean:	10.88	**Count:**	20	**-1s Warning Limit:**	1.044	**-2s Action Limit:**	-8.792
Sigma:	9.836	**CV:**	90.40%	**+1s Warning Limit:**	20.72	**+2s Action Limit:**	30.55

Quality Control Data

Point	Year	Month	Day	QC Data	Delta	Sigma	Warning	Action	Test ID	Analysis ID
1	2010	Dec	9	16.33	5.45	0.5541			11-6961-0538	
2			15	13.47	2.59	0.2633			19-0590-0808	
3	2011	Jan	6	26.41	15.53	1.579	(+)		19-6411-6735	
4			6	6.331	-4.549	-0.4625			13-9924-3103	
5			6	7.186	-3.694	-0.3756			13-6888-4572	
6			6	12.49	1.61	0.1637			03-2374-4587	
7			6	22.02	11.14	1.133	(+)		00-2805-9161	
8			13	35.57	24.69	2.51	(+)	(+)	03-0330-2750	
9			13	24.64	13.76	1.399	(+)		21-0624-5132	
10			13	15.64	4.76	0.4839			05-9744-8253	
11		May	26	5.405	-5.475	-0.5566			01-4238-5878	
12			26	8.656	-2.224	-0.2261			06-7406-7916	
13			26	1.957	-8.923	-0.9072			14-0794-4631	
14			26	2.026	-8.854	-0.9002			08-6056-6483	
15			26	3.273	-7.607	-0.7734			02-9870-0536	
16			26	1.681	-9.199	-0.9352			02-6305-8372	
17		Jun	2	3.273	-7.607	-0.7734			11-6590-4701	
18			2	0	-10.88	-1.106	(-)		05-1781-0260	
19		Aug	24	3.273	-7.607	-0.7734			08-7317-1377	
20			24	7.99	-2.89	-0.2938			17-9413-5277	
21			24	1.681	-9.199	-0.9352			09-4243-1992	

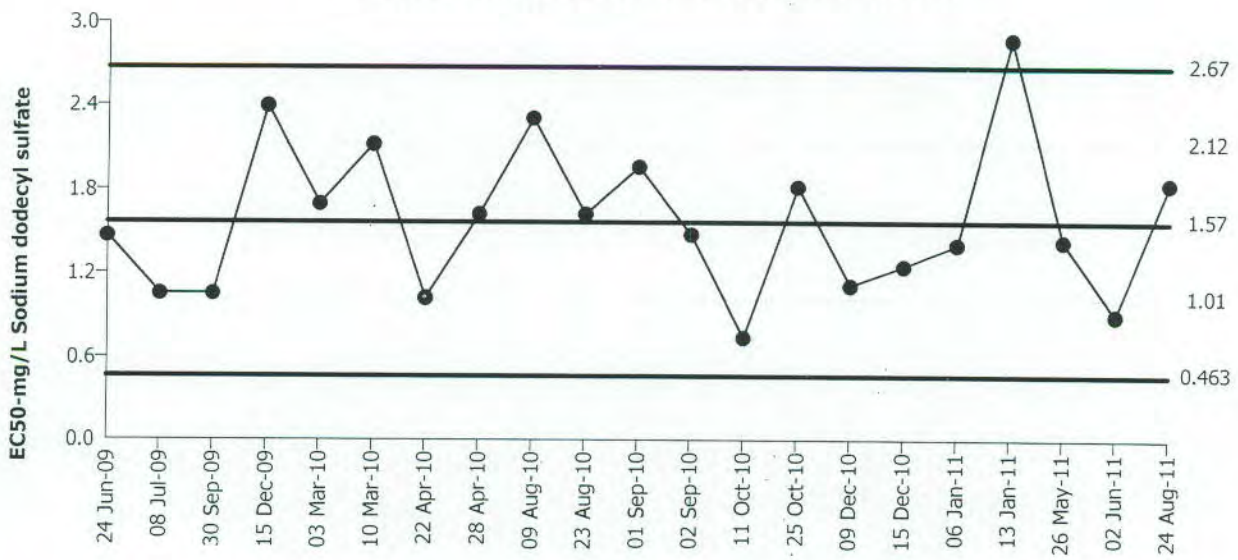
Appendix D
Reference Toxicant Test

Salmonid Embryo Survival and Development Test

Nautilus Environmental WA

Test Type:	Survival-Development	**Organism:**	Oncorhynchus mykiss (Rainbow Tro	**Material:**	Sodium dodecyl sulfate
Protocol:	EC/EPS 1/RM/28	**Endpoint:**	Combined Development	**Source:**	Reference Toxicant-REF

Salmonid Embryo Survival and Development Test



Mean:	1.567	**Count:**	20	**-1s Warning Limit:**	1.015	**-2s Action Limit:**	0.463
Sigma:	0.552	**CV:**	35.20%	**+1s Warning Limit:**	2.119	**+2s Action Limit:**	2.671

Quality Control Data

Point	Year	Month	Day	QC Data	Delta	Sigma	Warning	Action	Test ID	Analysis ID
1	2009	Jun	24	1.468	-0.0989	-0.1792			14-9518-3179	08-5131-4931
2		Jul	8	1.054	-0.5127	-0.9288			12-5239-8398	14-3555-6901
3		Sep	30	1.055	-0.5124	-0.9283			21-3690-2403	07-3566-4008
4		Dec	15	2.394	0.8267	1.498	(+)		13-8589-7680	12-0408-0107
5	2010	Mar	3	1.698	0.1308	0.237			04-2049-9485	15-9318-5719
6			10	2.122	0.5551	1.006	(+)		14-5366-5433	17-5366-0459
7		Apr	22	1.024	-0.5433	-0.9843			14-3935-8341	02-7510-4136
8			28	1.623	0.05613	0.1017			12-1077-1008	14-4510-6639
9		Aug	9	2.306	0.7385	1.338	(+)		07-8456-2911	19-1323-0255
10			23	1.623	0.05613	0.1017			14-3745-7869	09-9058-1844
11		Sep	1	1.961	0.3941	0.714			11-0756-8084	06-0529-0065
12			2	1.478	-0.08931	-0.1618			07-4392-7752	03-6365-5149
13		Oct	11	0.7433	-0.8237	-1.492	(-)		07-0606-3388	05-6680-1950
14			25	1.821	0.2539	0.4599			01-4882-1445	01-9014-5536
15		Dec	9	1.114	-0.4534	-0.8213			08-4175-8660	09-4729-9037
16			15	1.251	-0.3158	-0.5721			19-0590-0808	15-1777-9718
17	2011	Jan	6	1.405	-0.1621	-0.2936			00-2805-9161	17-1635-2519
18			13	2.87	1.303	2.36	(+)	(+)	21-0624-5132	06-2825-3209
19		May	26	1.429	-0.1377	-0.2495			01-4238-5878	12-9006-8038
20		Jun	2	0.9002	-0.6668	-1.208	(-)		11-6590-4701	18-4745-3619
21		Aug	24	1.843	0.2762	0.5004			09-4243-1992	12-2546-8094

CETIS Summary Report

Salmonid Embryo Survival and Development Test — **Nautilus Environmental WA**

Batch ID:	06-6745-3747	**Test Type:**	Survival-Development	**Analyst:**	Cat Curran
Start Date:	24 Aug-11 14:00	**Protocol:**	EC/EPS 1/RM/28	**Diluent:**	Mod-Hard Synthetic Water
Ending Date:	31 Aug-11 13:00	**Species:**	Oncorhynchus mykiss	**Brine:**	
Duration:	6d 23h	**Source:**	Trout Lodge Fish Farm	**Age:**	

Sample ID:	20-3880-1451	**Code:**	RT082411OME	**Client:**	Reference Toxicant Test
Sample Date:	24 Aug-11	**Material:**	Sodium dodecyl sulfate	**Project:**	
Receive Date:	24 Aug-11	**Source:**	Reference Toxicant		
Sample Age:	14h	**Station:**			

Comparison Summary

Analysis ID	Endpoint	NOEL	LOEL	TOEL	PMSD	TU	Method
12-5525-5399	Combined Development	1.25	2.5	1.768	9.96%		Steel Many-One Rank Test

Point Estimate Summary

Analysis ID	Endpoint	Level	mg/L	95% LCL	95% UCL	TU	Method
12-2546-8094	Combined Development	EC50	1.843	1.78	1.909		Trimmed Spearman-Kärber

Combined Development Summary

Conc-mg/L	Control Type	Count	Mean	95% LCL	95% UCL	Min	Max	Std Err	Std Dev	CV%	%Effect
0	Dilution Water	4	0.975	0.9631	0.9869	0.9333	1	0.01596	0.03191	3.27%	0.0%
0.625		4	0.95	0.9339	0.9661	0.9	1	0.02152	0.04303	4.53%	2.56%
1.25		4	0.975	0.9631	0.9869	0.9333	1	0.01596	0.03191	3.27%	0.0%
2.5		4	0.075	0.01899	0.131	0	0.3	0.075	0.15	200.0%	92.31%
5		4	0	0	0	0	0	0	0		100.0%
10		4	0	0	0	0	0	0	0		100.0%

Combined Development Detail

Conc-mg/L	Control Type	Rep 1	Rep 2	Rep 3	Rep 4
0	Dilution Water	0.9333	0.9667	1	1
0.625		0.9667	0.9333	0.9	1
1.25		1	0.9667	1	0.9333
2.5		0	0	0.3	0
5		0	0	0	0
10		0	0	0	0

Nautilus Environmental
Washington Laboratory

Client: Reference Toxicant
Sample ID. 10mg/L SDS
Test No: RT 082411 OME
Log-In#: — —

Initial and Final Chemistries
Seven Day Chronic Freshwater Bioassay
Start Date & Time: 8/24/11 1400
Stop Date & Time: 8/31/11 1300
Test Species: Oncorhynchus mykiss

Conc. or % CON	Days													
	0		1		2		3		4		5		6	
	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final
pH	7.15	7.38	7.28	7.31	7.28	7.47	7.45	7.50	7.35	7.29	7.25	7.37	7.55	7.23
DO (mg/l)	10.4	10.4	10.5	10.3	10.1	10.5	10.5	10.5	10.4	10.1	9.9	10.6	9.7	9.0
Cond. (µmhos-cm)	219	229	221	221	224	223	225	229	232	228	228	227	258	248
Temperature (°C)	14.4	14.4	14.5	14.6	14.8	14.3	13.9	14.3	14.2	14.3	14.4	14.6	14.1	14.1

0.625	Days													
	0		1		2		3		4		5		6	
	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final
pH	7.21	7.38	7.32	7.33	7.33	7.48	7.50	7.52	7.41	7.32	7.31	7.40	7.58	7.26
DO (mg/l)	10.6	10.5	10.5	10.4	10.2	10.4	10.5	10.5	10.5	10.2	10.4	10.6	9.6	9.5
Cond. (µmhos-cm)	220	227	223	220	225	218	221	225	233	226	227	226	259	244
Temperature (°C)	14.2	14.5	14.6	14.2	14.8	14.2	14.0	14.0	14.3	14.2	14.4	14.6	14.0	14.1

1.25	Days													
	0		1		2		3		4		5		6	
	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final
pH	7.26	7.39	7.35	7.37	7.35	7.49	7.50	7.52	7.27	7.34	7.35	7.42	7.59	7.27
DO (mg/l)	10.5	10.5	10.5	10.3	10.1	10.5	10.5	10.5	10.5	10.3	10.3	10.5	9.9	9.5
Cond. (µmhos-cm)	220	228	223	222	226	218	221	225	231	226	226	226	258	245
Temperature (°C)	14.4	14.4	14.6	14.1	14.8	13.8	13.8	14.1	13.9	14.2	14.4	14.5	14.0	14.1

2.5	Days													
	0		1		2		3		4		5		6	
	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final
pH	7.26	7.39	7.37	7.40	7.37	7.50	7.51	7.52	7.32	7.35	7.38	7.36	7.61	7.29
DO (mg/l)	10.3	10.4	10.5	10.2	10.1	10.4	10.4	10.4	10.5	10.2	10.4	9.4	9.8	8.9
Cond. (µmhos-cm)	225	233	224	224	226	222	222	226	233	231	226	232	260	248
Temperature (°C)	14.4	14.6	14.6	14.1	14.8	13.8	13.9	14.0	13.9	14.2	14.4	14.3	14.0	14.0

5	Days													
	0		1		2		3		4		5		6	
	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final
pH	7.29	7.40	7.40	7.42	7.38	7.50	7.50	7.51	7.32					
DO (mg/l)	10.3	10.5	10.4	10.3	10.2	10.3	10.4	10.5	10.5					
Cond. (µmhos-cm)	221	230	224	231	227	226	223	230	234					
Temperature (°C)	14.5	14.5	14.6	14.3	14.8	14.0	13.9	14.1	13.9					

10	Days													
	0		1		2		3		4		5		6	
	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final	init.	final
pH	7.30	7.41	7.42	7.42	7.40									
DO (mg/l)	10.6	10.4	10.5	10.3	10.1									
Cond. (µmhos-cm)	223	240	226	225	228									
Temperature (°C)	14.5	14.4	14.6	14.3	14.8									
Tech. Initials	SS	SS	SS	SS	SS	BP	BP	BP	BP	MF	MF	SS	SS	MF

Dilution Water Batch #: 063 MHSW
Test Chamber: Env Rm C

QA Check: MW

Sample Description:
Animal Source: Trout Lodge
Date Received: 8/24/11 Date of Hatch: —
Comments:

Nautilus Environmental
Washington Laboratory
5009 Pacific Hwy. E., Suite 2
Tacoma, WA 98424

Raw Data Sheet
Rainbow Trout
(Oncorhynchus mykiss)
Trout Embryo Test

Client Name: Reference Toxicant **Test No.:** RT082411OME

Sample ID: 10mg/L SDS

Embryos/Container

Conc.	Cont.	Rep.	Days 0	1	2	3	4	5	6	7	# Normal	# Abnormal	Mean % Viable
CON	101	1	30	30	30	30	30	30	29	28	28	0	
	102	2	30	30	30	30	30	30	30	30	29	1	
	103	3	30	30	30	30	30	30	30	30	30	0	
	104	4	30	30	30	30	30	30	30	30	30	0	
0.625	105	1	30	30	30	30	30	30	30	30	29	1	
	106	2	30	30	30	30	30	30	29	29	28	1	
	107	3	30	30	30	30	30	30	30	30	27	3	
	108	4	30	30	30	30	30	30	30	30	30	0	
1.25	109	1	30	30	30	30	30	30	30	30	30	0	
	110	2	30	30	30	30	30	30	29	29	29	0	
	111	3	30	30	30	30	30	30	30	30	30	0	
	112	4	30	30	30	30	30	30	29	29	28	1	
2.5	113	1	30	30	30	30	28	2	1	1	0	1	
	114	2	30	30	30	30	30	2	0	—	—	—	
	115	3	30	30	30	30	30	15	11	10	9	1	
	116	4	30	30	30	30	30	1	1	1	0	1	
5	117	1	30	30	2	0	0						
	118	2	30	30	15	0	0						
	119	3	30	30	29	7	0						
	120	4	30	30	14	2	0						
10	121	1	30	1	0								
	122	2	30	23	0								
	123	3	30	26	0								
	124	4	30	0	0								
		1											
		2											
		3											
		4											
		1											
		2											
		3											
		4											
Tech Initials			MF	SS	SS	BP	BP	MF	SS	MF	MF	MF	

QA Check: (M)

Comments:

Trout Embryo Development Fertilization Worksheet

Test Date: 8/24/11
Animal Source: Trout Lodge
Date Received: 8/24/11

Number of females: 4

Sperm condition: good

Egg ripeness and condition: A → D good

Eggs rinsed with 13.68g/L NaHCO3: NO

Number of fertilization dishes: 4

Fertilization start time: 1345

Amount of sperm per fertilization dish: 750µl

Fertilization end time: 1400

Test initiation time: 1400

Comments and additional notes:

Nautilus Environmental
Washington Laboratory
5009 Pacific Hwy. E., Suite 2
Tacoma, WA 98424
(253) 922-4296

Appendix E
Sample Check-In Sheet

Nautilus Environmental
5009 Pacific Hwy East, Ste. 2
Tacoma, WA 98424

Sample Check-In Information

Client: Pierce County

Tests Performed: Dm-e

Test ID No(s).: 1108-T033, 1108-T034

Sample Description:

Sample ID:	LoRes First Flush	UrbM First Flush	LoRes Hardness	UrbM Hardness
Log-in No. (10-xxxx):	11-237	11-238		
Sample Collection Date & Time:	8/23/11 1120	8/23/11 1255		
Sample Receipt Date & Time:	8/23/11 1410	8/23/11 1410		
Check-in Temperature (°C)	8.7	8.2		
Temperature OK?	(Y) N	(Y) N	Y N	Y N
DO (mg/L)	11.0	9.2		
pH (units)	7.06	6.92		
Conductivity (µS/cm)	104 86	86		
Salinity (ppt)	—	—		
Tit. Vol / Sam. Vol. / Alkalinity (mg/L)*	1.2 / 25 / 148	0.8 / 25 / 132	1 / 1	1 / 1
Tit. Vol. / Sam. Vol. / Hardness (mg/L)* [a]	1.4 / 25 / 156	1.1 / 25 / 144	1.7 / 25 / 15 60	0.4 / 25 / 16
Total Chlorine (mg/L)	<0.03	<0.03		
Total Ammonia (mg/L)	<1.0	<1.0		
Technician Initials	SB/ET	SB/ET	ET	ET

* = mg/L as $CaCO_3$, [a] = Measured for freshwater samples only, NA = Not Applicable, NM = Not Measured

COC Complete? Y or N
1 (Y) 2 ___ 3 ___

Filtration? Y (N)
Pore Size: ___
Organisms or Debris

Aeration? Y (N)
Length of Time: ___
Final DO: ___
Final pH: ___

Hardness Adjustment? Y (N)
If adjusted, please see worksheet for details.

Freshwater Tests:

Control/Dilution Water Source: test type: Dm-e 8:2 (DMW) (MHW) Other: ___ Alkalinity: ___ Hardness: ___

Control/Dilution Water Source: test type: ___ 8:2 (DMW) MHW Other: ___ Alkalinity: ___ Hardness: ___

Additional Control? Y (N) = ___ Alkalinity: ___ Hardness: ___

Marine Tests:

Control/Dilution Water Source: test type: ___ ART SW NAT SW Alkalinity: ___ Salinity: ___

Control/Dilution Water Source: test type: ___ ART SW NAT SW Alkalinity: ___ Salinity: ___

Additional Control? Y N = ___ Alkalinity: ___ Salinity: ___

Sample Salted w/ artificial salt? Y N If yes, what ppt? ___ test type: ___

Sample salted w/brine? Y N If yes, what ppt? ___ test type: ___

Sub-samples for additional chemistry:

Comments: Temperature for grab sample must be 0-20°C if received within 1 hour of collection time, 0-12°C if effluent received within 4 hours of collection time, and 0-6°C for all other samples.

QC Check: (M)

Appendix F
Chain-of-Custody Form

Nautilus Environmental

Chain of Custody

TESTING LOCATION (Please Check Box)

☐ **California**
5550 Morehouse Drive, Suite 150
San Diego, CA 92121
Phone 858.587.7333
Fax 858.587.3961

☑ **Washington**
5009 Pacific Highway East, Suite 2
Tacoma, WA 98424
Phone 253.922.4296
Fax 253.922.5814

☐ **British Columbia**
8664 Commerce Court
Burnaby, British Columbia, Canada V5A 4N3
Phone 604.420.8773
Fax 604.357.1361

Date 8/23/11 Page 1 of 1

Sample Collection By: Diane Hennebert, Berl Eldridge

Report to:
Company: Pierce Co.
Address:
City/State/Zip:
Contact:
Phone:
Email:

Invoice To:
Company:
Address:
City/State/Zip:
Contact:
Phone:
Email:

ANALYSES REQUIRED

	SAMPLE ID	DATE	TIME	MATRIX	CONTAINER TYPE	NO. OF CONTAINERS	COMMENTS	trout embryo									Receipt Temperature (°C)
1	LURES FIRST FLUSH	8/23/11	11:20	H_2O	20 L Cubitainer	3		X		11-237							8.7
2	COMM FIRST FLUSH	8/23/11	12:55	H_2O	20 L Cubitainer	2	Canyon receiving water	X		11-238							8.2
3																	
4																	
5																	
6																	
7																	
8																	
9																	
10																	

PROJECT INFORMATION
Client:
PO No.:
Shipped Via: Client

SAMPLE RECEIPT
Total No. of Containers:
Received Good Condition? Y
Matches Test Schedule? Y

RELINQUISHED BY (CLIENT)
(Signature) Diane Klavano (Time) 8/23/11
(Printed Name) Diane Klavano (Date) 8/23/11
(Company) Pierce Co

RELINQUISHED BY (COURIER)
(Signature) (Time)
(Printed Name) (Date)
(Company)

RECEIVED BY (COURIER)
(Signature) (Time)
(Printed Name) (Date)
(Company)

RECEIVED BY (LABORATORY)
(Signature) Cat Curran (Time) 1410
(Printed Name) Cat Curran (Date) 8.23.11
(Company) Nautilus

SPECIAL INSTRUCTIONS/COMMENTS:

Additional costs may be required for sample disposal or storage. Payment net 30 unless otherwise contracted.

DISTRIBUTION: WHITE - Nautilus Environmental, COLOR - Originator